EXAM PACK

EEE329

Basic Communication Engineering



Exam Pack

Session 2020–21 *Solutions and formula sheet of the 3rd Year 2nd Semester Final Examination, 2023 and Term Tests 1–2*

IN THIS PACK

1 PAST PAPER SOLUTIONS
Question Paper Solutions 2020–21 — pages 3--74 — 72 pages

2 FORMULA SHEET
Formula Sheet — pages 75--88 — 14 pages

Tap a document to open it. Each one keeps its own cover, contents and links.

Prepared by Nur Mohammad

November 2026

IN THIS PACK

Exam Pack

Scan to open the website

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1 PAST PAPER SOLUTIONS
Question Paper Solutions 2020–21

pages 3--74

Question Paper 5
1(a) Entropy of a Single-Bit Binary Memoryless Source 12
1(b) Major Bands of the EM Spectrum 14
1(c) Thermal Noise and Transit-Time Noise 16
1(d) AM Modulation Index and Side Frequencies . . . 17
1(e) AM Sidebands and Total Power 19
2(a) Multiplexing versus Multiple Access 21
2(b) Minimum Bandwidth of an FDM Link 22
2(c) Relation Between FM and PM 23
2(d) FM Frequency Deviation and Modulation Index 25
3(a) Slope Overload, Granular Noise and ADM . . . 27
3(b) Delta Modulation Transmitter and Receiver . . 29
3(c) Block Diagram of a PCM System 30
3(d) Information Rate and Channel Capacity 31
4(a) Superheterodyne Receiver 34
4(b) Recovering the Message from a VSB Signal . . 36
4(c) Nyquist Sampling Theorem and the Interpolation Formula 40
4(d) Carson's Rule and its Limitations 43
5(a) TDMA versus CDMA 45
5(b) Channel Capacity and Minimum SNR 46
5(c) Minimum Cable Bandwidth for the ECG TDM System 47
5(d) Coherent Detection of DSB-SC and the Lowest Carrier 49
6(a) FSK Generation, Spectrum and Bandwidth . . 53
6(b) Coherent Detection of QAM 55
6(c) Frequency Hopping Spread Spectrum 57
6(d) DPSK Encoder and Receiver Table for '1010011100' 58
7(a) Entropy of a Binary Memoryless Source and its Curve 61
7(b) Distortion and the Conditions for Distortionless Transmission 62
7(c) Attenuation, Channel Capacity and Required Signal Level 64
7(d) Equivalent Noise Resistance of Cascaded Amplifiers 65
8(a) DSB-WC and SSB-SC Spectra of the Triangular Message Spectrum 68
8(b) Effect of Phase and Frequency Error on Coherent Detection of AM 70
8(c) AM Antenna Current and Percentage Modulation 71
8(d) Pre-emphasis and De-emphasis 72
8(e) FM Bandwidth by Carson's Rule 74

2 FORMULA SHEET
Formula Sheet

pages 75--88

1 Information and channel capacity 78
2 Noise 79
3 Amplitude modulation 79
4 Angle modulation 80
5 Sampling, PCM and delta modulation 81
6 Multiplexing 82
7 Receivers and digital modulation 82
8 Key formulas 84
9 Symbol index 85

PAST PAPER SOLUTIONS

EEE329

Basic Communication Engineering

Question Paper Solutions

Session 2020–21

Final Examination, 2023 (Q1–Q6) and Term Tests 1–2 (Q7–Q8)

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	25 marks	25 marks

Q7	Q8
20 marks	25 marks

EACH ANSWER

Theory → Answer → Keyword Explanation → Notes and Remarks → Final Concise Answer

Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

The paper at a glance

3 h	100	4 of 6	45 min
Total time	Total marks	Questions to answer	Per question

Time = marks × 1.8 min (180 min for 100 marks); a part with its own time (a term test) uses that. Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
Part-A · Answer any two of the following questions				
Q1	(a) Entropy of a Single-Bit Binary Memoryless Source	5	9 min	☐
	(b) Major Bands of the EM Spectrum	6	11 min	
	(c) Thermal Noise and Transit-Time Noise	4	7 min	
	(d) AM Modulation Index and Side Frequencies	4	7 min	
	(e) AM Sidebands and Total Power	6	11 min	
Q2	(a) Multiplexing versus Multiple Access	5	9 min	☐
	(b) Minimum Bandwidth of an FDM Link	5	9 min	
	(c) Relation Between FM and PM	5	9 min	
	(d) FM Frequency Deviation and Modulation Index	10	18 min	
Q3	(a) Slope Overload, Granular Noise and ADM	5	9 min	☐
	(b) Delta Modulation Transmitter and Receiver	5	9 min	
	(c) Block Diagram of a PCM System	5	9 min	
	(d) Information Rate and Channel Capacity	10	18 min	
Part-B · Answer any two of the following questions				
Q4	(a) Superheterodyne Receiver	7	13 min	☐
	(b) Recovering the Message from a VSB Signal	7	13 min	
	(c) Nyquist Sampling Theorem and the Interpolation Formula	7	13 min	
	(d) Carson's Rule and its Limitations	4	7 min	
Q5	(a) TDMA versus CDMA	5	9 min	☐
	(b) Channel Capacity and Minimum SNR	6	11 min	
	(c) Minimum Cable Bandwidth for the ECG TDM System	7	13 min	
	(d) Coherent Detection of DSB-SC and the Lowest Carrier	7	13 min	
Q6	(a) FSK Generation, Spectrum and Bandwidth	8	14 min	☐
	(b) Coherent Detection of QAM	7	13 min	
	(c) Frequency Hopping Spread Spectrum	6	11 min	
	(d) DPSK Encoder and Receiver Table for '1010011100'	4	7 min	
Term Test 1 · 4 marks · 40 min				
Q7	(a) Entropy of a Binary Memoryless Source and its Curve	5	50 min	☐
	(b) Distortion and the Conditions for Distortionless Transmission	5	50 min	
	(c) Attenuation, Channel Capacity and Required Signal Level	5	50 min	
	(d) Equivalent Noise Resistance of Cascaded Amplifiers	5	50 min	
Term Test 2 · 4 marks · 40 min				
Q8	(a) DSB-WC and SSB-SC Spectra of the Triangular Message Spectrum	5	50 min	☐
	(b) Effect of Phase and Frequency Error on Coherent Detection of AM	5	50 min	
	(c) AM Antenna Current and Percentage Modulation	5	50 min	
	(d) Pre-emphasis and De-emphasis	5	50 min	
	(e) FM Bandwidth by Carson's Rule	5	50 min	

Formulas and conventions follow Lathi and Ding, Modern Digital and Analog Communication Systems (4th ed.); 1(b), 2(a), 2(b), 5(a), 5(b), 6(c) and 7(c) follow Forouzan, Data Communications and Networking; 1(c)(ii) and 7(d) follow Kennedy and Davis, Electronic Communication Systems.

Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
3rd Year 2nd Semester Final Examination, 2023
Course Code: EEE 329, Session: 2020-2021
Course Title: Basic Communication Engineering

Total Marks: 100 **Credits: 3.0** **Total Time: 3 Hours**

Part-A

[Answer any two of the following questions]

1. (a) Find the entropy of single bit binary memoryless source. Also draw the curve of entropy Vs. probability for that source. 5

   (b) Briefly explain the three major bands of the electromagnetic (EM) spectrum-radio waves, microwaves and infra-red waves-in terms of their bandwidth, propagation mode and applications. 6

   (c) Write short note on-

   (i) Thermal Noise
   (ii) Transit Time Noise 4

   (d) A carrier wave of frequency 10 MHz and peak value of 10 V is amplitude modulated by a 5 kHz sine wave of amplitude 6 V. Determine the modulation index, side frequencies and amplitude of side frequencies. 4

   (e) An audio frequency signal $10\sin(2\pi 500t)$ is used to amplitude modulated a carrier of $50\sin(2\pi 105t)$ [*sic*: should be $2\pi 10^5 t$]. Calculate:

   (i) Modulation index
   (ii) Side-band frequencies
   (iii) Total power delivered to the load of 600 Ω 6

2. (a) Differentiate between multiplexing and multiple access technologies. 5

   (b) Five channels, each with 100 kHz bandwidth, are to be frequency division multiplexing together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channel to prevent interference? 5

   (c) With proper block diagram explain the relation between frequency modulation and phase modulation. 5

   (d) In an FM system, when the audio frequency is 500 Hz and modulating voltage 2.5 V, the deviation produced is 5 kHz. If the modulating voltage is now increased to 7.5 V, calculate the new value of frequency deviation produced. If the AF voltage is raised to 10 V while the modulating frequency dropped to 250 Hz, what is the frequency deviation? Calculate the modulation index in each case. 10

3. (a) Define slope overload distortion and granular noise in delta modulation and how it is removed in ADM? 5

   (b) Draw the transmitter and receiver circuits block diagram of delta modulation. 5

   (c) Explain the basic block diagram of PCM system. 5

   (d) An analog signal having 4 kHz bandwidth is sampled at 1.25 times the Nyquist rate, and each sample is quantized into one of 256 equally likely level. Assuming the samples to be statistically independent:

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(i) What is the information rate of the source?
(ii) Can the output of this source be transmitted without error over the AWGN channel with a bandwidth of 10 kHz and S/N ratio of 20 dB?
(iii) Find S/N ratio required for error-free transmission of part ii.
(iv) Find the bandwidth required for an AWGN channel for an error free transmission of output of this source if S/N ratio is 20 dB 10

Part-B

[Answer any two of the following questions]

4. (a) With appropriate block diagram, describe the operation of superheterodyne receiver. 7
(b) Explain how the message signal is reconstructed from VSB signal. 7
(c) State and prove Nyquist sampling theorem. Also find the interpolation formula for the reconstruction of analog signals from its uniform samples. 7
(d) State Carson's rule and its limitations. 4

5. (a) Write down the comparison between TDMA and CDMA. 5
(b) A voice grade telephone channel has a bandwidth of 3400 Hz. If the signal to noise ratio (SNR) on the channel is 30 dB, determine the capacity of the channel. If the above channel is to be used to transmit 4.8 kbps of data, determine the minimum SNR required on the channel. 6
(c) It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patents [*sic*: should be "patients"] are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitude is 0.25% of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data. 7
(d) Consider a message signal m(t) with the spectrum shown in Fig. 1. The message bandwidth W = 1 kHz. This signal is applied to a product modulator, together with a carrier wave $A_c \cos(2\pi f_c t)$, producing the DSBSC modulated wave s(t). This modulated wave is next applied to a coherent detector. Assuming perfect synchronism between the carrier waves in the modulator and detector, determine the spectrum of the detector output when:

(i) the carrier frequency $f_c = 1.25$ kHz and
(ii) the carrier frequency $f_c = 0.75$ kHz.

What is the lowest carrier frequency for which each component of the modulated wave s(t) is uniquely determined by m(t)? 7

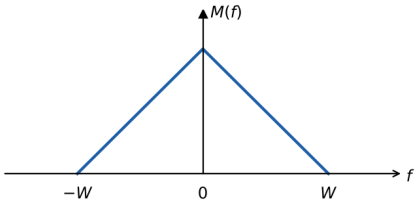


Fig. 1 of the paper (redrawn): the message spectrum M(f)

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6. (a) Briefly describe FSK generation with the necessary block diagram. Include its spectrum and discuss the bandwidth requirements, along with the advantages and disadvantages of FSK. 8

   (b) With proper block diagram explain the principle of coherent detection of QAM. Mention its advantages and drawbacks. 7

   (c) Explain the principle of frequency hopping spread spectrum technique. 6

   (d) Tabulate the values of signal at each stage of DPSK encoder and receiver for message signal '1010011100' 4

Term Test 1 (its questions 1 to 4 are 7(a) to 7(d))

[EEE 329 (2020-21), Total Marks: 4 × 5 = 20, Total time: 40 minutes]

7. (a) Find the entropy of single bit binary memoryless source. Also draw the curve of entropy vs probability for that source. 5

   (b) Explain distortion as a transmission impairment in communication systems. What are the conditions required for distortionless transmission? 5

   (c) We have a communication channel with a bandwidth of 5 MHz. A signal with an amplitude of 10V is transmitted, and the receiver detects a signal with an amplitude of 2V and noise with an amplitude of 5 mV.

   (1) Calculate the attenuation.
   (2) Determine the maximum channel capacity.
   (3) To achieve a data rate of 100 Mbps, find the required signal level. 5

   (d) Find the equivalent noise resistance for several amplifiers in cascade. 5

Term Test 2 (its questions 1 to 5 are 8(a) to 8(e))

[EEE 329 (2020-21), Total Marks: 4 × 5 = 20 (Answer any 4), Total time: 40 minutes]

8. (a) Amplitude spectrum of a message signal shown in figure-1. Draw the spectrum of its DSB-WC and SSB-SC modulated signal. 5

   (b) Explain how phase and frequency error affect the coherent detection of AM. 5

   (c) The antenna current of an AM transmitter is 8 Amps, when only the carrier is sent, but it increases to 8.93A, when the carrier is modulated by a single sine wave. Find percentage modulation. Determine the antenna current when the percentage modulation changes to 0.8 5

   (d) Explain principle of preemphasis and deemphasis 5

   (e) Consider an FM signal obtained from a modulating signal frequency of 2000 Hz and maximum Amplitude of 5 volts. The frequency sensitivity of modulator is 2 kHz/V. Find the bandwidth of the FM using Carson's rule. 5

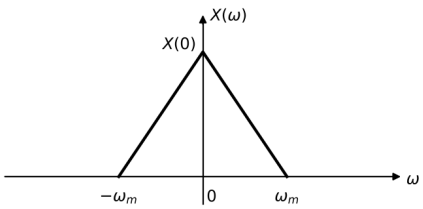


Figure-1 of the paper (redrawn): the amplitude spectrum X(ω) of the message

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Contents

1(a) Entropy of a Single-Bit Binary Memoryless Source **10**
1 Theory . . . 10
2 Answer . . . 11
3 Notes and Remarks . . . 12

1(b) Major Bands of the EM Spectrum **12**
1 Theory . . . 12
2 Answer . . . 13
3 Notes and Remarks . . . 13

1(c) Thermal Noise and Transit-Time Noise **14**
1 Theory . . . 14
2 Answer . . . 14
3 Notes and Remarks . . . 15

1(d) AM Modulation Index and Side Frequencies **15**
1 Theory . . . 15
2 Answer . . . 15

1(e) AM Sidebands and Total Power **17**
1 Theory . . . 17
2 Answer . . . 17
3 Notes and Remarks . . . 18
4 Final Concise Answer . . . 18

2(a) Multiplexing versus Multiple Access **19**
1 Theory . . . 19
2 Answer . . . 19

2(b) Minimum Bandwidth of an FDM Link **20**
1 Theory . . . 20
2 Answer . . . 20

2(c) Relation Between FM and PM **21**
1 Theory . . . 21
2 Answer . . . 22

2(d) FM Frequency Deviation and Modulation Index **23**
1 Theory . . . 23
2 Answer . . . 24
3 Notes and Remarks . . . 24
4 Final Concise Answer . . . 24

3(a) Slope Overload, Granular Noise and ADM **25**
1 Theory . . . 25
2 Answer . . . 26
3 Notes and Remarks . . . 26

3(b) Delta Modulation Transmitter and Receiver **27**
1 Theory . . . 27
2 Answer . . . 27

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3(c) Block Diagram of a PCM System **28**
1 Theory . . . 28
2 Answer . . . 28

3(d) Information Rate and Channel Capacity **29**
1 Theory . . . 29
2 Answer . . . 30
3 Notes and Remarks . . . 31
4 Final Concise Answer . . . 31

4(a) Superheterodyne Receiver **32**
1 Theory . . . 32
2 Answer . . . 32
3 Notes and Remarks . . . 33

4(b) Recovering the Message from a VSB Signal **34**
1 Theory . . . 34
2 Answer . . . 35

4(c) Nyquist Sampling Theorem and the Interpolation Formula **38**
1 Theory . . . 38
2 Answer . . . 38
3 Notes and Remarks . . . 41

4(d) Carson's Rule and its Limitations **41**
1 Theory . . . 41
2 Answer . . . 42
3 Final Concise Answer . . . 42

5(a) TDMA versus CDMA **43**
1 Theory . . . 43
2 Answer . . . 44

5(b) Channel Capacity and Minimum SNR **44**
1 Theory . . . 44
2 Answer . . . 45

5(c) Minimum Cable Bandwidth for the ECG TDM System **45**
1 Theory . . . 45
2 Answer . . . 46
3 Notes and Remarks . . . 47

5(d) Coherent Detection of DSB-SC and the Lowest Carrier **47**
1 Theory . . . 47
2 Answer . . . 48
3 Notes and Remarks . . . 50
4 Final Concise Answer . . . 50

6(a) FSK Generation, Spectrum and Bandwidth **51**
1 Theory . . . 51
2 Answer . . . 51
3 Notes and Remarks . . . 53

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6(b) Coherent Detection of QAM **53**
1 Theory . . . 53
2 Answer . . . 54
3 Notes and Remarks . . . 55

6(c) Frequency Hopping Spread Spectrum **55**
1 Theory . . . 55
2 Answer . . . 55
3 Notes and Remarks . . . 56

6(d) DPSK Encoder and Receiver Table for '1010011100' **56**
1 Theory . . . 56
2 Answer . . . 57
3 Notes and Remarks . . . 58
4 Final Concise Answer . . . 58

7(a) Entropy of a Binary Memoryless Source and its Curve **59**
1 Theory . . . 59
2 Answer . . . 59
3 Notes and Remarks . . . 60

7(b) Distortion and the Conditions for Distortionless Transmission **60**
1 Theory . . . 60
2 Answer . . . 61
3 Notes and Remarks . . . 62

7(c) Attenuation, Channel Capacity and Required Signal Level **62**
1 Theory . . . 62
2 Answer . . . 63
3 Notes and Remarks . . . 63

7(d) Equivalent Noise Resistance of Cascaded Amplifiers **63**
1 Theory . . . 63
2 Answer . . . 64
3 Notes and Remarks . . . 65
4 Final Concise Answer . . . 65

8(a) DSB-WC and SSB-SC Spectra of the Triangular Message Spectrum **66**
1 Theory . . . 66
2 Answer . . . 67
3 Notes and Remarks . . . 68

8(b) Effect of Phase and Frequency Error on Coherent Detection of AM **68**
1 Theory . . . 68
2 Answer . . . 68
3 Notes and Remarks . . . 69

8(c) AM Antenna Current and Percentage Modulation **69**
1 Theory . . . 69
2 Answer . . . 70
3 Notes and Remarks . . . 70

8(d) Pre-emphasis and De-emphasis **70**

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1 Theory . . . 70
2 Answer . . . 70
3 Notes and Remarks . . . 71

8(e) FM Bandwidth by Carson's Rule **72**
1 Theory . . . 72
2 Answer . . . 72
3 Notes and Remarks . . . 72
4 Final Concise Answer . . . 72

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1. (a) Find the entropy of single bit binary memoryless source. Also draw the curve of entropy Vs. probability for that source. Marks: 5

   (b) Briefly explain the three major bands of the electromagnetic (EM) spectrum-radio waves, microwaves and infra-red waves-in terms of their bandwidth, propagation mode and applications. Marks: 6

   (c) Write short note on-

   (i) Thermal Noise
   (ii) Transit Time Noise Marks: 4

   (d) A carrier wave of frequency 10 MHz and peak value of 10 V is amplitude modulated by a 5 kHz sine wave of amplitude 6 V. Determine the modulation index, side frequencies and amplitude of side frequencies. Marks: 4

   (e) An audio frequency signal $10\sin(2\pi 500t)$ is used to amplitude modulated a carrier of $50\sin(2\pi 105t)$ [*sic*: should be $2\pi 10^5 t$]. Calculate:

   (i) Modulation index
   (ii) Side-band frequencies
   (iii) Total power delivered to the load of 600 Ω Marks: 6

1.(a) Find the entropy of single bit binary memoryless source. Also draw the curve of entropy Vs. probability for that source. Marks: 5

1. Theory

1. What is a source? (Core Idea)

A **source** is anything that produces messages: a keyboard, a sensor, a microphone after sampling. We model it as a machine that sends one **symbol** at a time, picked from a fixed list. A **binary** source has only two symbols, 0 and 1. A **memoryless** source picks each symbol independently: what came before has no effect on what comes next. A coin toss is memoryless. So a single-bit binary memoryless source is a machine that keeps tossing a (possibly unfair) coin.

2. How much information is in one symbol?

Information measures **surprise**. "The sun rose today" tells you nothing new. "It snowed in Sylhet" tells you a lot. So a rare symbol carries more information than a common one. A symbol of probability P carries

$$I = \log_2 \frac{1}{P} \quad \text{bits}$$

Why a logarithm? Two independent symbols have probability $P_1 P_2$, and we want their information to add. The logarithm turns the product into a sum. The check is simple: a fair coin toss has $P = 1/2$, so $I = \log_2 2 = 1$ bit. That is where the word "bit" comes from.

3. What is entropy?

Entropy H is the **average** information per symbol. Each symbol's information is weighted by how often the symbol comes:

$$H = \sum_i P_i \log_2 \frac{1}{P_i} \quad \text{bits/symbol}$$

It is also the smallest average number of binary digits per symbol that any lossless code can reach.

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2. Answer

Step 1: Entropy from the definition

Let the source send 1 with probability p and 0 with probability $1-p$ (the two must add to 1). Put $P_1 = p$ and $P_2 = 1-p$ into $H = \sum_i P_i \log_2(1/P_i)$:

$$H(p) = p \log_2 \frac{1}{p} + (1-p) \log_2 \frac{1}{1-p}$$

Since $\log_2(1/x) = -\log_2 x$:

$$\boxed{H(p) = -p \log_2 p - (1-p) \log_2(1-p) \quad \text{bits/symbol}}$$

Step 2: Where is the maximum?

Differentiate term by term. Write $\log_2 x = \ln x / \ln 2$, so $\frac{d}{dx} \log_2 x = \frac{1}{x \ln 2}$. By the product rule:

$$\frac{d}{dp}\left[-p \log_2 p\right] = -\left(\log_2 p + p \cdot \frac{1}{p \ln 2}\right) = -\log_2 p - \frac{1}{\ln 2}$$

For the second term, the chain rule gives $\frac{d}{dp} \log_2(1-p) = \frac{-1}{(1-p)\ln 2}$, so

$$\frac{d}{dp}\left[-(1-p)\log_2(1-p)\right] = -\left[(-1)\log_2(1-p) + (1-p) \cdot \frac{-1}{(1-p)\ln 2}\right] = \log_2(1-p) + \frac{1}{\ln 2}$$

Add the two. The $\frac{1}{\ln 2}$ terms cancel:

$$\frac{dH}{dp} = \log_2(1-p) - \log_2 p = \log_2 \frac{1-p}{p}$$

Set it to zero. A logarithm is zero only when its argument is 1:

$$\log_2 \frac{1-p}{p} = 0 \;\Rightarrow\; \frac{1-p}{p} = 1 \;\Rightarrow\; 1-p = p \;\Rightarrow\; p = \frac{1}{2}$$

It is a maximum, not a minimum. Differentiate once more:

$$\frac{d^2 H}{dp^2} = \frac{-1}{(1-p)\ln 2} - \frac{1}{p \ln 2} < 0 \quad \text{for } 0 < p < 1$$

The second derivative is negative, so the curve bends downward and $p = \frac{1}{2}$ is a maximum. Its value is

$$H_{max} = \frac{1}{2}\log_2 2 + \frac{1}{2}\log_2 2 = \frac{1}{2} + \frac{1}{2} = 1 \text{ bit/symbol}$$

Step 3: The end points

At $p = 0$ the term $p \log_2 p$ has the form $0 \times (-\infty)$. Write it as a fraction and use L'Hôpital's rule:

$$\lim_{p \to 0} p \ln p = \lim_{p \to 0} \frac{\ln p}{1/p} = \lim_{p \to 0} \frac{1/p}{-1/p^2} = \lim_{p \to 0}(-p) = 0$$

So $p \log_2 p \to 0$, and the other term is $1 \cdot \log_2 1 = 0$. Hence $H(0) = 0$, and in the same way $H(1) = 0$: the output is certain, so it carries no information.

Step 4: The curve

Computing $H(p)$ at a few points:

p	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
$H(p)$	0	0.469	0.722	0.881	0.971	1	0.971	0.881	0.722	0.469	0

Table 1: Entropy of a binary memoryless source

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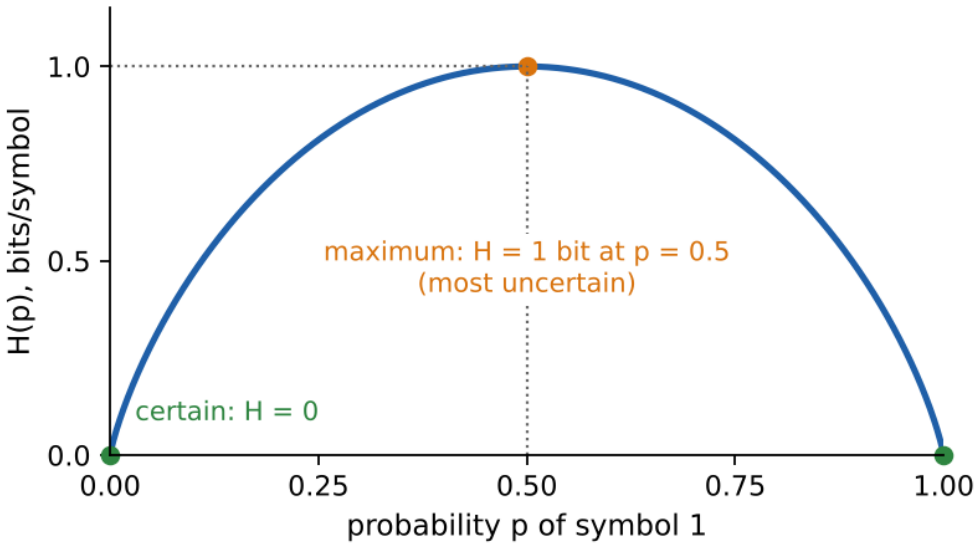


Figure 1: Entropy of a binary memoryless source against probability

Reading the figure. The curve is a symmetric arch. It is 0 at $p = 0$ and $p = 1$, where the output is certain (green dots), and it peaks at 1 bit when $p = 0.5$ (orange dot), where both symbols are equally likely. Swapping the names of 0 and 1 changes nothing, so $H(p) = H(1 - p)$.

3. Notes and Remarks

- A source that sends one binary digit per symbol carries *less* than one bit of information unless $p = 0.5$. The binary digit (a container) and the bit (the unit of information) are different things.

1.(b) Briefly explain the three major bands of the electromagnetic (EM) spectrum-radio waves, microwaves and infra-red waves-in terms of their bandwidth, propagation mode and applications. Marks: 6

1. Theory

1. What is the EM spectrum? (Core Idea)

Electromagnetic (EM) waves are travelling electric and magnetic fields. They all move at the speed of light $c = 3 \times 10^8$ m/s and differ only in frequency f. The wavelength is $\lambda = c/f$. The **spectrum** is the list of all these waves in order of frequency.

2. Why does the band matter?

Three things change as the frequency rises:

- **Bandwidth:** a band at high frequency is wider in Hz, so it can carry more data.
- **Propagation:** low frequencies bend around the Earth and pass through walls. High frequencies travel in straight lines and are blocked easily.
- **Antenna size:** an antenna is a fraction of λ, so high frequencies use small antennas that can focus the beam.

A **propagation mode** is the path a wave takes: **ground wave** (following the Earth's surface), **sky wave** (bent back by the ionosphere) or **line of sight** (a straight path).

2. Answer

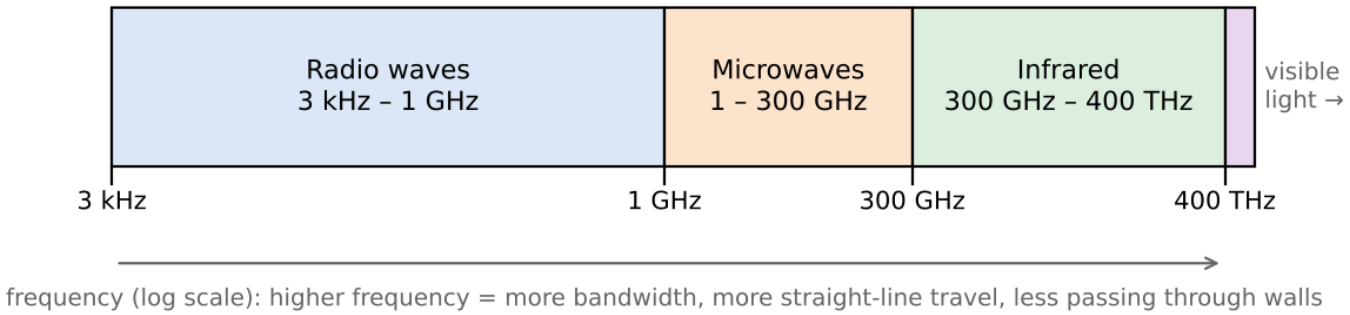


Figure 2: The three bands on a frequency scale

Reading the figure. The axis is logarithmic, so each band's width on the page shows its ratio of frequencies, not its width in Hz. Radio waves stop at 1 GHz, microwaves run to 300 GHz, and infrared runs to 400 THz, just below visible light.

	Radio waves	Microwaves	Infrared
Frequency band	3 kHz to 1 GHz	1 GHz to 300 GHz	300 GHz to 400 THz
Bandwidth	smallest: narrow channels, low data rates	wide: high data rates	very wide, but used over short distances
Propagation	ground wave at low frequencies, sky wave in the HF band, line of sight at VHF and UHF; mostly omnidirectional	line of sight; unidirectional, narrow beams, so the antennas must be aligned	short-range line of sight; cannot pass through walls
Antennas	simple omnidirectional antennas (dipole, monopole)	parabolic dish and horn	LED and photodiode
Applications	AM and FM radio, television, maritime radio, paging	satellite links, cellular phones, wireless LANs, radar	TV remote controls, short-range device links, wireless keyboards and mice

Table 2: The three major bands

Key Point: going up in frequency gives more bandwidth but a shorter, straighter path.

3. Notes and Remarks

- The band edges follow Forouzan. Other books place the edge between radio waves and microwaves differently; the idea stays the same.
- Infrared cannot pass through walls, which is an advantage too: one room's link does not disturb the next. Sunlight, however, contains infrared and can interfere outdoors.

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1.(c) Write short note on- (i) Thermal Noise (ii) Transit Time Noise Marks: 4

1. Theory

1. What is noise? (Core Idea)

Noise is any unwanted random signal added to the wanted one. The receiver cannot predict it, so it cannot simply subtract it. Some noise comes from outside (lightning, motors). Some is made inside the receiver's own parts, and that is the type asked here.

Electric current is a stream of separate electrons. They never move perfectly smoothly, and that tiny randomness is the source of both types of noise below.

2. Where the two types sit

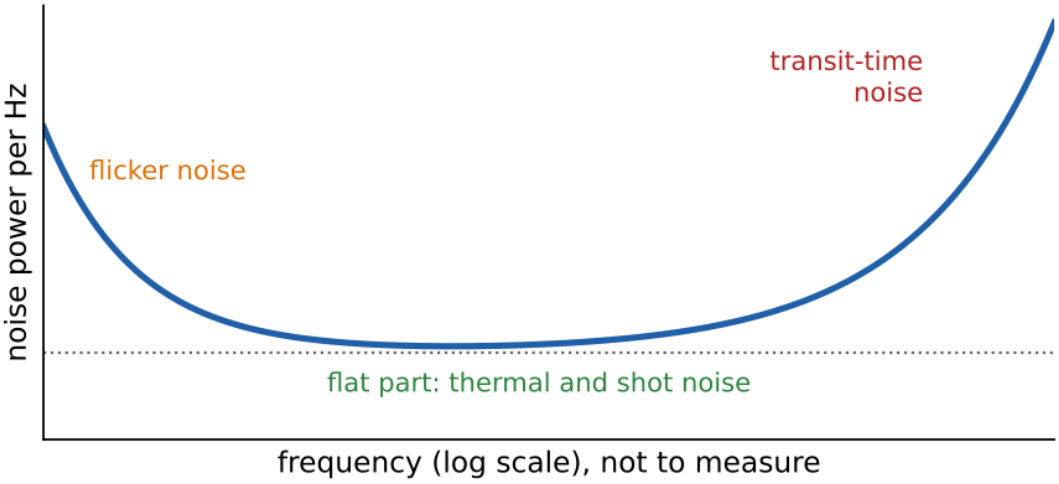


Figure 3: How the noise of a transistor changes with frequency

Reading the figure. This sketch is qualitative. At very low frequency flicker noise is large. In the middle the curve is flat, set by thermal and shot noise. At the far right the curve rises again: this is transit-time noise.

2. Answer

i. Thermal Noise

Thermal (Johnson) noise is the random voltage produced by the random motion of free electrons in any conductor or resistor, caused by heat. It exists even with no current flowing, and it vanishes only at absolute zero.

Key Points:

- Its power grows with temperature and bandwidth, not with the signal:

$$P_n = kTB$$

where $k = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant, T is the temperature in kelvin and B is the bandwidth in Hz.

- A resistor R acts as a noise source of r.m.s. voltage

$$V_n = \sqrt{4kTBR}$$

- It is **white**: its power per Hz is the same over the whole range used in communication. Its amplitude is Gaussian.

Practical Note: at $T = 290$ K with $B = 1$ MHz and $R = 1$ kΩ, $P_n = 4.0 \times 10^{-15}$ W (-114 dBm) and $V_n \approx 4$ μV. Reducing the bandwidth is the simple way to reduce it.

ii. Transit-Time Noise

Transit-time noise is the noise that appears in a transistor at very high frequencies, when the time an electron takes to cross the device becomes significant compared with one period of the signal.

Key Points:

- **Cause:** at the upper VHF range and beyond, carriers are still crossing while the signal changes. Random fluctuations in the output current then induce small currents in the input, which act as random noise.
- **Effect:** the noise rises with frequency above the flat part of the noise curve, so it sets a limit on low-noise operation at high frequency.
- **Remedy:** use devices with very short transit times (small geometry and high carrier mobility), chosen to have their cut-off far above the working frequency.

3. Notes and Remarks

- Thermal noise follows Lathi. Transit-time noise follows Kennedy and Davis, since neither Lathi nor Forouzan treats it.

1.(d) A carrier wave of frequency 10 MHz and peak value of 10 V is amplitude modulated by a 5 kHz sine wave of amplitude 6 V. Determine the modulation index, side frequencies and amplitude of side frequencies. Marks: 4

1. Theory

1. What is amplitude modulation? (Core Idea)

A message (audio) has low frequency and cannot travel far by itself. So we use a high-frequency **carrier** and make its amplitude follow the message. That is **amplitude modulation** (AM):

$$s(t) = [E_c + E_m \sin \omega_m t] \sin \omega_c t$$

The bracket is the **envelope**: the slowly changing outline of the fast carrier.

2. Modulation index

The **modulation index** μ tells how deeply the message moves the carrier's amplitude:

$$\mu = \frac{E_m}{E_c}$$

$\mu = 0$ means no modulation, and $\mu = 1$ means the envelope just reaches zero. Above 1 the envelope folds over, and a simple detector distorts the message.

3. Where the side frequencies come from

Put $E_m = \mu E_c$ and use $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$:

$$s(t) = E_c \sin \omega_c t + \frac{\mu E_c}{2} \cos(\omega_c - \omega_m)t - \frac{\mu E_c}{2} \cos(\omega_c + \omega_m)t$$

So AM is three plain sine waves: the carrier and two **side frequencies** at $f_c \pm f_m$, each of amplitude $\mu E_c/2$.

2. Answer

Given:

- Carrier frequency $f_c = 10$ MHz, peak value $E_c = 10$ V
- Message frequency $f_m = 5$ kHz $= 0.005$ MHz, amplitude $E_m = 6$ V

Step 1: Modulation index

$$\mu = \frac{E_m}{E_c} = \frac{6}{10}$$

$$\boxed{\mu = 0.6 \ (60\%)}$$

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Step 2: Side frequencies

$$f_{USB} = f_c + f_m = 10 + 0.005 = 10.005 \text{ MHz}$$

$$f_{LSB} = f_c - f_m = 10 - 0.005 = 9.995 \text{ MHz}$$

$$\boxed{f_{USB} = 10.005 \text{ MHz}, \quad f_{LSB} = 9.995 \text{ MHz}}$$

Step 3: Amplitude of the side frequencies

$$\frac{\mu E_c}{2} = \frac{0.6 \times 10}{2}$$

$$\boxed{\text{each side frequency} = 3 \text{ V}}$$

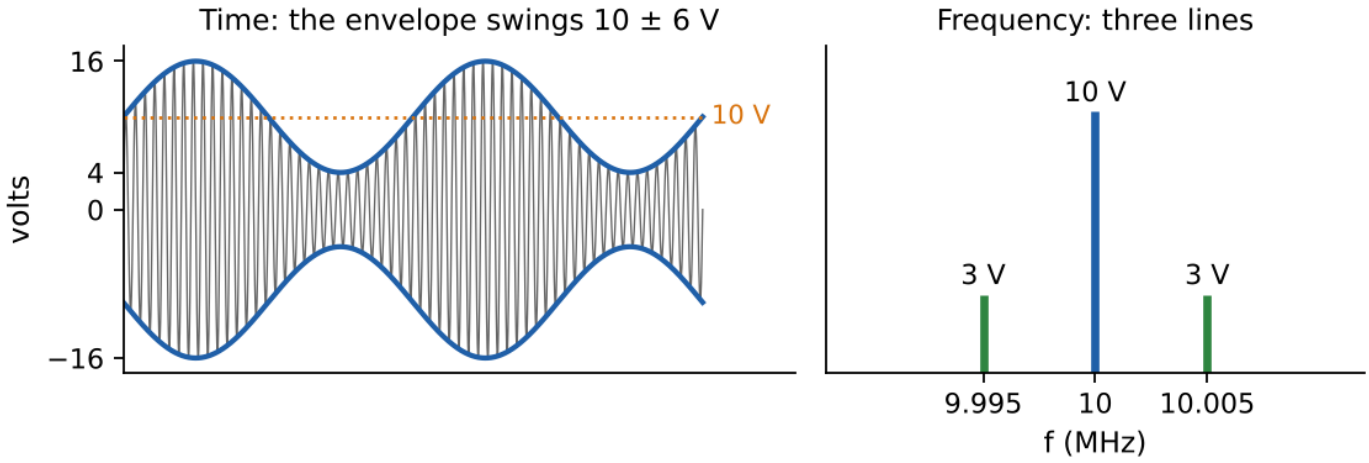


Figure 4: The AM wave in time and in frequency

Reading the figure. Left: the solid blue envelope swings between $10 - 6 = 4$ V and $10 + 6 = 16$ V around the dotted orange 10 V level. Right: the same signal in frequency is one 10 V carrier with a 3 V line on each side, 5 kHz away.

Final Answer: $\mu = 0.6$; side frequencies 10.005 MHz and 9.995 MHz; each of amplitude 3 V.

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1.(e) An audio frequency signal $10\sin(2\pi 500t)$ is used to amplitude modulated a carrier of $50\sin(2\pi 105t)$ [*sic*: should be $2\pi 10^5 t$]. Calculate: (i) Modulation index (ii) Side-band frequencies (iii) Total power delivered to the load of 600 Ω Marks: 6

1. Theory

1. Reading a signal from its formula

A signal written as $A\sin(2\pi ft)$ has peak amplitude A and frequency f. So the amplitude and frequency can be read straight from the formula. The modulation index and side frequencies then follow as in 1(d).

2. Power in an AM wave

A sine wave of peak A has r.m.s. value $A/\sqrt{2}$, so across a resistance R its average power is

$$P = \frac{(A/\sqrt{2})^2}{R} = \frac{A^2}{2R}$$

AM is three sine waves at different frequencies. To see them, write the AM wave with $E_m = \mu E_c$ and multiply out:

$$s(t) = [E_c + \mu E_c \sin\omega_m t]\sin\omega_c t = E_c\sin\omega_c t + \mu E_c \sin\omega_m t\sin\omega_c t$$

Use $\sin A\sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$ with $A = \omega_c t$, $B = \omega_m t$:

$$s(t) = E_c\sin\omega_c t + \frac{\mu E_c}{2}\cos(\omega_c - \omega_m)t - \frac{\mu E_c}{2}\cos(\omega_c + \omega_m)t$$

So there is a carrier of amplitude E_c and two side frequencies of amplitude $\mu E_c/2$. Sine waves of different frequencies do not interfere on average, so their powers simply add. With the carrier power $P_c = E_c^2/2R$, each side frequency of amplitude $\mu E_c/2$ has power

$$P_{SB} = \frac{(\mu E_c/2)^2}{2R} = \frac{\mu^2}{4}\cdot\frac{E_c^2}{2R} = \frac{\mu^2}{4}P_c$$

Adding the carrier and the two side frequencies:

$$P_t = P_c + 2\cdot\frac{\mu^2}{4}P_c = P_c\left(1 + \frac{\mu^2}{2}\right)$$

2. Answer

Note on the paper

The carrier is printed as $50\sin(2\pi 105t)$, which would make it 105 Hz, below the 500 Hz message. AM needs a carrier far above the message, so the exponent has been lost: it is read as $50\sin(2\pi 10^5 t)$, a 100 kHz carrier.

Given:

- Message $10\sin(2\pi 500t)$: $E_m = 10$ V, $f_m = 500$ Hz
- Carrier $50\sin(2\pi 10^5 t)$: $E_c = 50$ V, $f_c = 100$ kHz
- Load $R = 600\ \Omega$

Step 1: Modulation index

$$\mu = \frac{E_m}{E_c} = \frac{10}{50}$$

$$\boxed{\mu = 0.2}$$

Step 2: Side-band frequencies

$$f_{USB} = 100 + 0.5 = 100.5 \text{ kHz}, \qquad f_{LSB} = 100 - 0.5 = 99.5 \text{ kHz}$$

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$$\boxed{f_{USB} = 100.5 \text{ kHz}, \quad f_{LSB} = 99.5 \text{ kHz}}$$

Step 3: Total power

$$P_c = \frac{E_c^2}{2R} = \frac{50^2}{2 \times 600} = 2.0833 \text{ W}$$

$$P_t = P_c \left(1 + \frac{\mu^2}{2}\right) = 2.0833 \left(1 + \frac{0.2^2}{2}\right) = 2.0833 \times 1.02$$

$$\boxed{P_t = 2.125 \text{ W}}$$

The sidebands carry only $2 \times 0.02083 = 0.0417$ W, about 2% of the total. The rest is carrier, which carries no information.

Final Answer: $\mu = 0.2$; side-band frequencies 100.5 kHz and 99.5 kHz; total power 2.125 W.

3. Notes and Remarks

- The power does not depend on the exact carrier frequency, so the reading of the misprint changes only the side-band frequencies.
- The amplitudes in the formulas are peak values, which is why $E_c^2/2R$ is used and not E_c^2/R.

4. Final Concise Answer

(a) $H(p) = -p \log_2 p - (1-p) \log_2(1-p)$; 0 at $p = 0$ and $p = 1$, maximum 1 bit at $p = 0.5$; the curve is a symmetric arch.

(b) Radio waves: 3 kHz to 1 GHz, ground, sky or line of sight, radio and TV. Microwaves: 1 to 300 GHz, line of sight, satellite and cellular. Infrared: 300 GHz to 400 THz, short-range line of sight, remote controls.

(c) Thermal noise: random electron motion from heat, $P_n = kTB$, white. Transit-time noise: appears at very high frequency when the electron crossing time becomes significant compared with the signal period; it rises with frequency.

(d) $\mu = 0.6$; 10.005 MHz and 9.995 MHz; 3 V each.

(e) $\mu = 0.2$; 100.5 kHz and 99.5 kHz; $P_t = 2.125$ W.

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2. (a) Differentiate between multiplexing and multiple access technologies. Marks: 5

(b) Five channels, each with 100 kHz bandwidth, are to be frequency division multiplexing together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channel to prevent interference? Marks: 5

(c) With proper block diagram explain the relation between frequency modulation and phase modulation. Marks: 5

(d) In an FM system, when the audio frequency is 500 Hz and modulating voltage 2.5 V, the deviation produced is 5 kHz. If the modulating voltage is now increased to 7.5 V, calculate the new value of frequency deviation produced. If the AF voltage is raised to 10 V while the modulating frequency dropped to 250 Hz, what is the frequency deviation? Calculate the modulation index in each case. Marks: 10

2.(a) Differentiate between multiplexing and multiple access technologies. Marks: 5

1. Theory

1. The shared-resource problem (Core Idea)

A cable or a radio band is expensive, and many signals want to use it. Both multiplexing and multiple access let several signals share one resource. They divide it in the same ways: by frequency, by time or by code. The difference is **who is sharing and from where**.

2. Multiplexing

Multiplexing is like a bus at a terminal. All the passengers board at one stop, ride one road and get off at one stop. Several signals are combined at a single point by a **multiplexer (MUX)**, sent along one link, and separated at the far end by a **demultiplexer (DEMUX)**.

3. Multiple access

Multiple access is like many people in different houses taking turns on one shared telephone line. The users are in different places, each sending its own signal, and no single box combines them. They must follow a rule so that they do not collide, and the signals come together in the medium itself.

2. Answer

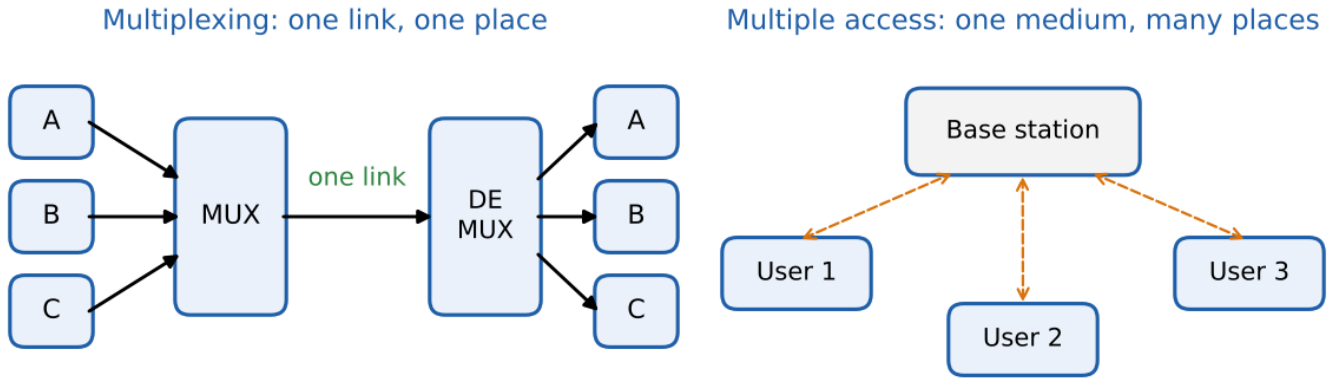


Figure 5: Multiplexing and multiple access

Reading the figure. Left: inputs A, B and C are combined by one MUX, carried on one link and separated by one DEMUX. Right: three separate users reach the same base station over the shared medium (dashed links), with no MUX between them.

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	Multiplexing	Multiple access
What is shared	one link between two points	one medium (such as the air around a tower) by many separate users
Where signals are combined	at one place, by a MUX	in the channel; each user transmits from its own place
Control	one MUX/DEMUX pair; allocation is usually fixed	a protocol gives resources to users, who join and leave
Techniques	FDM, TDM, WDM	FDMA, TDMA, CDMA
Examples	telephone trunks (T-1, E-1), FDM radio	mobile networks, satellite access

Table 3: Multiplexing versus multiple access

In short, multiplexing combines several signals at one point onto one link. Multiple access lets distant users share one medium.

2.(b) Five channels, each with 100 kHz bandwidth, are to be frequency division multiplexing together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channel to prevent interference? Marks: 5

1. Theory

1. Frequency division multiplexing (Core Idea)

Radio stations already show the idea: each one has its own frequency, and they all share the air. **FDM** gives each channel its own slice of the link's bandwidth, using a separate carrier for each.

2. Guard bands

Real filters do not cut sharply, so a channel's edge leaks a little into its neighbour. To stop this we leave an empty gap, a **guard band**, between neighbouring channels.

3. Counting the bandwidth

With N channels side by side there are only $N-1$ gaps **between** them, just as N posts in a line have $N-1$ spaces. So

$$B_{link} = NB_{ch} + (N-1)B_g$$

2. Answer

Given:

- Number of channels $N = 5$
- Channel bandwidth $B_{ch} = 100$ kHz
- Guard band $B_g = 10$ kHz, between channels

$$B_{link} = 5 \times 100 + (5-1) \times 10 = 500 + 40$$

$$\boxed{B_{link} = 540 \text{ kHz}}$$

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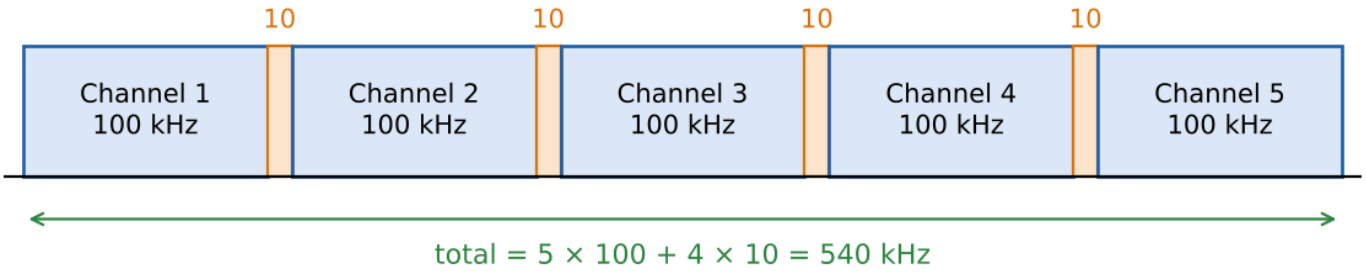


Figure 6: Five channels and four guard bands

Reading the figure. The five channels of 100 kHz are separated by four narrow guard bands of 10 kHz, and the double arrow marks the total of 540 kHz.

Assumption: guard bands are needed only between channels, as the question says. If guard bands were also kept at the two outer edges, there would be six and the total would be 560 kHz.

2.(c) With proper block diagram explain the relation between frequency modulation and phase modulation. Marks: 5

1. Theory

1. Angle modulation (Core Idea)

AM changes the height of the carrier. **Angle modulation** keeps the height fixed and changes the carrier's angle instead. Write the carrier as

$$\varphi(t) = A\cos\theta(t)$$

The rate at which the angle turns is the **instantaneous frequency**:

$$\omega_i(t) = \frac{d\theta}{dt}$$

So angle and frequency are tied together: frequency is the derivative of angle, and angle is the integral of frequency. This one fact explains the whole question.

2. Phase modulation

In **PM** the angle follows the message directly:

$$\theta(t) = \omega_c t + k_p m(t), \qquad \omega_i = \omega_c + k_p \dot{m}(t)$$

3. Frequency modulation

In **FM** the frequency follows the message directly:

$$\omega_i = \omega_c + k_f m(t), \qquad \theta(t) = \omega_c t + k_f \int m(\alpha)\, d\alpha$$

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4. Seeing the difference

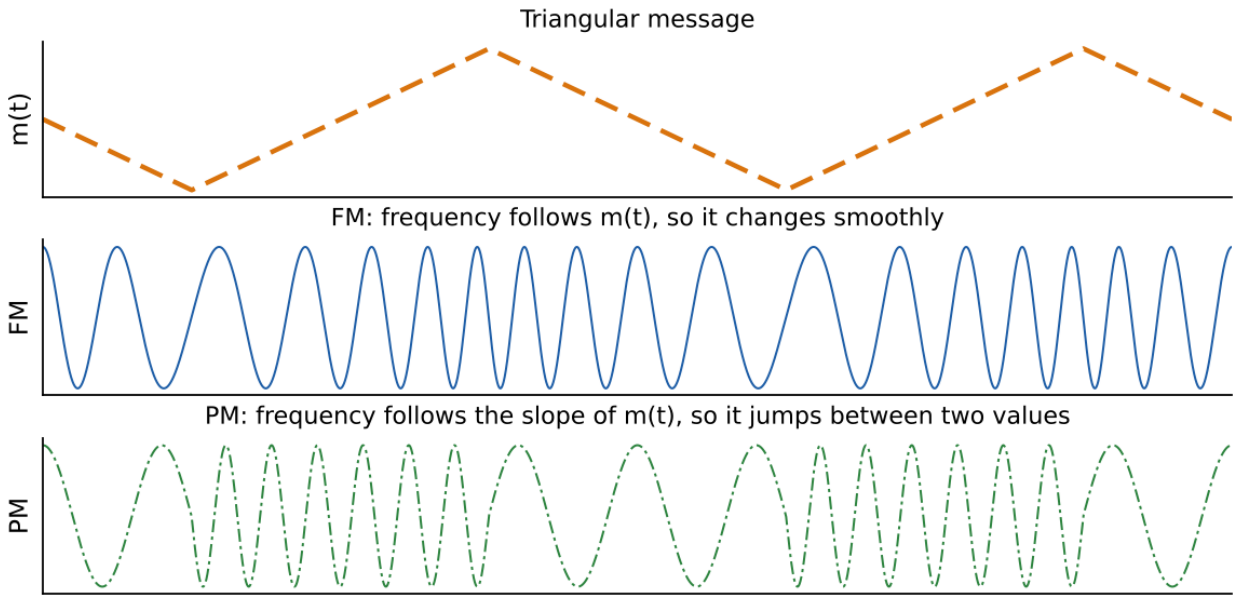


Figure 7: FM and PM for the same triangular message

Reading the figure. The dashed orange triangle is the message. The solid blue FM wave packs tighter as $m(t)$ rises and spreads as it falls, so its frequency changes smoothly with the message. The dash-dot green PM wave depends on the slope of $m(t)$: a rising slope gives one fixed high frequency and a falling slope one fixed low frequency, so it jumps at each corner.

2. Answer

FM and PM are two forms of angle modulation, and each can be made from the other.

$$\varphi_{PM}(t) = A\cos[\omega_c t + k_p m(t)]$$

$$\varphi_{FM}(t) = A\cos\left[\omega_c t + k_f \int m(\alpha)\, d\alpha\right]$$

Compare the two by feeding each modulator a changed message.

FM from PM. Give the phase modulator the integral $a(t) = \int m(\alpha)\, d\alpha$ in place of $m(t)$:

$$A\cos[\omega_c t + k_p a(t)] = A\cos\left[\omega_c t + k_p \int m(\alpha)\, d\alpha\right]$$

This is exactly $\varphi_{FM}(t)$ with $k_f = k_p$. So FM of $m(t)$ is the same as PM of $\int m\, dt$.

PM from FM. Give the frequency modulator the derivative $\dot{m}(t)$ in place of $m(t)$:

$$A\cos\left[\omega_c t + k_f \int \dot{m}(\alpha)\, d\alpha\right] = A\cos[\omega_c t + k_f m(t)]$$

because integrating a derivative gives back the function (taking $m = 0$ at the start). This is exactly $\varphi_{PM}(t)$ with $k_p = k_f$. So PM of $m(t)$ is the same as FM of $\dot{m}(t)$.

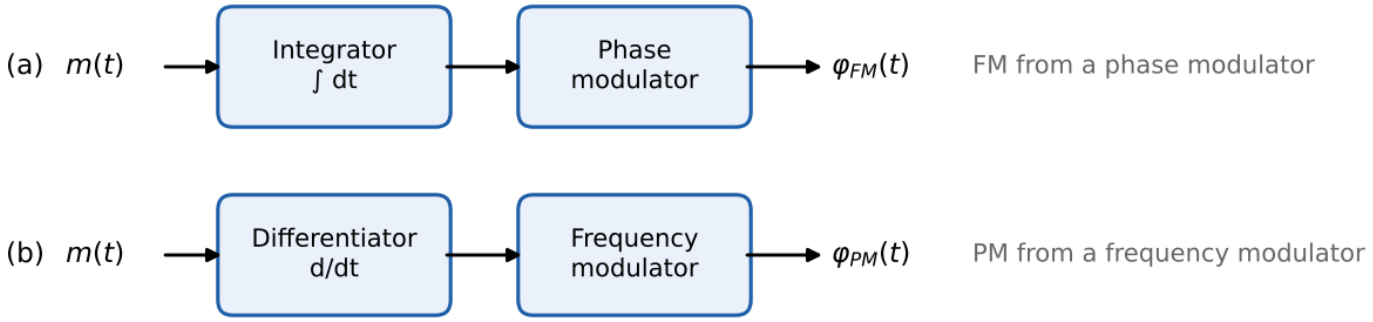


Figure 8: FM and PM made from each other

Reading the figure. (a) The message is integrated and then phase modulated, and the output is FM. (b) The message is differentiated and then frequency modulated, and the output is PM.

Key Points:

- **FM from PM:** integrate $m(t)$ and pass it to a phase modulator. This is the Armstrong (indirect) method of making FM.
- **PM from FM:** differentiate $m(t)$ and pass it to a frequency modulator.
- In PM the frequency deviation follows the **slope** of $m(t)$. In FM it follows $m(t)$ itself.
- Both have a constant amplitude A, so the power stays $A^2/2$, whatever the message.
- Just by looking at a modulated wave, you cannot tell FM from PM without knowing the message.

2.(d) In an FM system, when the audio frequency is 500 Hz and modulating voltage 2.5 V, the deviation produced is 5 kHz. If the modulating voltage is now increased to 7.5 V, calculate the new value of frequency deviation produced. If the AF voltage is raised to 10 V while the modulating frequency dropped to 250 Hz, what is the frequency deviation? Calculate the modulation index in each case. Marks: 10

1. Theory

1. Frequency deviation (Core Idea)

In FM the carrier's frequency moves up and down around f_c. The largest shift from f_c is the **frequency deviation** Δf. Because the frequency follows the message's *voltage*, the deviation is proportional to the modulating voltage:

$$\Delta f = k_f V_m$$

Here k_f (Hz/V) is the sensitivity of the modulator. It is a fixed property of the circuit. **The modulating frequency does not appear in this formula.** A louder tone swings the carrier further. A higher-pitched tone only swings it faster.

2. Modulation index

The **FM modulation index** β compares how far the carrier swings with how fast:

$$\beta = \frac{\Delta f}{f_m}$$

Unlike AM's μ, it can be much greater than 1, and it has no unit.

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2. Answer

Given (Case 1): $f_{m1} = 500$ Hz, $V_1 = 2.5$ V, $\Delta f_1 = 5$ kHz.
Sensitivity:

$$k_f = \frac{\Delta f_1}{V_1} = \frac{5000}{2.5} = 2000 \text{ Hz/V} = 2 \text{ kHz/V}$$

Case 1: 2.5 V at 500 Hz

$$\beta_1 = \frac{\Delta f_1}{f_{m1}} = \frac{5000}{500}$$

$$\boxed{\Delta f_1 = 5 \text{ kHz}, \quad \beta_1 = 10}$$

Case 2: 7.5 V at 500 Hz

$$\Delta f_2 = k_f V_2 = 2 \times 7.5 = 15 \text{ kHz}$$

$$\beta_2 = \frac{15000}{500}$$

$$\boxed{\Delta f_2 = 15 \text{ kHz}, \quad \beta_2 = 30}$$

The voltage rose three times, so the deviation and the index each rose three times.

Case 3: 10 V at 250 Hz

$$\Delta f_3 = k_f V_3 = 2 \times 10 = 20 \text{ kHz}$$

$$\beta_3 = \frac{20000}{250}$$

$$\boxed{\Delta f_3 = 20 \text{ kHz}, \quad \beta_3 = 80}$$

The drop in frequency had no effect on the deviation, but it doubled β on top of the rise in voltage.
Final Summary:

Case	V_m (V)	f_m (Hz)	Δf (kHz)	β
1	2.5	500	5	10
2	7.5	500	15	30
3	10	250	20	80

Table 4: Deviation and modulation index in each case

3. Notes and Remarks

- Δf depends only on the amplitude; β depends on both the amplitude and the frequency.
- The word "deviation" here means the peak shift on one side of f_c. The total swing (carrier swing) is $2\Delta f$.

4. Final Concise Answer

(a) Multiplexing combines signals at one point onto one link (FDM, TDM, WDM). Multiple access lets separate users share one medium (FDMA, TDMA, CDMA).
(b) $5 \times 100 + 4 \times 10 = 540$ kHz.
(c) $\omega_i = d\theta/dt$, so FM of $m(t)$ equals PM of $\int m\,dt$ and PM of $m(t)$ equals FM of $\dot{m}(t)$; use an integrator or a differentiator before the modulator.
(d) $k_f = 2$ kHz/V; $\Delta f = 5, 15, 20$ kHz; $\beta = 10, 30, 80$.

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3. (a) Define slope overload distortion and granular noise in delta modulation and how it is removed in ADM? Marks: 5

(b) Draw the transmitter and receiver circuits block diagram of delta modulation. Marks: 5

(c) Explain the basic block diagram of PCM system. Marks: 5

(d) An analog signal having 4 kHz bandwidth is sampled at 1.25 times the Nyquist rate, and each sample is quantized into one of 256 equally likely level. Assuming the samples to be statistically independent:

(i) What is the information rate of the source?

(ii) Can the output of this source be transmitted without error over the AWGN channel with a bandwidth of 10 kHz and S/N ratio of 20 dB?

(iii) Find S/N ratio required for error-free transmission of part ii.

(iv) Find the bandwidth required for an AWGN channel for an error free transmission of output of this source if S/N ratio is 20 dB Marks: 10

3.(a) Define slope overload distortion and granular noise in delta modulation and how it is removed in ADM? Marks: 5

1. Theory

1. What is delta modulation? (Core Idea)

Neighbouring samples of a signal are usually close in value. So instead of sending each full sample, **delta modulation (DM)** sends only one bit per sample: "go up" or "go down". The receiver draws a **staircase** $\hat{m}(t)$, moving one fixed step σ up or down at each sample. If the staircase stays close to the signal $m(t)$, the receiver has the signal.

Think of walking behind a moving friend with steps of one fixed length, taken at a fixed rate. You compare your place with your friend's, and step toward them.

2. How fast can the staircase climb?

Each sample takes $T_s = 1/f_s$ seconds and moves the staircase by σ, so the steepest slope it can follow is

$$\text{largest slope} = \frac{\sigma}{T_s} = \sigma f_s$$

3. The two errors

If your friend runs faster than you can walk, you fall behind. If your friend stands still, you keep stepping past and back. These are the two errors below, and they pull in opposite directions: a big step cures the first and worsens the second.

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2. Answer

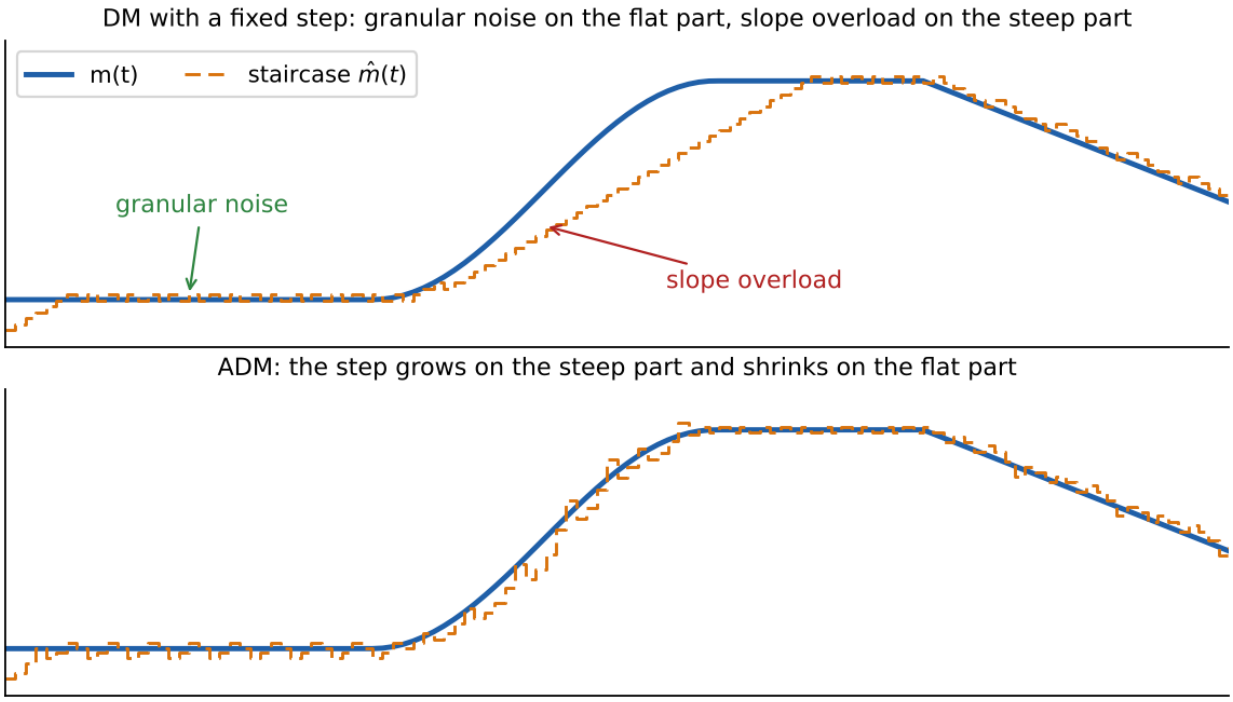


Figure 9: DM and ADM following the same signal

Reading the figure. Top: with a fixed step, the dashed orange staircase rattles around the flat part of the solid blue signal (granular noise) and falls behind on the steep rise (slope overload). Bottom: with ADM, on the same signal, the steps grow on the rise, so the staircase keeps up, and then shrink on the flat part.

Slope overload distortion: the error that occurs when the signal rises or falls faster than the staircase can follow, so that the staircase lags behind it. It occurs when

$$\left|\frac{dm}{dt}\right|_{max} > \sigma f_s$$

For a tone $m(t) = A\cos\omega_m t$, the slope is $\dot{m}(t) = -A\omega_m \sin\omega_m t$, whose largest size is $A\omega_m$ (when $|\sin\omega_m t| = 1$). So the condition for no overload is $A\omega_m \leq \sigma f_s$.

Granular noise: the error that occurs when the signal is flat or changes slowly. The staircase cannot stay still, so it hunts up and down by one step around the signal. This is like quantization noise, and it grows with σ.

Removal by adaptive delta modulation (ADM): ADM changes the step size to suit the signal.

- A **run of equal bits** (…1111 or …0000) shows that the staircase is lagging, so the step is **increased**. This removes slope overload.
- **Alternating bits** (…1010) show that the signal is flat, so the step is **decreased**. This reduces granular noise.

The step is worked out from the bit stream itself, so the receiver can follow the same rule with no extra information.

3. Notes and Remarks

- For fixed-step DM, the best σ is a compromise between the two errors. ADM avoids that compromise.
- The step cannot grow or shrink without limit. Real ADM sets a smallest and a largest step, as in the sketch above.

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3.(b) Draw the transmitter and receiver circuits block diagram of delta modulation. Marks: 5

1. Theory

1. The plan (Core Idea)

The transmitter must decide "up" or "down" at each sample. To do that it needs to know where the receiver's staircase is now, so **it builds an exact copy of the receiver inside itself**. It compares the true signal with that copy and sends only the sign of the difference.

2. The accumulator

An **accumulator** keeps a running total: each incoming $+\sigma$ or $-\sigma$ is added to what it already holds. Its output is the staircase. It acts as a discrete integrator, which is why it is often called the integrator.

2. Answer

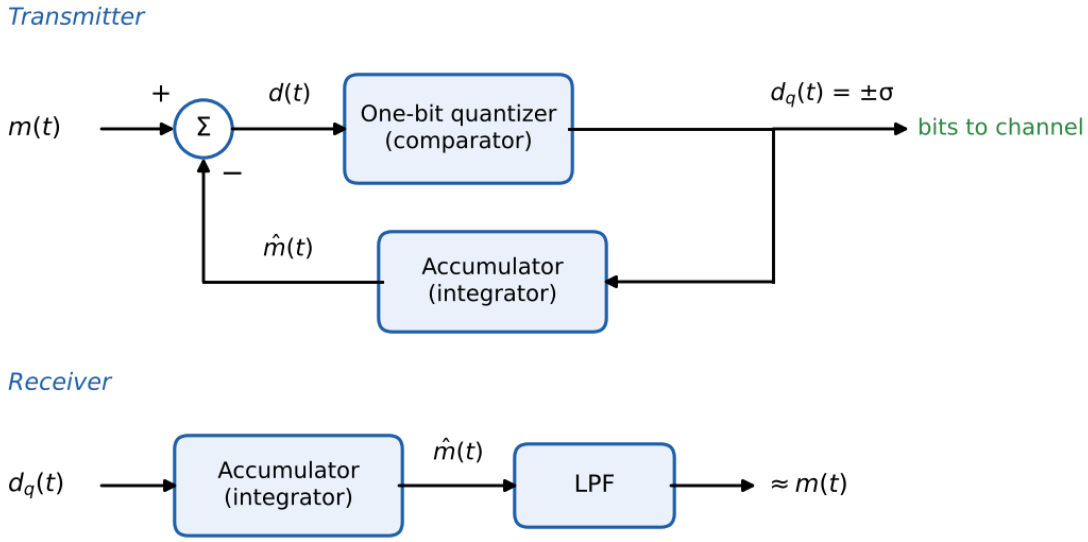


Figure 10: DM transmitter and receiver

Reading the figure. Top: $m(t)$ and the feedback $\hat{m}(t)$ meet at the summer, which gives $d(t)$. The one-bit quantizer sends $\pm\sigma$ to the channel, and also feeds the accumulator that builds $\hat{m}(t)$. Bottom: the received pulses go through an identical accumulator and an LPF.

Transmitter:

1. The **summer** finds the error $d(t) = m(t) - \hat{m}(t)$.
2. The **one-bit quantizer (comparator)** gives $d_q(t) = +\sigma$ if $d > 0$ and $-\sigma$ if $d < 0$, once per sample. This is sent as bit 1 or 0.
3. The **accumulator** in the feedback path adds up $d_q(t)$ to give the staircase $\hat{m}(t)$.

Receiver:

1. The **accumulator** adds up the received $\pm\sigma$ pulses and rebuilds the same staircase $\hat{m}(t)$.
2. The **low-pass filter** removes the sharp steps and out-of-band noise, leaving a smooth copy of $m(t)$.

The transmitter's feedback loop is what keeps errors from building up: the staircase is always steered back toward $m(t)$.

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3.(c) Explain the basic block diagram of PCM system. Marks: 5

1. Theory

1. Why go digital? (Core Idea)

An analog signal picks up noise on the way, and an amplifier boosts that noise along with the signal. A digital signal has only two levels, so a repeater can decide "0 or 1" and send out a clean new pulse. Noise is thrown away at every stage. **Pulse code modulation (PCM)** turns an analog signal into a stream of bits for this reason.

2. The three steps of PCM

- **Sampling** takes values at regular instants. Time becomes discrete.
- **Quantizing** rounds each value to the nearest of L allowed levels. Amplitude becomes discrete.
- **Encoding** writes each level number in binary, using $n = \log_2 L$ bits.

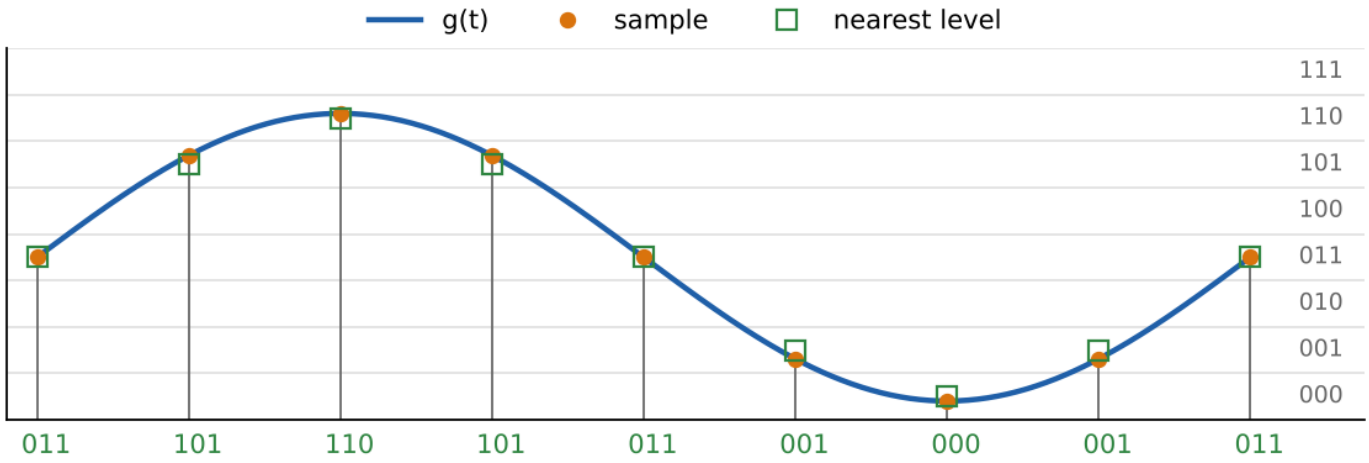


Figure 11: Sampling, quantizing and encoding

Reading the figure. The solid blue curve is $g(t)$. Each round orange dot is a sample, and each green square is the nearest of the 8 levels. The 3-bit code of each level is shown on the right, and the code sent for each sample is shown along the bottom. The small gap between a dot and its square is the quantization error.

2. Answer

PCM converts an analog signal into binary pulses, sends them, and rebuilds the signal at the receiver.

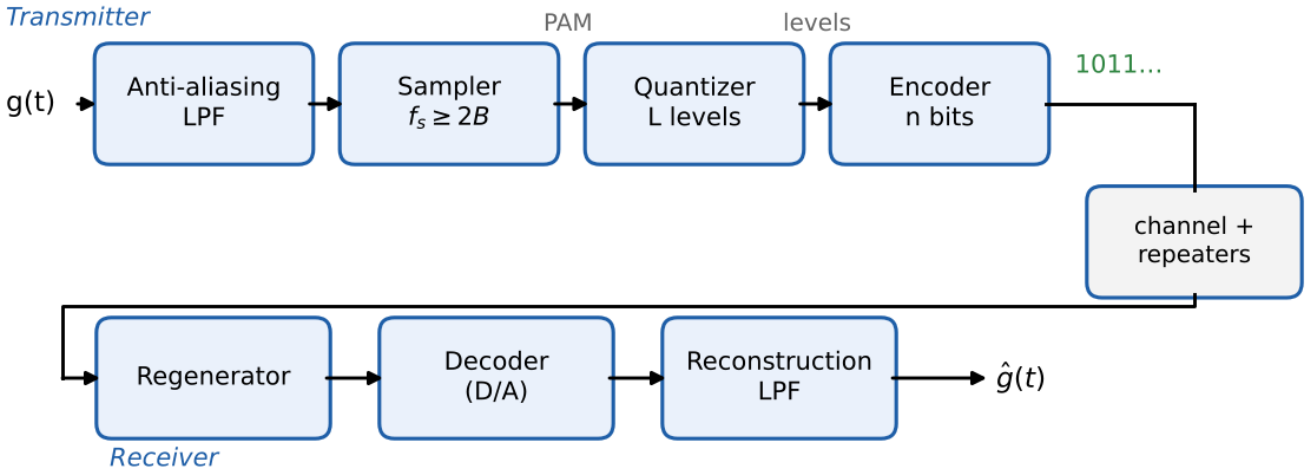


Figure 12: Block diagram of a PCM system

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Reading the figure. The top row is the transmitter, from $g(t)$ to the bit stream. The bits pass through the channel and repeaters, and the bottom row is the receiver, which ends with the rebuilt $\hat{g}(t)$.

Transmitter:

1. **Anti-aliasing LPF:** limits the signal to bandwidth B, so that sampling does not cause aliasing.
2. **Sampler:** takes samples at $f_s \geq 2B$ and gives PAM pulses.
3. **Quantizer:** rounds each sample to one of L levels. The rounding error is **quantization noise**, the only noise PCM adds on purpose.
4. **Encoder:** turns each level into n bits ($L = 2^n$).

Channel: **repeaters** along the path detect each pulse and send a fresh one, so noise does not add up.

Receiver:

1. **Regenerator:** decides 0 or 1 and restores clean pulses.
2. **Decoder (D/A):** turns each n-bit word back into a level.
3. **Reconstruction LPF:** smooths the levels into the analog signal $\hat{g}(t)$.

Bit rate: $R_b = nf_s$. With $f_s = 8$ kHz and $n = 8$, as in telephony, $R_b = 64$ kbit/s.

3.(d) An analog signal having 4 kHz bandwidth is sampled at 1.25 times the Nyquist rate, and each sample is quantized into one of 256 equally likely level. Assuming the samples to be statistically independent: (i) What is the information rate of the source? (ii) Can the output of this source be transmitted without error over the AWGN channel with a bandwidth of 10 kHz and S/N ratio of 20 dB? (iii) Find S/N ratio required for error-free transmission of part ii. (iv) Find the bandwidth required for an AWGN channel for an error free transmission of output of this source if S/N ratio is 20 dB Marks: 10

1. Theory

1. The Nyquist rate (Core Idea)

To capture a signal of bandwidth B completely, it must be sampled at least $2B$ times per second. This minimum, $f_N = 2B$, is the **Nyquist rate**. It is proved in 4(c).

2. Information rate

Each sample is one symbol from a source. If there are L equally likely levels, each one has probability $1/L$, so the entropy (see 1(a)) is

$$H = \sum_{i=1}^{L} \frac{1}{L} \log_2 L = \log_2 L \quad \text{bits/sample}$$

The **information rate** is the number of samples per second times the information per sample: $R = f_s H$ bits/s. "Statistically independent" means that samples add no information about each other, so nothing is saved by looking back.

3. Channel capacity

An **AWGN channel** adds white Gaussian noise, like thermal noise. The **Shannon-Hartley law** gives its capacity C, the highest rate at which information can pass with as few errors as we like:

$$C = B \log_2 \left(1 + \frac{S}{N}\right)$$

- If $R \leq C$, error-free transmission is possible with suitable coding.
- If $R > C$, it is impossible, whatever the code.

S/N in this formula is a plain power ratio, so a value in dB is first converted: $S/N = 10^{\text{dB}/10}$.

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2. Answer

Given:

- Signal bandwidth $B_m = 4$ kHz
- Sampling rate $= 1.25 \times$ Nyquist rate
- $L = 256$ equally likely levels, independent samples
- Channel: $B = 10$ kHz, $S/N = 20$ dB

(i) Information rate

$$f_N = 2B_m = 2 \times 4 = 8 \text{ kHz}$$

$$f_s = 1.25 \times 8 = 10 \text{ kHz}$$

$$H = \log_2 256 = 8 \text{ bits/sample}$$

$$R = f_s H = 10\,000 \times 8$$

$$\boxed{R = 80 \text{ kbit/s}}$$

(ii) Can it pass over the 10 kHz channel?

$$\frac{S}{N} = 10^{20/10} = 100$$

$$C = B \log_2 \left(1 + \frac{S}{N}\right) = 10\,000 \log_2 101 = 10\,000 \times 6.658$$

$$\boxed{C = 66.6 \text{ kbit/s}}$$

Since $R = 80$ kbit/s $> C = 66.6$ kbit/s, **error-free transmission is not possible** on this channel.

(iii) S/N needed on the 10 kHz channel

Set $C = R$:

$$10\,000 \log_2 \left(1 + \frac{S}{N}\right) = 80\,000 \;\Rightarrow\; \log_2 \left(1 + \frac{S}{N}\right) = 8$$

$$\frac{S}{N} = 2^8 - 1 = 255$$

$$\boxed{\frac{S}{N} = 255 = 24.1 \text{ dB}}$$

(iv) Bandwidth needed at 20 dB

Set $C = R$ with $S/N = 100$:

$$B = \frac{R}{\log_2(1 + S/N)} = \frac{80\,000}{\log_2 101} = \frac{80\,000}{6.658}$$

$$\boxed{B = 12.02 \text{ kHz}}$$

Final Answer: (i) 80 kbit/s; (ii) no, $C = 66.6$ kbit/s is less than 80 kbit/s; (iii) $S/N = 255$ (24.1 dB); (iv) 12.02 kHz.

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3. Notes and Remarks

- These are limits. Shannon's law shows that the rate is possible, but not which code reaches it.
- Parts (iii) and (iv) show the trade-off between bandwidth and power: the channel can be fixed with about 4 dB more signal power or 20% more bandwidth.

4. Final Concise Answer

(a) Slope overload: the signal's slope exceeds σf_s, and the staircase lags. Granular noise: the staircase hunts by $\pm\sigma$ around a flat signal. ADM increases the step on runs of equal bits and decreases it on alternating bits.
(b) Transmitter: summer, one-bit quantizer, accumulator in the feedback path. Receiver: accumulator and LPF.
(c) LPF, sampler, quantizer, encoder; channel with repeaters; regenerator, decoder, LPF. $R_b = nf_s$.
(d) $R = 80$ kbit/s; $C = 66.6$ kbit/s, so no; $S/N = 255$ (24.1 dB); $B = 12.02$ kHz.

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4. (a) With appropriate block diagram, describe the operation of superheterodyne receiver. Marks: 7
   (b) Explain how the message signal is reconstructed from VSB signal. Marks: 7
   (c) State and prove Nyquist sampling theorem. Also find the interpolation formula for the reconstruction of analog signals from its uniform samples. Marks: 7
   (d) State Carson's rule and its limitations. Marks: 4

4.(a) With appropriate block diagram, describe the operation of superheterodyne receiver. Marks: 7

1. Theory

1. The problem with a simple receiver (Core Idea)

A radio receiver must pick one station out of many and amplify it strongly. A filter with very high gain and sharp edges is hard to build when it must also be **tunable** over a wide range of frequencies. Each station would need the filter to be retuned perfectly.

The **superheterodyne** idea is clever: do not move the filter to the station. Move the station to the filter. Every station is shifted down to one fixed **intermediate frequency (IF)**, where one fixed, sharp, high-gain amplifier does all the hard work.

2. How a mixer shifts frequency

A **mixer** multiplies two signals. Since $2\cos A\cos B = \cos(A-B)+\cos(A+B)$, multiplying a station at f_s by a local tone at f_{LO} gives the difference and the sum frequencies. The IF filter keeps the difference, and the message moves with it. For AM radio $f_{IF} = 455$ kHz.

2. Answer

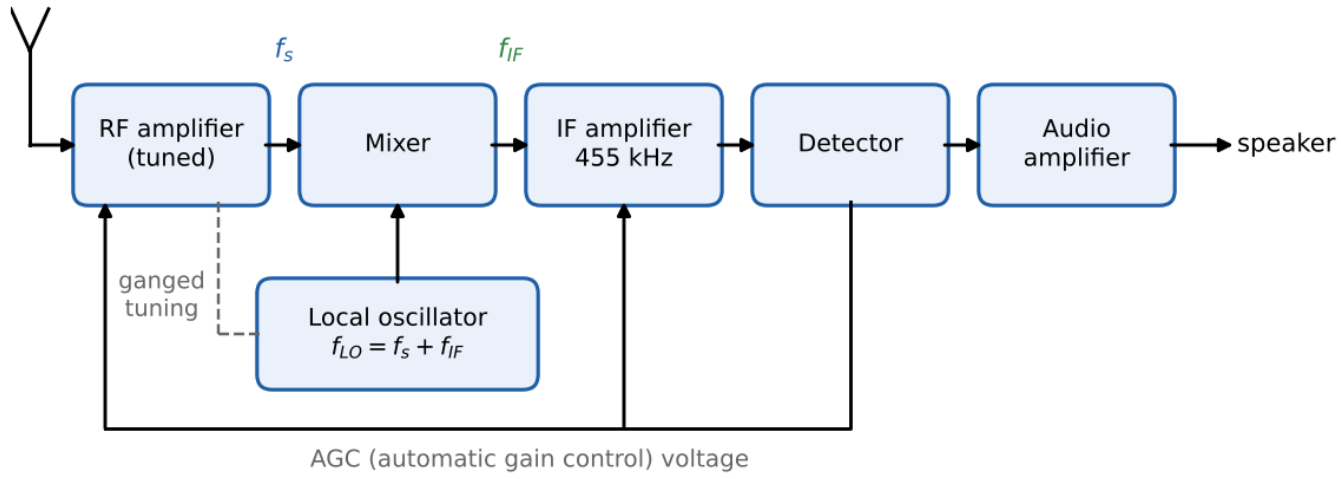


Figure 13: Block diagram of a superheterodyne receiver

Reading the figure. The signal passes from the antenna along the top row to the speaker. The local oscillator feeds the mixer and is tuned together with the RF stage (dashed "ganged tuning" line). The AGC line runs from the detector back to the RF and IF amplifiers.

The superheterodyne receiver converts every received station to one fixed intermediate frequency before amplifying and detecting it.

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Operation:

1. **RF amplifier (tuned):** selects the wanted band around f_s, gives some gain and rejects the image frequency.
2. **Local oscillator:** tuned together with the RF stage (ganged), so that it is always above the station by f_{IF}:

$$f_{LO} = f_s + f_{IF}$$

3. **Mixer:** multiplies the two and gives $f_{LO} - f_s = f_{IF}$ (and the unwanted sum, which is filtered out). The message is unchanged, but now sits at 455 kHz.
4. **IF amplifier:** fixed-tuned, sharp filters and high gain. It gives most of the receiver's **selectivity** and **sensitivity**.
5. **Detector:** recovers the message (an envelope detector for AM).
6. **Audio amplifier:** raises the message to drive the **speaker**.
7. **AGC (automatic gain control):** a dc voltage from the detector reduces the RF and IF gain on strong stations, so the output stays steady.

Example

for a station at $f_s = 1000$ kHz, $f_{LO} = 1000 + 455 = 1455$ kHz.

Image frequency: the mixer output depends only on the *difference*, so a station on the other side of the oscillator also gives 455 kHz:

$$f_{si} = f_s + 2f_{IF}$$

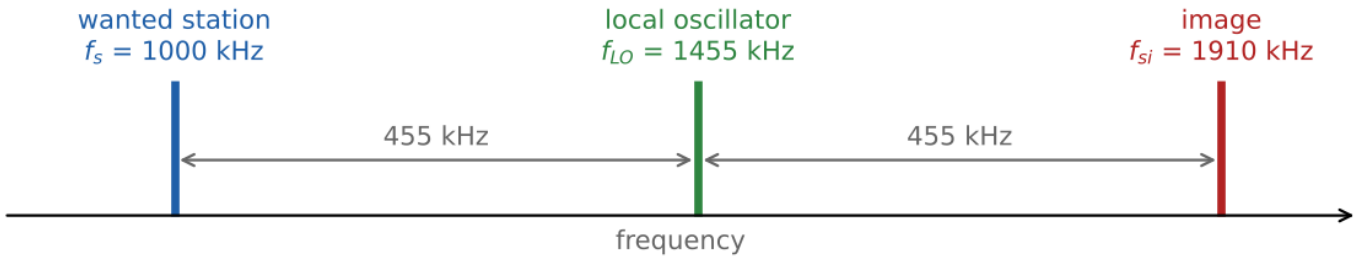


Figure 14: The image frequency

Reading the figure. The wanted station at 1000 kHz and the image at 1910 kHz are both 455 kHz from the oscillator at 1455 kHz. The IF stage cannot tell them apart, so the RF stage must remove the image before the mixer.

Advantages: high and uniform gain and selectivity across the band, with one fixed IF section. **Drawback:** image interference, and more circuits than a simple receiver.

3. Notes and Remarks

- The oscillator is put above the station because the required tuning ratio is then small. For the AM band (540 to 1600 kHz), it covers 995 to 2055 kHz (ratio about 2). Below the station, it would need 85 to 1145 kHz (ratio about 13), which is hard to tune.
- FM broadcast receivers use the same structure, with $f_{IF} = 10.7$ MHz and an FM detector.

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4.(b) Explain how the message signal is reconstructed from VSB signal. Marks: 7

1. Theory

1. What is VSB? (Core Idea)

An AM or DSB signal has two sidebands that carry the same information, so one is a waste of bandwidth. **SSB** removes one sideband, but that needs a filter with a nearly vertical edge, which cannot be built when the message has energy near 0 Hz. **Vestigial sideband (VSB)** is the compromise: it keeps one sideband and a small **vestige** (trace) of the other, using a filter with a gentle slope. Its bandwidth is only a little more than B. Broadcast television used it.

2. The trick that makes it work

The filter's slope cuts into the signal around f_c, so the message seems to be damaged there. But the slope is shaped with **odd symmetry** about f_c. After demodulation, the part lost from the main sideband is exactly filled in by the vestige of the other one.

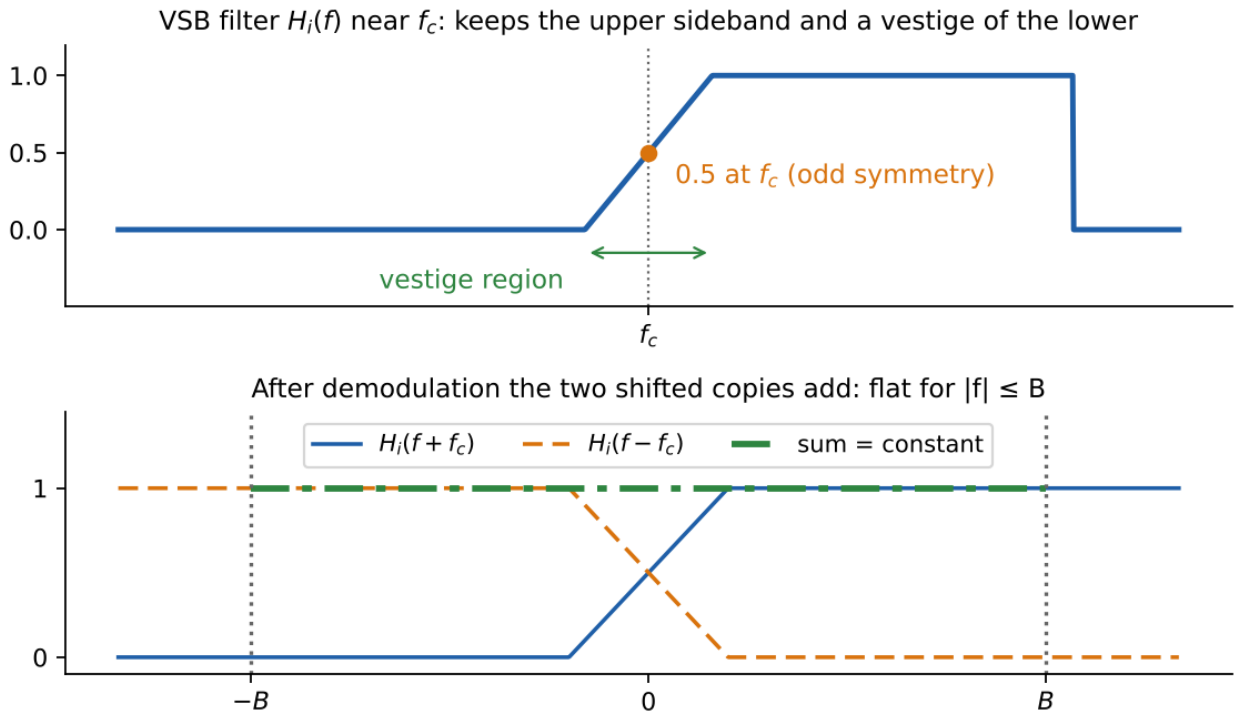


Figure 15: VSB filter and how the two shifted copies add

Reading the figure. Top: the filter passes the upper sideband fully, and its slope crosses 0.5 exactly at f_c. Bottom: demodulation shifts the filter by $+f_c$ (solid blue) and $-f_c$ (dashed orange). Their two slopes are mirror images, so the sum (dash-dot green) stays flat across $|f| \le B$.

3. Two facts about filters that the answer uses

- **A filter multiplies the spectrum.** If a signal with spectrum $X(f)$ goes through a filter with transfer function $H(f)$, the output spectrum is $Y(f) = X(f)\,H(f)$. Each frequency is simply scaled by the filter's value at that frequency.
- **A real filter has a mirror-image response.** A filter built from real parts has a real impulse response $h(t)$. Then

$$H(-f) = \int_{-\infty}^{\infty} h(t)\, e^{+j2\pi ft}\, dt = \left[\int_{-\infty}^{\infty} h(t)\, e^{-j2\pi ft}\, dt\right]^* = H^*(f)$$

The middle step holds because $h(t)$ is real, so taking the complex conjugate only flips the sign of j. For a filter drawn with a real (zero-phase) response, as in the figures here, this just means $H(-f) = H(f)$: the part of H_i near $-f_c$ is the mirror image of the part near $+f_c$.

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2. Answer

A VSB signal is demodulated **coherently**: multiplying by the carrier and low-pass filtering recovers $m(t)$, provided the filters meet the condition derived below. The steps marked **Basic** build tools from first principles; a student who already knows them can skip straight to the result of that step. The notation follows Lathi (Sec. 4.5, Eqs. 4.23 to 4.26).

Step 1: The VSB system

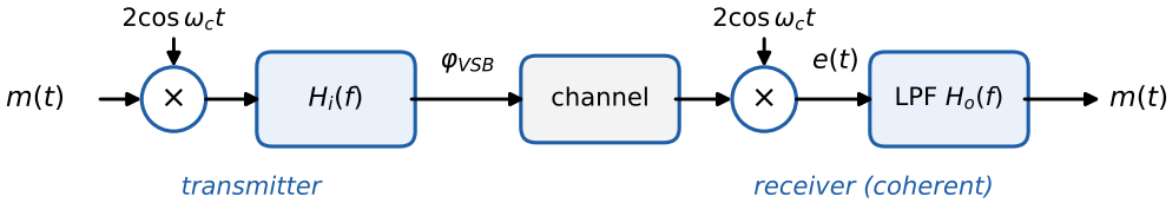


Figure 16: VSB transmitter and receiver

Reading the figure. Transmitter: the message $m(t)$ is multiplied by $2\cos\omega_c t$ (this makes DSB-SC), and the DSB-SC signal goes through the VSB shaping filter $H_i(f)$, giving $\varphi_{VSB}(t)$, which is sent through the channel. Receiver: $\varphi_{VSB}(t)$ is multiplied by the same carrier $2\cos\omega_c t$, giving $e(t)$, which goes through the low-pass filter $H_o(f)$.

The carrier is written as $2\cos\omega_c t$ (Lathi's choice) at both ends. The factor 2 makes the factors of $\frac{1}{2}$ in the steps below cancel; it does not change the result.

We must find the receiver filter $H_o(f)$ that makes the output exactly $m(t)$.

Step 2 (Basic): Multiplying by a cosine shifts a spectrum

Start from the definition of the Fourier transform, $X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$. Take the transform of $x(t)\, e^{j2\pi f_0 t}$:

$$\int_{-\infty}^{\infty} x(t)\, e^{j2\pi f_0 t}\, e^{-j2\pi ft}\, dt = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi(f-f_0)t}\, dt = X(f - f_0)$$

The two exponentials were joined into one ($e^a e^b = e^{a+b}$), and the last integral is the definition of X with f replaced by $f - f_0$. So

$$x(t)\, e^{j2\pi f_0 t} \Longleftrightarrow X(f - f_0)$$

Multiplying by $e^{j2\pi f_0 t}$ moves the whole spectrum to the right by f_0. In the same way, multiplying by $e^{-j2\pi f_0 t}$ moves it to the left: $X(f + f_0)$.

Now use Euler's formula, $e^{j\theta} + e^{-j\theta} = 2\cos\theta$, with $\theta = 2\pi f_c t = \omega_c t$:

$$2\cos\omega_c t = e^{j2\pi f_c t} + e^{-j2\pi f_c t}$$

So multiplying by $2\cos\omega_c t$ is the sum of one right shift and one left shift:

$$x(t) \cdot 2\cos\omega_c t \Longleftrightarrow X(f - f_c) + X(f + f_c)$$

This one result is used twice: at the transmitter (Step 3) and at the receiver (Step 4).

Step 3: The transmitted spectrum

The DSB-SC signal. Apply Step 2 with $x(t) = m(t)$:

$$2m(t)\cos\omega_c t \Longleftrightarrow M(f - f_c) + M(f + f_c)$$

The message spectrum appears twice, centred at $+f_c$ and at $-f_c$.

The shaping filter. The DSB-SC signal passes through $H_i(f)$, and a filter multiplies the spectrum (Theory 3):

$$\Phi_{VSB}(f) = [M(f+f_c) + M(f-f_c)]\, H_i(f) \qquad \text{(Lathi, Eq. 4.23)}$$

Step 4: After the receiver's multiplier

The receiver forms $e(t) = 2\varphi_{VSB}(t)\cos\omega_c t$. Apply Step 2 again, now with $x(t) = \varphi_{VSB}(t)$:

$$E(f) = \Phi_{VSB}(f+f_c) + \Phi_{VSB}(f-f_c)$$

Work out each shifted spectrum by writing $(f+f_c)$, and then $(f-f_c)$, in place of every f in the result of Step 3.

Left shift, $f \to f+f_c$:

$$\Phi_{VSB}(f+f_c) = [M(f+f_c+f_c) + M(f+f_c-f_c)]\, H_i(f+f_c) = [M(f+2f_c) + M(f)]\, H_i(f+f_c)$$

Right shift, $f \to f-f_c$:

$$\Phi_{VSB}(f-f_c) = [M(f-f_c+f_c) + M(f-f_c-f_c)]\, H_i(f-f_c) = [M(f) + M(f-2f_c)]\, H_i(f-f_c)$$

Add the two. Multiply out the brackets:

$$E(f) = M(f+2f_c)H_i(f+f_c) + M(f)H_i(f+f_c) + M(f)H_i(f-f_c) + M(f-2f_c)H_i(f-f_c)$$

The second and third terms both contain $M(f)$, so take $M(f)$ out as a common factor:

$$E(f) = \underbrace{M(f)\,[H_i(f+f_c) + H_i(f-f_c)]}_{\text{near } f=0} + \underbrace{M(f+2f_c)H_i(f+f_c) + M(f-2f_c)H_i(f-f_c)}_{\text{near } f=\mp 2f_c}$$

Step 5: After the low-pass filter

$M(f)$ is non-zero only for $|f| \le B$. So $M(f-2f_c)$, which is M moved to the right by $2f_c$, is non-zero only for

$$2f_c - B \le f \le 2f_c + B$$

Its lowest frequency, $2f_c - B$, is above B whenever $f_c > B$, which is always true (the carrier is far above the highest message frequency). So this term lies completely outside the message band $|f| \le B$. In the same way, $M(f+2f_c)$ lies around $-2f_c$, completely below $-B$.

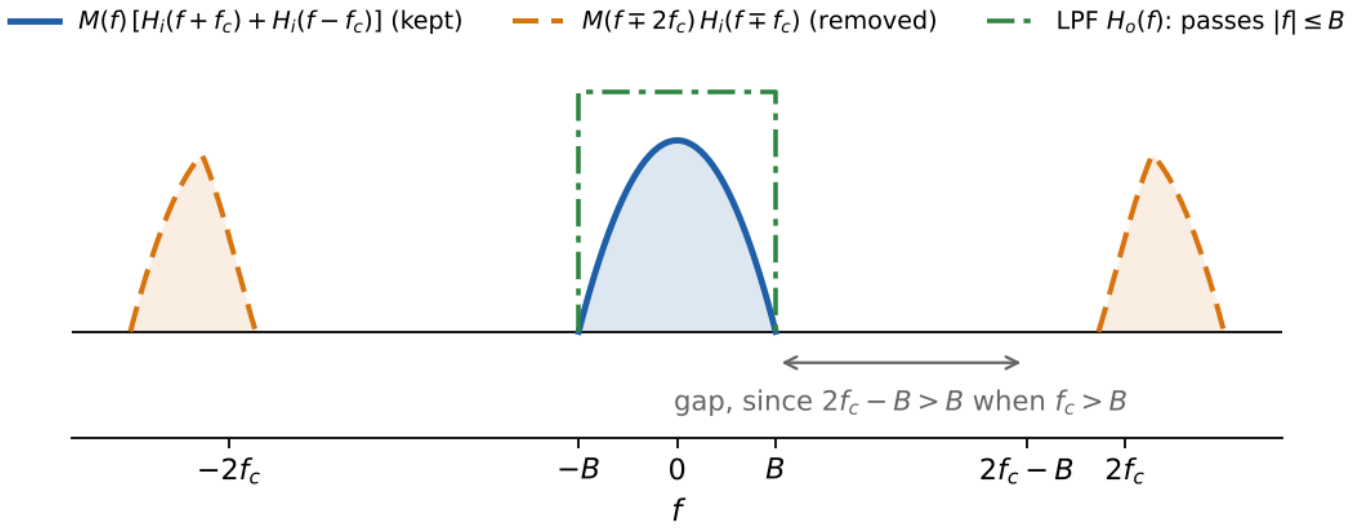


Figure 17: Spectrum after the receiver's multiplier (reused from the 2021-22 paper)

Reading the figure. The solid blue hump in the middle is the wanted term $M(f)[H_i(f+f_c)+H_i(f-f_c)]$, inside $|f| \le B$. The dashed orange humps are the copies around $\pm 2f_c$; each keeps the VSB shape (one sideband and a vestige). The dash-dot green box is the low-pass filter $H_o(f)$, which passes only

$|f| \le B$. The grey arrow shows the empty gap between B and $2f_c - B$, so the filter can keep the middle term and reject the copies.

The low-pass filter $H_o(f)$ passes $|f| \le B$ and rejects everything else, so its output spectrum is

$$E_o(f) = M(f)\,[H_i(f + f_c) + H_i(f - f_c)]\,H_o(f), \qquad |f| \le B$$

Step 6: The condition for perfect recovery

We want the output to be the message itself, $E_o(f) = M(f)$. Comparing with Step 5, everything that multiplies $M(f)$ must equal 1:

$$[H_i(f + f_c) + H_i(f - f_c)]\,H_o(f) = 1$$

so, over the message band,

$$\boxed{H_o(f) = \frac{1}{H_i(f + f_c) + H_i(f - f_c)}, \quad |f| \le B} \qquad \text{(Lathi, Eq. 4.25)}$$

The receiver filter is an **equalizer**: it undoes whatever gain the bracket gives each message frequency, so every frequency of $m(t)$ comes out with the same gain (the distortionless condition).

Step 7: When is H_o just a plain low-pass filter?

If the shaping filter is chosen so that the bracket is the same for every message frequency (Lathi takes it as 1, the **complementary VSB filter**, Eq. 4.26),

$$H_i(f + f_c) + H_i(f - f_c) = 1, \qquad |f| \le B$$

then Step 6 gives $H_o(f) = 1$ for $|f| \le B$: a plain low-pass filter. This is what the figure in Theory 2 shows. What does this condition mean for the shape of H_i?

For $|f| \le B$, the term $H_i(f + f_c)$ reads the filter near $+f_c$, and $H_i(f - f_c)$ reads it near $-f_c$. Rename the message frequency f as x (so $|x| \le B$, the distance from the carrier): $H_i(f_c + x) + H_i(x - f_c) = 1$.

Bring both terms to the side near $+f_c$. By the mirror property of a real filter (Theory 3), $H_i(x - f_c) = H_i(-(f_c - x)) = H_i^*(f_c - x)$. For the real (zero-phase) filter drawn here, $H_i^* = H_i$. So the condition becomes

$$H_i(f_c + x) + H_i(f_c - x) = 1, \qquad |x| \le B$$

The value at the carrier. Put $x = 0$: $2H_i(f_c) = 1$, so $H_i(f_c) = \frac{1}{2}$.

The symmetry. Write the filter near f_c as half plus a deviation, $H_i(f_c + x) = \frac{1}{2} + g(x)$. Then $H_i(f_c - x) = \frac{1}{2} + g(-x)$, and the condition gives

$$\tfrac{1}{2} + g(x) + \tfrac{1}{2} + g(-x) = 1 \quad \Rightarrow \quad g(-x) = -g(x)$$

So g is odd: the filter has **odd symmetry about the point** $(f_c, \frac{1}{2})$. As far as it rises above $\frac{1}{2}$ at $f_c + x$, it falls below $\frac{1}{2}$ at $f_c - x$. After demodulation the vestige of the unwanted sideband exactly fills in what the slope cut from the wanted one.

Key Points:

- With a general (not complementary) H_i, the receiver still recovers $m(t)$ exactly by using the equalizer H_o of Step 6.
- With a large carrier added (VSB + C, as in TV), a simple envelope detector can be used instead.

Steps to remember

DSB-SC → shape with H_i → multiply by $2\cos\omega_c t$ (two shifts by $\pm f_c$) → add and collect $M(f)$ → LPF removes the $\pm 2f_c$ copies → $H_o = 1/[H_i(f + f_c) + H_i(f - f_c)]$ → a plain LPF is enough when H_i has half gain and odd symmetry at f_c.

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4.(c) State and prove Nyquist sampling theorem. Also find the interpolation formula for the reconstruction of analog signals from its uniform samples. Marks: 7

1. Theory

1. What is sampling? (Core Idea)

Sampling keeps the value of $g(t)$ only at instants T_s apart. It seems that the values in between are lost forever. The surprising result is that nothing is lost, if the signal's frequencies are limited and the samples are close enough together. A smooth curve cannot wiggle much between close points.

2. Sampling as multiplication by impulses

Mathematically, sampling multiplies $g(t)$ by a train of unit impulses $\delta_{T_s}(t)$, one every T_s seconds. This gives

$$\bar{g}(t) = g(t)\delta_{T_s}(t) = \sum_n g(nT_s)\delta(t - nT_s)$$

3. Two tools

- **Fourier series:** any periodic signal is a sum of harmonics. An impulse train of period T_s contains every harmonic of $f_s = 1/T_s$ equally:

$$\delta_{T_s}(t) = \frac{1}{T_s}\sum_{n=-\infty}^{\infty} e^{jn2\pi f_s t}$$

- **Frequency shifting:** multiplying by $e^{j2\pi f_0 t}$ moves a spectrum by f_0, so $G(f)$ becomes $G(f - f_0)$.

2. Answer

Statement: a signal bandlimited to B Hz (with no spectral content above B) is completely determined by its samples, and can be recovered from them exactly, if the sampling rate satisfies

$$\boxed{f_s \geq 2B}$$

The minimum rate $2B$ is the **Nyquist rate**, and $1/2B$ is the Nyquist interval (Lathi, Eqs. 6.5 and 6.6).

Proof.

Step 1: Write sampling as a multiplication

Let $g(t)$ have spectrum $G(f)$ with $G(f) = 0$ for $|f| > B$. Taking a sample every T_s seconds is the same as multiplying $g(t)$ by a train of unit impulses, $\delta_{T_s}(t) = \sum_n \delta(t - nT_s)$. Each impulse picks out the value of g at its own instant:

$$\bar{g}(t) = g(t)\,\delta_{T_s}(t) = \sum_n g(nT_s)\,\delta(t - nT_s)$$

This is Lathi's Eq. 6.1.

Step 2 (Basic): Fourier series of the impulse train

The train repeats every T_s seconds, so it is periodic and has an exponential Fourier series with fundamental frequency $f_s = 1/T_s$:

$$\delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} D_n\, e^{jn2\pi f_s t}, \qquad D_n = \frac{1}{T_s}\int_{-T_s/2}^{T_s/2} \delta_{T_s}(t)\, e^{-jn2\pi f_s t}\, dt$$

Inside the limits $-T_s/2$ to $T_s/2$ the train has only one impulse, the one at $t = 0$. The sifting property, $\int \delta(t)x(t)\,dt = x(0)$, picks out the exponential at $t = 0$, where it equals $e^0 = 1$. So $D_n = 1/T_s$ for every n:

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$$\delta_{T_s}(t) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} e^{jn2\pi f_s t}$$

This is Lathi's Eq. 6.2: the train holds every harmonic of f_s with the same strength.

Step 3: Put the series into Step 1

$$\bar{g}(t) = g(t) \cdot \frac{1}{T_s} \sum_{n=-\infty}^{\infty} e^{jn2\pi f_s t} = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} g(t)\, e^{jn2\pi f_s t}$$

(Lathi, Eq. 6.3). Each term is $g(t)$ multiplied by a complex exponential.

Step 4 (Basic): The frequency-shift property

Take the Fourier transform of $g(t)e^{j2\pi f_0 t}$ for any frequency f_0:

$$\int_{-\infty}^{\infty} g(t)e^{j2\pi f_0 t}e^{-j2\pi f t}\, dt = \int_{-\infty}^{\infty} g(t)e^{-j2\pi (f-f_0) t}\, dt$$

The right side is the definition of G with f replaced by $f - f_0$. So

$$g(t)e^{j2\pi f_0 t} \Longleftrightarrow G(f - f_0)$$

Multiplying by $e^{j2\pi f_0 t}$ moves the whole spectrum to the right by f_0.

Step 5: Transform the sum term by term

Use Step 4 on each term of Step 3, with $f_0 = nf_s$:

$$\bar{G}(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} G(f - nf_s)$$

This is Lathi's Eq. 6.4.

Step 6: Read the spectrum and find the condition

The sampled spectrum is $G(f)$, scaled by $1/T_s$, repeated every f_s Hz. The copy centred at 0 spans $-B$ to $+B$. The next copy is centred at f_s, so it starts at $f_s - B$. The two do not overlap if the second starts at or after the end of the first:

$$f_s - B \geq B \quad \Longrightarrow \quad f_s \geq 2B$$

Step 7: Recover the signal

An ideal LPF of cut-off B and gain T_s passes only the $n = 0$ copy, $\frac{1}{T_s}G(f)$, and multiplies it by T_s, which leaves $G(f)$ itself. So $g(t)$ is recovered exactly, which proves the theorem.

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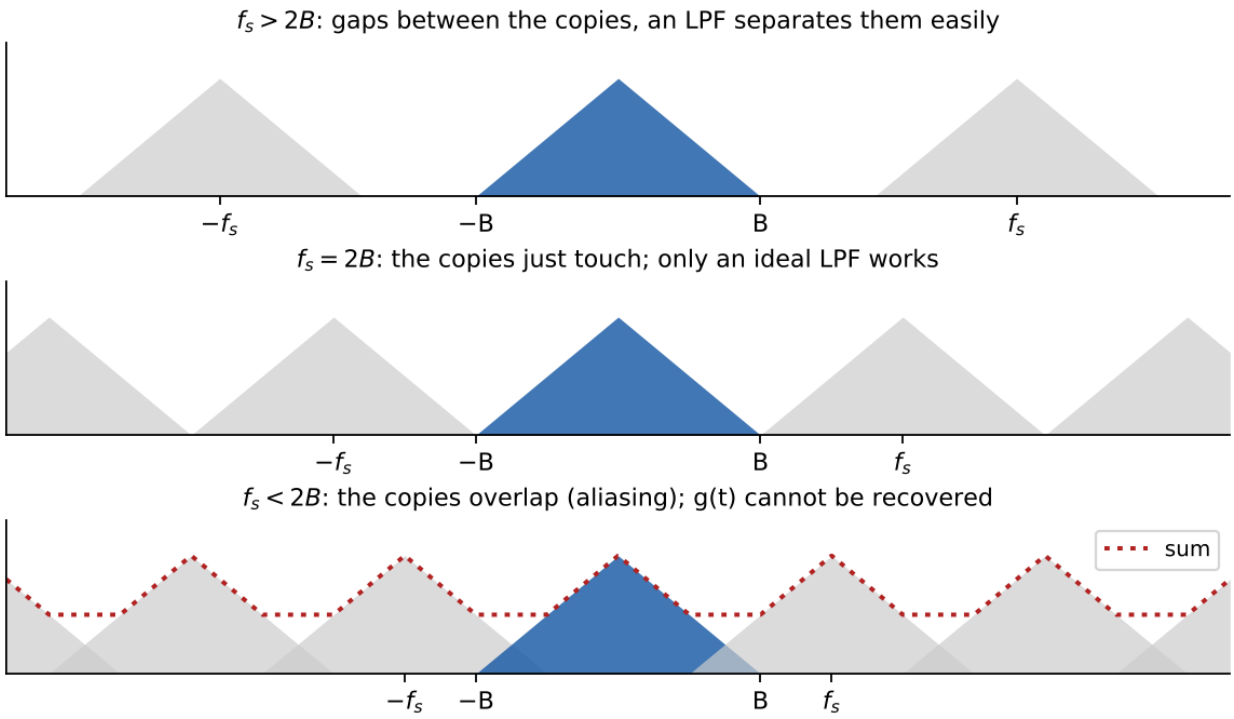


Figure 18: Spectrum of a sampled signal for three sampling rates

Reading the figure. The dark triangle is the original spectrum and the pale triangles are its copies. Top: copies are well apart. Middle: at exactly $2B$ they just touch. Bottom: below $2B$ they overlap, and the dotted red sum no longer shows the dark triangle. This overlap is **aliasing**.

Interpolation formula. The proof showed that the filter recovers $g(t)$; the next steps find what the filter does in time, which gives $g(t)$ directly from its samples.

Step 8: Impulse response of the LPF

Take $f_s = 2B$, so that $T_s = 1/2B$. The LPF has $H(f) = T_s$ for $|f| \le B$ and 0 elsewhere (Lathi, Eq. 6.7). Its impulse response is the inverse Fourier transform:

$$h(t) = \int_{-B}^{B} T_s\, e^{j2\pi ft}\, df = T_s \left[\frac{e^{j2\pi ft}}{j2\pi t}\right]_{-B}^{B} = T_s\, \frac{e^{j2\pi Bt} - e^{-j2\pi Bt}}{j2\pi t}$$

Euler's formula gives $e^{jx} - e^{-jx} = 2j \sin x$, so

$$h(t) = T_s\, \frac{2j \sin(2\pi Bt)}{j2\pi t} = T_s\, \frac{\sin(2\pi Bt)}{\pi t}$$

Multiply the top and bottom by $2B$, and write $\operatorname{sinc} x = \sin x / x$:

$$h(t) = 2BT_s\, \frac{\sin(2\pi Bt)}{2\pi Bt} = 2BT_s \operatorname{sinc}(2\pi Bt) = \operatorname{sinc}(2\pi Bt)$$

since $2BT_s = 1$ (Lathi, Eqs. 6.8 and 6.9). It is 1 at $t = 0$ and 0 at every other multiple of $T_s = 1/2B$, because there $\sin(2\pi Bt) = \sin(k\pi) = 0$.

Step 9: Add the outputs of all the impulses

The filter is linear and time-invariant. Each input impulse $g(kT_s)\delta(t - kT_s)$ therefore produces the same pulse h, shifted to kT_s and scaled by $g(kT_s)$. Adding all of them,

$$g(t) = \sum_k g(kT_s)\, h(t - kT_s)$$

Step 10: Put in $h(t)$

With $2\pi BkT_s = 2\pi Bk/(2B) = k\pi$:

$$\boxed{g(t) = \sum_k g(kT_s)\,\mathrm{sinc}[2\pi B(t - kT_s)] = \sum_k g(kT_s)\,\mathrm{sinc}(2\pi Bt - k\pi)}$$

This is the **interpolation formula**, Lathi's Eqs. 6.10a and 6.10b.

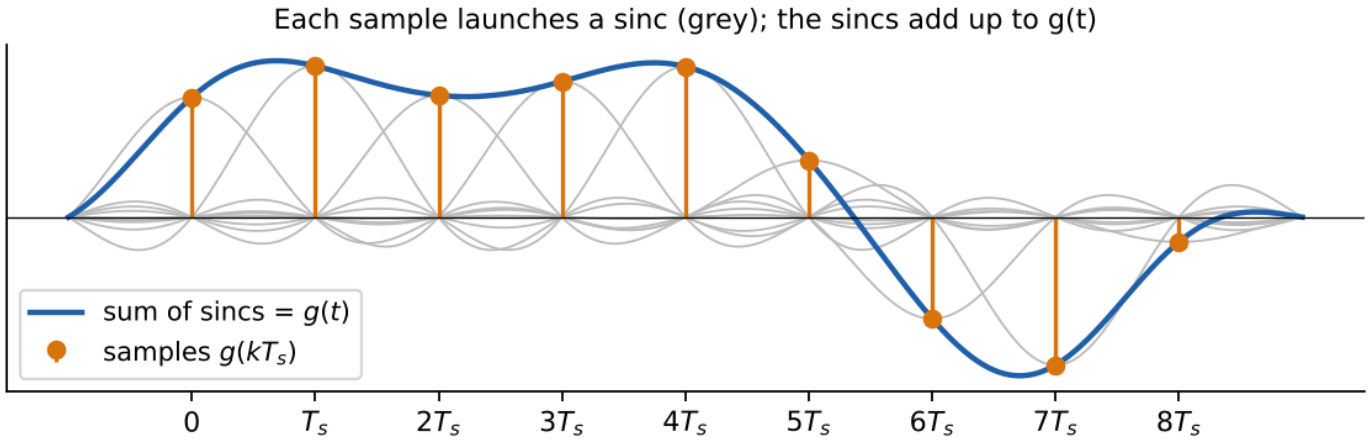


Figure 19: Reconstruction from samples

Reading the figure. Each round orange sample starts one thin grey sinc pulse. A sinc is 1 at its own sample and 0 at every other sample instant, so at each instant only one pulse counts. Between the instants the pulses add up to the thick solid blue curve, $g(t)$.

Steps to remember

Sampling = multiply by an impulse train → write the train as a Fourier series ($1/T_s$ for every harmonic) → each term shifts $G(f)$ by nf_s → copies every f_s, no overlap if $f_s \geq 2B$ → an LPF of gain T_s keeps $G(f)$ → its impulse response is $\mathrm{sinc}(2\pi Bt)$ → add one sinc per sample.

3. Notes and Remarks

- In practice f_s is kept above $2B$. The gap between copies (the **guard band**) allows a real filter with a gradual slope.
- For $f_s > 2B$ with an LPF of cut-off f_0 ($B \leq f_0 \leq f_s - B$) and gain T_s, $h(t) = 2f_0T_s\,\mathrm{sinc}(2\pi f_0t)$, and the same sum holds.

4.(d) State Carson's rule and its limitations. Marks: 4

1. Theory

1. Why is FM's bandwidth a problem? (Core Idea)

A single-tone FM wave does not have just two side frequencies, as AM does. In theory it has infinitely many, at $f_c \pm nf_m$, with sizes given by Bessel functions $J_n(\beta)$. But far from the carrier they become tiny, so a practical bandwidth can be chosen.

2. Two simple limits

- For small β (narrowband FM), only the first side frequencies count, so $B_T \approx 2f_m$.
- For large β (wideband FM), the frequency sweeps slowly across $\pm\Delta f$, so $B_T \approx 2\Delta f$.

Carson's rule is one formula that covers both limits.

2. Answer

Carson's rule: the bandwidth of an FM signal is approximately twice the sum of the frequency deviation and the highest modulating frequency:

$$\boxed{B_T \approx 2(\Delta f + f_m) = 2(\beta + 1) f_m}$$

Example

FM broadcast has $\Delta f = 75$ kHz and $f_m = 15$ kHz, so $B_T \approx 2(75 + 15) = 180$ kHz.

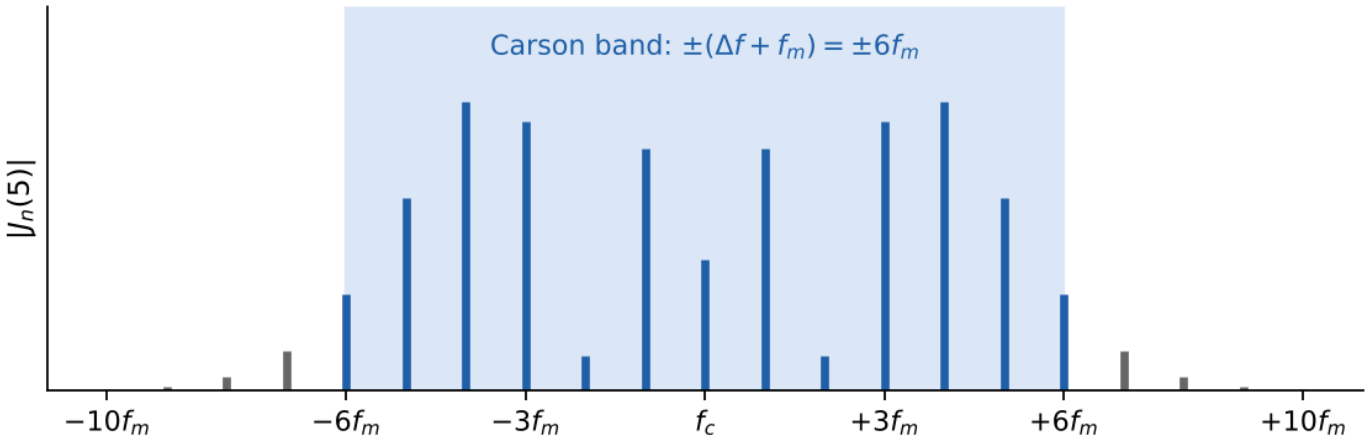


Figure 20: FM spectrum for a modulation index of 5

Reading the figure. For $\beta = 5$, the shaded Carson band covers $\pm 6 f_m$. It holds all the large lines (99.4% of the power). The grey lines outside it are small, but they are not zero.

Limitations:

1. It is an **approximation**. FM's true bandwidth is infinite, and the rule leaves out the small side frequencies outside the band.
2. It **underestimates** the bandwidth for intermediate values of β (about 2 to 10). Designers then use a larger value, such as $2(\Delta f + 2 f_m)$, which gives 210 kHz for FM broadcast.
3. It was built for a **single tone**. For a real message, Δf and f_m are worst-case values that rarely occur together, so the result is only an estimate.
4. It does not say how much distortion the cut-off causes, so it cannot be used where a precise distortion limit is needed.

3. Final Concise Answer

(a) RF amplifier, local oscillator ($f_{LO} = f_s + f_{IF}$), mixer, fixed IF amplifier at 455 kHz, detector, audio amplifier, AGC. Image frequency $f_{si} = f_s + 2 f_{IF}$.

(b) Multiply by $2\cos\omega_c t$ (shifts by $\pm f_c$) and low-pass filter (removes the $\pm 2 f_c$ copies). $E_o(f) = M(f)[H_i(f + f_c) + H_i(f - f_c)]H_o(f)$, so $H_o(f) = 1/[H_i(f + f_c) + H_i(f - f_c)]$ for $|f| \le B$; a plain LPF is enough when H_i has half gain and odd symmetry about f_c.

(c) $f_s \ge 2B$; $\bar{G}(f) = (1/T_s)\sum G(f - n f_s)$, so the copies do not overlap. $g(t) = \sum_k g(kT_s)\,\mathrm{sinc}(2\pi B t - k\pi)$.

(d) $B_T \approx 2(\Delta f + f_m)$. It is approximate, underestimates at intermediate β, is built for one tone, and does not give the distortion.

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5. (a) Write down the comparison between TDMA and CDMA. Marks: 5

(b) A voice grade telephone channel has a bandwidth of 3400 Hz. If the signal to noise ratio (SNR) on the channel is 30 dB, determine the capacity of the channel. If the above channel is to be used to transmit 4.8 kbps of data, determine the minimum SNR required on the channel. Marks: 6

(c) It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patents [*sic*: should be "patients"] are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitude is 0.25% of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data. Marks: 7

(d) Consider a message signal m(t) with the spectrum shown in Fig. 1. The message bandwidth W = 1 kHz. This signal is applied to a product modulator, together with a carrier wave $A_c \cos(2\pi f_c t)$, producing the DSBSC modulated wave s(t). This modulated wave is next applied to a coherent detector. Assuming perfect synchronism between the carrier waves in the modulator and detector, determine the spectrum of the detector output when:

(i) the carrier frequency $f_c = 1.25$ kHz and
(ii) the carrier frequency $f_c = 0.75$ kHz.

What is the lowest carrier frequency for which each component of the modulated wave s(t) is uniquely determined by m(t)? Marks: 7

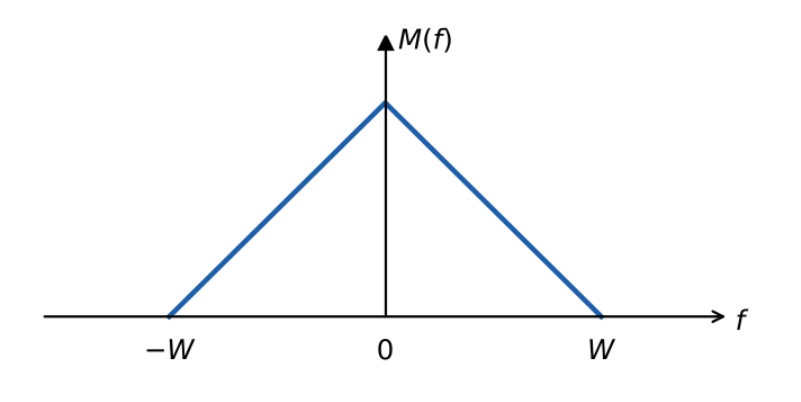


Figure 21: Fig. 1 of the paper (redrawn): the message spectrum M(f)

5.(a) Write down the comparison between TDMA and CDMA. Marks: 5

1. Theory

1. Multiple access again (Core Idea)

Many users must share one radio channel without disturbing each other (see 2(a)). Imagine a room of people who must all talk. In **TDMA** they take turns speaking. In **CDMA** they all speak at once, but each pair uses a different language, so a listener hears its own language clearly and the rest as background murmur.

2. How the two methods work

- **Time division multiple access (TDMA):** time is divided into repeating frames, and each frame into slots. Each user sends only in its own slot, as a short burst, using the whole band. Everyone must keep exact time.
- **Code division multiple access (CDMA):** each user multiplies its data by a fast, unique **code**. All users send at the same time, in the same wide band. The receiver multiplies by the wanted

code: that user's data comes out, and the others stay as low-level noise.

2. Answer

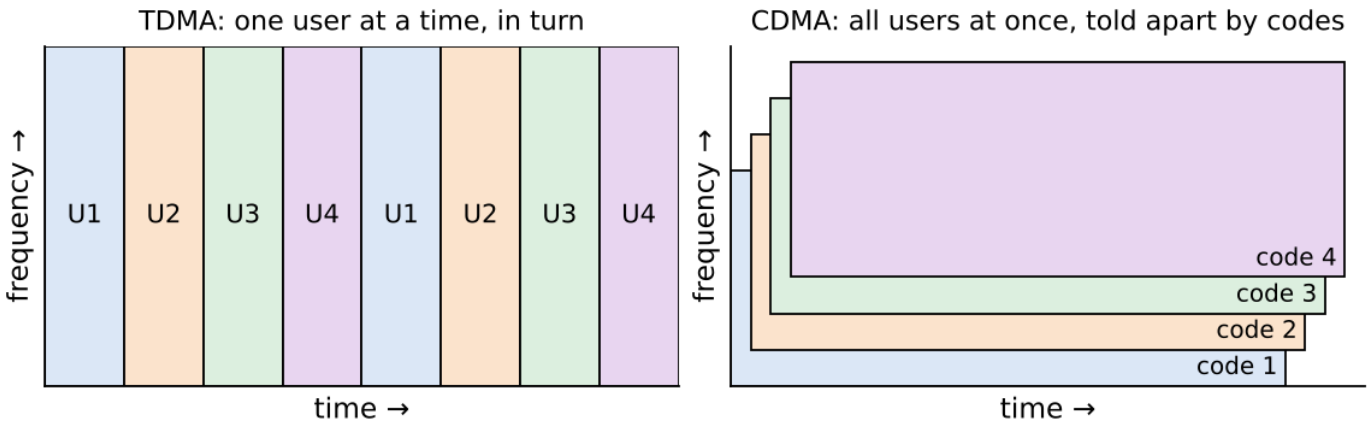


Figure 22: TDMA and CDMA in time and frequency

Reading the figure. Left: in TDMA, users U1 to U4 use the full band in turn, one slot at a time. Right: in CDMA, all four users occupy all of the time and frequency together, and only their codes separate them.

	TDMA	CDMA
What separates users	time slots	unique codes
Band and time used	whole band, in its own slot only	whole band, all the time
Transmission	bursts	continuous
Synchronization	exact timing and guard times are essential	code timing is needed at the receiver; no shared time slots
Capacity	hard limit: fixed number of slots	soft limit: each new user adds a little noise
Interference	little between users, when timing is good	other users appear as noise, so power control is essential
Bandwidth of each signal	that of the data burst	spread far wider than the data
Security and jamming	low, without encryption	high, since a signal looks like noise without its code
Examples	GSM (with FDMA), digital satellite links	IS-95, CDMA2000, GPS

Table 5: TDMA versus CDMA

5.(b) A voice grade telephone channel has a bandwidth of 3400 Hz. If the signal to noise ratio (SNR) on the channel is 30 dB, determine the capacity of the channel. If the above channel is to be used to transmit 4.8 kbps of data, determine the minimum SNR required on the channel. Marks: 6

1. Theory

1. Decibels (Core Idea)

The signal-to-noise ratio is often very large, so it is written on a log scale: $\mathrm{SNR}_{\mathrm{dB}} = 10 \log_{10} \mathrm{SNR}$. Every 10 dB is a factor of 10. So 30 dB is a power ratio of 1000.

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2. Shannon capacity

The largest error-free rate of a noisy channel is

$$C = B \log_2(1 + \text{SNR})$$

Here B is the bandwidth in Hz, and the SNR is a plain ratio. **Bandwidth** sets how many symbols can be sent per second. **SNR** sets how many different levels can be told apart in each symbol.

3. Minimum SNR for a required rate

The channel can carry a rate R only if $R \leq C$. At the limit $R = C$, solving for the SNR gives

$$\text{SNR}_{min} = 2^{R/B} - 1$$

2. Answer

Given:

- Bandwidth $B = 3400$ Hz
- SNR $= 30$ dB
- Data rate $R = 4.8$ kbps $= 4800$ bit/s

Step 1: Capacity

$$\text{SNR} = 10^{30/10} = 1000$$

$$C = 3400 \log_2(1 + 1000) = 3400 \times 9.967$$

$$\boxed{C = 33.89 \text{ kbit/s}}$$

Step 2: Minimum SNR for 4.8 kbps

$$\frac{R}{B} = \frac{4800}{3400} = 1.412$$

$$\text{SNR}_{min} = 2^{1.412} - 1 = 2.661 - 1$$

$$\boxed{\text{SNR}_{min} = 1.66 = 2.2 \text{ dB}}$$

Final Answer: $C = 33.89$ kbit/s; minimum SNR 1.66 (2.2 dB).

5.(c) It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patents [*sic*: should be "patients"] **are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitude is 0.25% of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data. Marks: 7**

1. Theory

1. The plan (Core Idea)

Each patient's analog ECG must become bits, the bits of all 10 patients must share one cable in turn, and the cable must carry the total rate. So we find four things in order: samples per second, bits per sample, total bits per second and, finally, the bandwidth.

2. Sampling rate

The Nyquist rate of a signal of bandwidth B is $2B$ (see 4(c)).

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3. Quantization error

A signal between $-m_p$ and $+m_p$ is divided into L levels, so one step is

$$\Delta = \frac{2m_p}{L}$$

A sample is rounded to the nearest level, so the largest error is half a step:

$$e_{max} = \frac{\Delta}{2} = \frac{m_p}{L}$$

More levels mean less error, but more bits. With n bits, $L = 2^n$, and L must be a power of 2.

4. Time division multiplexing

TDM gives each input a turn on the same line. In each round, one n-bit word is sent for each of the N channels, and there are f_s rounds per second:

$$R_b = N f_s n$$

5. Bandwidth of a binary stream

Nyquist's result for pulses is that a bandwidth B_T carries at most $2B_T$ pulses per second without pulses disturbing each other. So the minimum bandwidth is $B_T = R_b/2$.

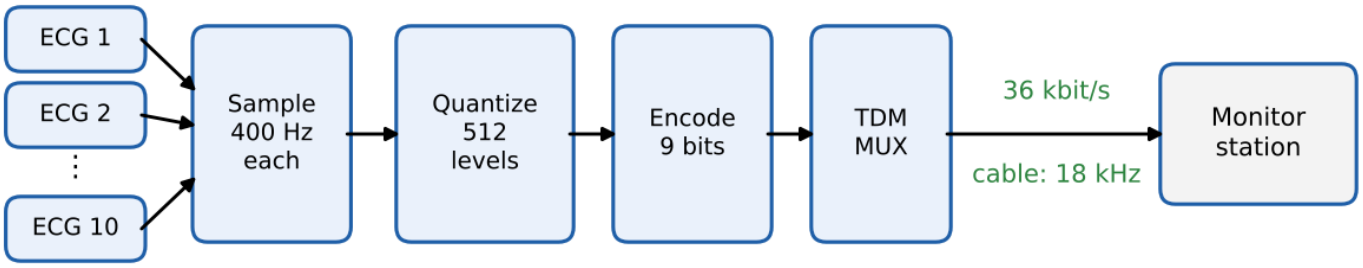


Figure 23: The ECG monitoring chain

Reading the figure. The ten ECG signals pass through sampling, quantizing and encoding, the TDM multiplexer combines them, and one cable takes the result to the monitoring station. The labels on the cable are the results of the answer below.

2. Answer

Note on the paper

"10 patents" in the second sentence is a misprint for "10 patients".

Given:

- Number of channels $N = 10$
- ECG bandwidth $B = 100$ Hz
- Maximum error $= 0.25\% = 0.0025\, m_p$
- Sampling rate at least $2\times$ the Nyquist rate

Step 1: Sampling rate

$$f_N = 2B = 200 \text{ Hz}, \qquad f_s = 2f_N = 400 \text{ Hz}$$

Step 2: Number of bits

The error is at most half a step:

$$\frac{m_p}{L} \le 0.0025\, m_p \;\Rightarrow\; L \ge 400$$

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The next power of 2 is $L = 512 = 2^9$, so

$$\boxed{n = 9 \text{ bits/sample}}$$

Step 3: Bit rate

$$R_b = N f_s n = 10 \times 400 \times 9 = 36\,000 \text{ bit/s}$$

Step 4: Minimum cable bandwidth

$$B_T = \frac{R_b}{2} = \frac{36\,000}{2}$$

$$\boxed{B_T = 18 \text{ kHz}}$$

Final Answer: the cable needs a minimum bandwidth of 18 kHz, for a total rate of 36 kbit/s.

3. Notes and Remarks

- A common slip is to take a full step as the error, which gives $L \geq 800$ and $n = 10$. Rounding to the *nearest* level gives half a step.
- With $L = 512$ the real error limit is $100/512 = 0.195\%$, within the 0.25% allowed.
- Framing bits are ignored. Real TDM adds a few, which raises the rate slightly.
- The same system is also solved in 6(b) of the 2021-22 paper.

5.(d) Consider a message signal m(t) with the spectrum shown in Fig. 1. The message bandwidth W = 1 kHz. This signal is applied to a product modulator, together with a carrier wave $A_c \cos(2\pi f_c t)$, producing the DSBSC modulated wave s(t). This modulated wave is next applied to a coherent detector. Assuming perfect synchronism between the carrier waves in the modulator and detector, determine the spectrum of the detector output when: (i) the carrier frequency $f_c = 1.25$ kHz and (ii) the carrier frequency $f_c = 0.75$ kHz. What is the lowest carrier frequency for which each component of the modulated wave s(t) is uniquely determined by m(t)? Marks: 7

1. Theory

1. DSB-SC modulation (Core Idea)

A **product modulator** simply multiplies the message by the carrier:

$$s(t) = A_c m(t) \cos(2\pi f_c t)$$

This is **double-sideband suppressed-carrier (DSB-SC)**. There is no separate carrier line, only the two sidebands. Since $\cos(2\pi f_c t) = \frac{1}{2}(e^{j2\pi f_c t} + e^{-j2\pi f_c t})$, and a complex exponential shifts a spectrum,

$$S(f) = \frac{A_c}{2}[M(f - f_c) + M(f + f_c)]$$

The message spectrum appears twice, centred at $+f_c$ and $-f_c$, each with half the height.

2. Coherent detection

The **coherent detector** multiplies the received wave by a local copy of the carrier, then low-pass filters. With a perfectly synchronized local carrier $\cos(2\pi f_c t)$, and $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$,

$$v(t) = s(t)\cos(2\pi f_c t) = \frac{A_c}{2} m(t) + \frac{A_c}{2} m(t)\cos(4\pi f_c t)$$

The first term is the message. The second is the message moved up to $2f_c$. In frequency, the first term is simply $\frac{A_c}{2} M(f)$. For the second, write $\cos(4\pi f_c t) = \frac{1}{2}(e^{j2\pi(2f_c)t} + e^{-j2\pi(2f_c)t})$; each exponential shifts the spectrum by $\pm 2f_c$, and the $\frac{1}{2}$ halves the height again:

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$$V(f) = \frac{A_c}{2}M(f) + \frac{A_c}{2}\cdot\frac{1}{2}[M(f-2f_c) + M(f+2f_c)] = \frac{A_c}{2}M(f) + \frac{A_c}{4}[M(f-2f_c) + M(f+2f_c)]$$

3. When does it fail?

The LPF keeps $|f| \le W$. It separates the wanted term from the copy at $2f_c$ only if they do not overlap. The copy starts at $2f_c - W$, so we need $2f_c - W \ge W$, or

$$f_c \ge W$$

So the carrier must be at least as high as the top frequency of the message.

2. Answer

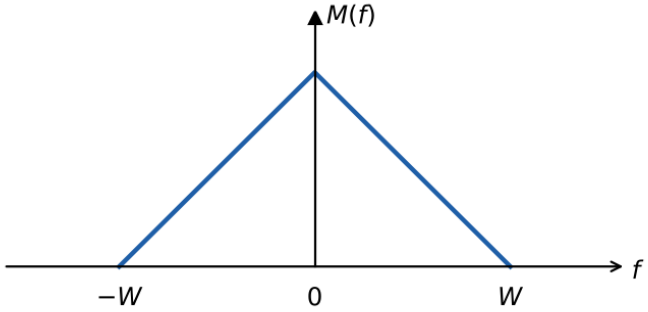


Figure 24: Repeated from the question: Fig. 1 of the paper (redrawn)

Given: $M(f)$ is a triangle of height $M_0 = M(0)$ on $|f| \le W$, with $W = 1$ kHz; synchronism is perfect. **Assumption:** the local carrier is $\cos(2\pi f_c t)$ and the detector's LPF has cut-off W. Then

$$S(f) = \frac{A_c}{2}[M(f-f_c) + M(f+f_c)], \qquad V(f) = \frac{A_c}{2}M(f) + \frac{A_c}{4}[M(f-2f_c) + M(f+2f_c)]$$

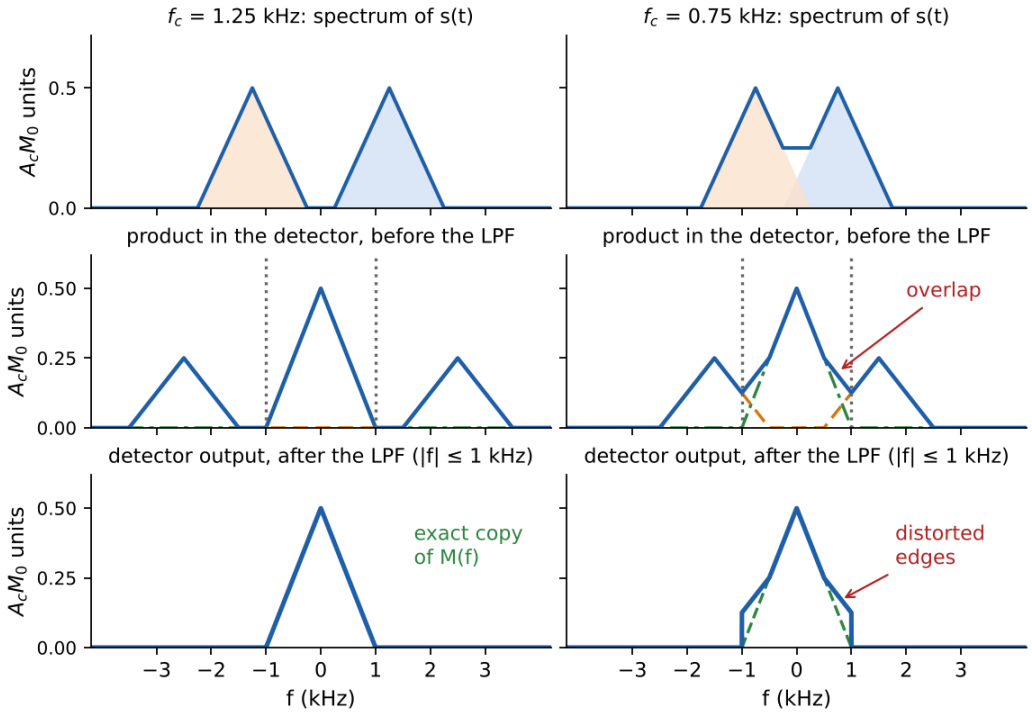


Figure 25: Spectra for the two carrier frequencies

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Reading the figure. Each column is one carrier frequency, and the rows follow the signal. Top: the spectrum of $s(t)$, with the two shifted copies tinted. Middle: the spectrum in the detector, where the solid blue line is the total $V(f)$, and the dotted lines mark the LPF edges $\pm W$. Bottom: the output after the LPF. On the right, the dashed green line is the ideal triangle and the solid blue line is the real output.

Case (i): $f_c = 1.25$ kHz

The copies in $V(f)$ are centred at ± 2.5 kHz and span $1.5 \le |f| \le 3.5$ kHz. They lie fully outside $|f| \le 1$ kHz, so the LPF removes them:

$$\boxed{V_o(f) = \frac{A_c}{2} M(f)}$$

The output is an undistorted triangle of height $A_c M_0/2$ on $|f| \le 1$ kHz, so $m(t)$ is recovered exactly.

Case (ii): $f_c = 0.75$ kHz

The copies are centred at ± 1.5 kHz and span $0.5 \le |f| \le 2.5$ kHz, which overlaps the message for $0.5 \le |f| \le 1$ kHz. The LPF cannot separate them:

$$\boxed{V_o(f) = \frac{A_c}{2} M(f) + \frac{A_c}{4}[M(f-1.5) + M(f+1.5)], \quad |f| \le 1 \text{ kHz}}$$

(f in kHz.) The triangle is $M(f) = M_0(1 - |f|)$ for $|f| \le 1$ and 0 outside. Take $f \ge 0$ (the spectrum is even, so $f < 0$ is the mirror image) and look at each copy inside the pass band $0 \le f \le 1$:

- **The copy at $+1.5$ kHz:** $M(f - 1.5) = M_0(1 - |f - 1.5|)$, non-zero only where $|f - 1.5| \le 1$, i.e. $0.5 \le f \le 2.5$. For $0.5 \le f \le 1$, $f - 1.5$ is negative, so $|f - 1.5| = 1.5 - f$ and

$$M(f - 1.5) = M_0[1 - (1.5 - f)] = M_0(f - 0.5)$$

- **The copy at -1.5 kHz:** $M(f + 1.5)$ is non-zero only for $-2.5 \le f \le -0.5$, so it is zero for $f \ge 0$.

Putting these into $V_o(f)$:

- For $0 \le f \le 0.5$ kHz: no copy reaches here, so $V_o = \dfrac{A_c M_0}{2}(1 - f)$, the correct triangle.
- For $0.5 \le f \le 1$ kHz:

$$V_o = \frac{A_c M_0}{2}(1 - f) + \frac{A_c M_0}{4}(f - 0.5) = \frac{A_c M_0}{4}[2(1 - f) + (f - 0.5)] = \frac{A_c M_0}{4}(1.5 - f)$$

By symmetry, for negative f replace f by $|f|$.
At $|f| = 1$ kHz the output is $A_c M_0/8$ instead of 0. The edges are distorted, so $m(t)$ is *not* recovered exactly.

Lowest carrier frequency

For each component of $s(t)$ to be uniquely determined by $m(t)$, the lobes centred at $+f_c$ and $-f_c$ must not overlap. They span $f_c - W$ to $f_c + W$ and the mirror image, so we need $f_c - W \ge 0$:

$$\boxed{f_{c,min} = W = 1 \text{ kHz}}$$

Final Answer: (i) $V_o(f) = (A_c/2)M(f)$, undistorted; (ii) the same plus overlapping tails for $0.5 \le |f| \le 1$ kHz, distorted; lowest carrier frequency 1 kHz.

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3. Notes and Remarks

- In case (ii), $f_c < W$ overlaps $S(f)$ itself around $f = 0$ (top right of the figure, for $|f| < 0.25$ kHz). That is the same failure, seen one step earlier.
- The scale $A_c/2$ depends on the local carrier's amplitude. With $2\cos(2\pi f_c t)$ the output is $A_c M(f)$. The shape is the same.
- At exactly $f_c = W$ the lobes just touch at $f = 0$, and only an ideal LPF separates the terms. Real systems use $f_c \gg W$.

4. Final Concise Answer

(a) TDMA separates users by time slots, needs exact timing, and has a hard capacity limit. CDMA separates users by codes in the same band at the same time, needs power control, and has a soft capacity limit.

(b) $C = 3400 \log_2(1001) = 33.89$ kbit/s; $\text{SNR}_{min} = 2^{4800/3400} - 1 = 1.66$ (2.2 dB).

(c) $f_s = 400$ Hz; $L \geq 400$, so $n = 9$; $R_b = 36$ kbit/s; $B_T = 18$ kHz.

(d) (i) $(A_c/2)M(f)$; (ii) $(A_c/2)M(f)$ plus tails overlapping for $0.5 \leq |f| \leq 1$ kHz; lowest carrier frequency $f_c = W = 1$ kHz.

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6. (a) Briefly describe FSK generation with the necessary block diagram. Include its spectrum and discuss the bandwidth requirements, along with the advantages and disadvantages of FSK. Marks: 8
   (b) With proper block diagram explain the principle of coherent detection of QAM. Mention its advantages and drawbacks. Marks: 7
   (c) Explain the principle of frequency hopping spread spectrum technique. Marks: 6
   (d) Tabulate the values of signal at each stage of DPSK encoder and receiver for message signal '1010011100' Marks: 4

6.(a) Briefly describe FSK generation with the necessary block diagram. Include its spectrum and discuss the bandwidth requirements, along with the advantages and disadvantages of FSK. Marks: 8

1. Theory

1. What is FSK? (Core Idea)

To send bits by radio, each bit must change something about a carrier. **Frequency shift keying (FSK)** changes its frequency: one frequency f_1 for bit 1 and another frequency f_0 for bit 0. The amplitude stays constant.

$$s(t) = \begin{cases} A\cos 2\pi f_1 t & \text{bit } 1 \\ A\cos 2\pi f_0 t & \text{bit } 0 \end{cases} \qquad f_{1,0} = f_c \pm \Delta f$$

Here f_c is the centre frequency and Δf is the shift on each side.

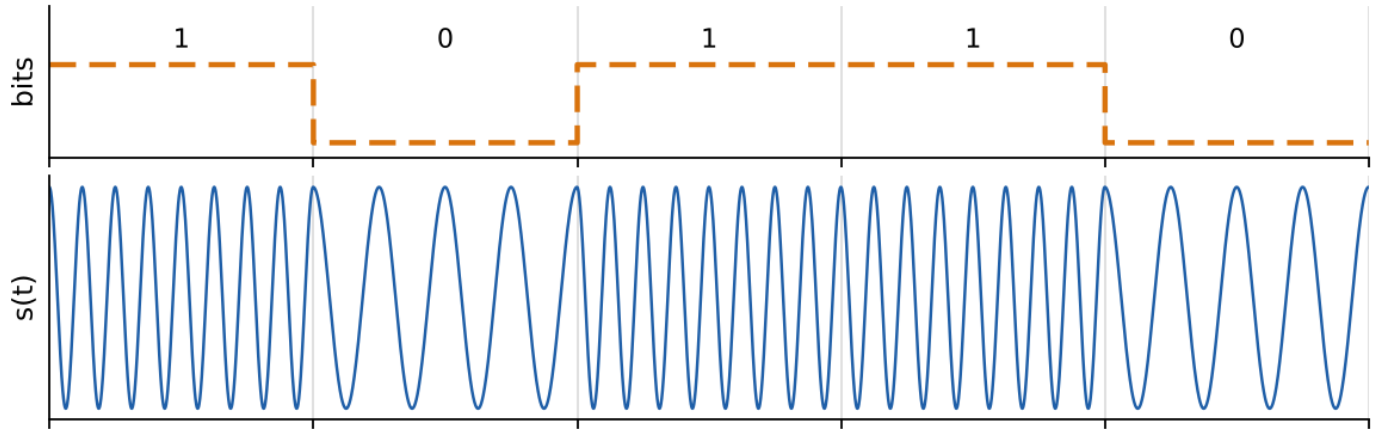


Figure 26: FSK waveform for the bits 10110

Reading the figure. The dashed orange line is the data. In the 1-bit intervals the solid blue wave has the higher frequency, and in the 0-bit intervals the lower one. Its height never changes.

2. FSK as two ASK signals

Consider only the bits equal to 1. The f_1 carrier is on during them and off otherwise: that is **amplitude shift keying (ASK)**. The same is true of the f_0 carrier and the 0 bits. So FSK is two ASK signals added together, and its spectrum is two ASK spectra, one around each frequency. This one idea gives the bandwidth.
An ASK signal of bit rate R_b with rectangular pulses has a sinc^2-shaped spectrum with a main lobe reaching $\pm R_b$ around its carrier.

2. Answer

FSK represents bits by two carrier frequencies, $f_1 = f_c + \Delta f$ for 1 and $f_0 = f_c - \Delta f$ for 0.

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Generation:

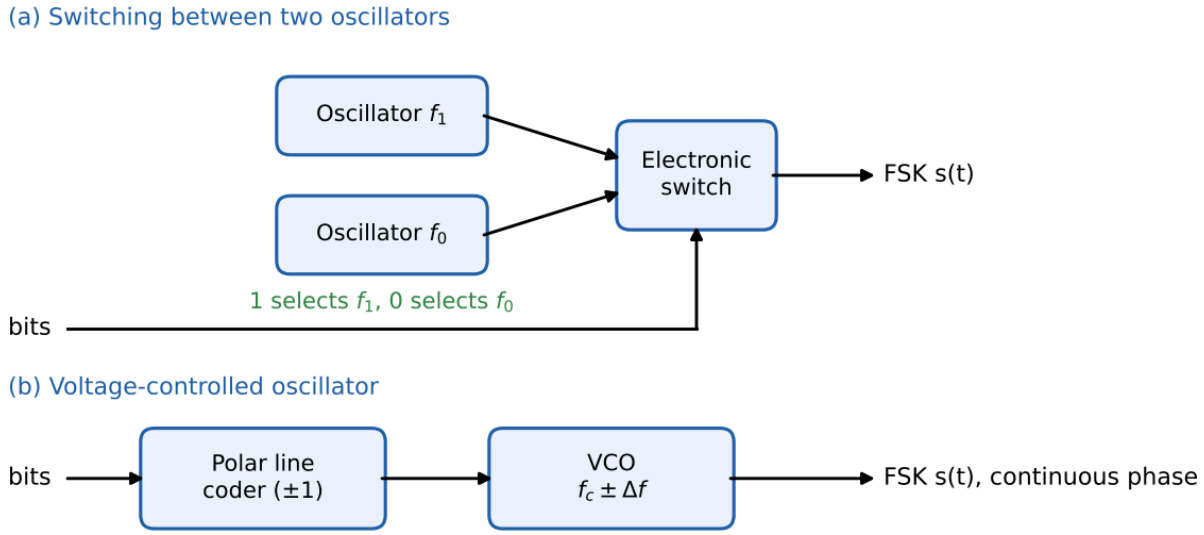


Figure 27: Two ways to generate FSK

Reading the figure. (a) Two free-running oscillators feed an electronic switch, and the bit stream selects which one reaches the output. (b) The bits become a polar ±1 wave, which drives a VCO, so the frequency moves to $f_c \pm \Delta f$.

- **(a) Switching:** simple, but since the two oscillators are independent, the phase can jump at bit edges. This widens the spectrum.
- **(b) VCO:** the phase stays continuous (CPFSK), so the spectrum is more compact. FSK is then FM with a digital message.

Spectrum and bandwidth:

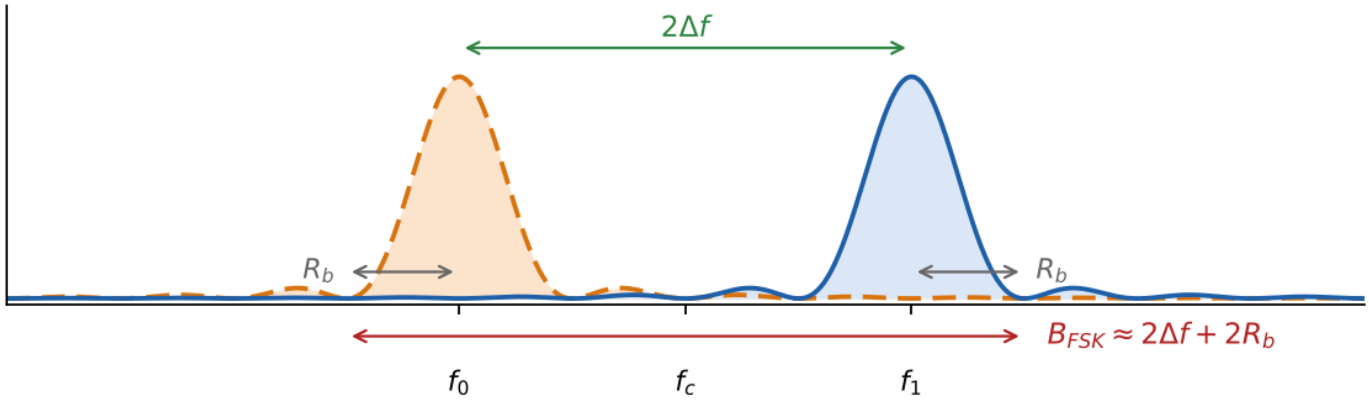


Figure 28: Spectrum of FSK

Reading the figure. The dashed orange lobe is around f_0 and the solid blue lobe is around f_1, $2\Delta f$ apart. Each main lobe reaches R_b on either side. The long arrow below the axis marks the total. The two ASK lobes are $2\Delta f$ apart, each extending R_b beyond its centre, so

$$\boxed{B_{FSK} \approx 2\Delta f + 2R_b}$$

For example, with $R_b = 1$ kbit/s and $\Delta f = 1$ kHz, $B_{FSK} \approx 2(1) + 2(1) = 4$ kHz. With filtered pulses, Forouzan writes $B = (1 + d)S + 2\Delta f$, where S is the baud rate and $0 \le d \le 1$ depends on the filter. It gives the same 4 kHz at $d = 1$.

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Advantages:

- Constant envelope, so noise and fading in amplitude have little effect, and efficient non-linear power amplifiers can be used.
- Simple, and can be detected non-coherently, with no local carrier.
- Better than ASK against noise.

Disadvantages:

- Needs more bandwidth than ASK or PSK.
- Coherent FSK needs 3 dB more power than PSK for the same error rate.
- Its spectrum is not used efficiently at high data rates.

3. Notes and Remarks

- The bandwidth uses main-lobe reach. Other authors count less of the spectrum and give a smaller value, so state the rule used.

6.(b) With proper block diagram explain the principle of coherent detection of QAM. Mention its advantages and drawbacks. Marks: 7

1. Theory

1. Why QAM? (Core Idea)

A DSB signal uses twice the message's bandwidth. **Quadrature amplitude modulation (QAM)** recovers this waste: it sends *two* messages in one band, on two carriers of the same frequency that are 90° apart, $\cos \omega_c t$ and $\sin \omega_c t$.

$$\varphi_{QAM}(t) = m_1(t) \cos \omega_c t + m_2(t) \sin \omega_c t$$

They overlap fully in frequency, yet they can be separated. The reason is that sine and cosine are **orthogonal**: over a full cycle, the average of their product is zero.

2. How the separation works

Two identities do the job:

$$2 \cos^2 x = 1 + \cos 2x, \qquad 2 \sin x \cos x = \sin 2x$$

Multiplying by $2 \cos \omega_c t$ turns the cos term into a baseband term, and leaves the sin term with only a $2\omega_c$ component. A low-pass filter then removes everything at $2\omega_c$.

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2. Answer

Coherent detection of QAM multiplies the received signal by two local carriers in quadrature and low-pass filters each product, which separates the two messages.

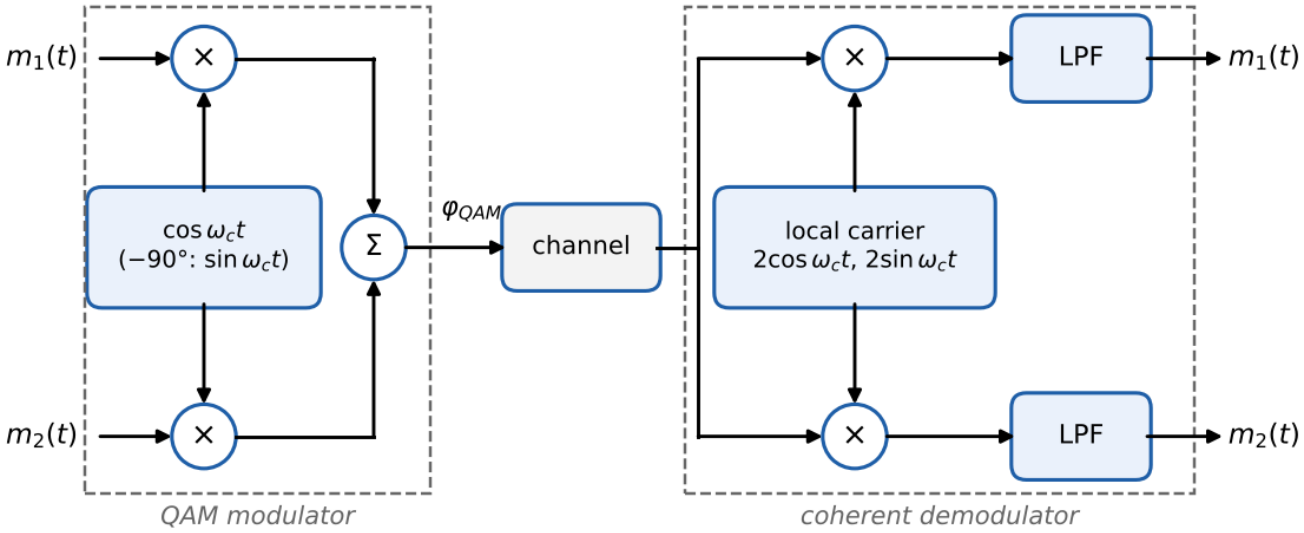


Figure 29: QAM modulator and coherent demodulator

Reading the figure. Left: m_1 and m_2 modulate carriers 90° apart, and the summer adds them. Right: the received signal goes to two multipliers, driven by the same pair of local carriers, and each LPF gives back one message.

Upper branch: multiply by $2\cos\omega_c t$ and multiply out:

$$\varphi_{QAM}(t) \cdot 2\cos\omega_c t = m_1(t) \cdot 2\cos^2\omega_c t + m_2(t) \cdot 2\sin\omega_c t\cos\omega_c t$$

Use $2\cos^2 x = 1 + \cos 2x$ and $2\sin x\cos x = \sin 2x$ with $x = \omega_c t$:

$$= m_1(t)[1 + \cos 2\omega_c t] + m_2(t)\sin 2\omega_c t = m_1(t) + m_1(t)\cos 2\omega_c t + m_2(t)\sin 2\omega_c t$$

The LPF removes the $2\omega_c$ terms, leaving $m_1(t)$.

Lower branch: multiply by $2\sin\omega_c t$:

$$\varphi_{QAM}(t) \cdot 2\sin\omega_c t = m_1(t) \cdot 2\sin\omega_c t\cos\omega_c t + m_2(t) \cdot 2\sin^2\omega_c t$$

Use $2\sin x\cos x = \sin 2x$ and $2\sin^2 x = 1 - \cos 2x$:

$$= m_1(t)\sin 2\omega_c t + m_2(t)[1 - \cos 2\omega_c t] = m_2(t) - m_2(t)\cos 2\omega_c t + m_1(t)\sin 2\omega_c t$$

The LPF leaves $m_2(t)$.

Effect of a phase error: let the upper local carrier be $2\cos(\omega_c t + \theta)$:

$$\varphi_{QAM}(t) \cdot 2\cos(\omega_c t + \theta) = m_1(t) \cdot 2\cos\omega_c t\cos(\omega_c t + \theta) + m_2(t) \cdot 2\sin\omega_c t\cos(\omega_c t + \theta)$$

Use $2\cos A\cos B = \cos(A-B)+\cos(A+B)$ and $2\sin A\cos B = \sin(A-B)+\sin(A+B)$, with $A = \omega_c t$ and $B = \omega_c t + \theta$, so $A - B = -\theta$ and $A + B = 2\omega_c t + \theta$. Since $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$:

$$= m_1(t)[\cos\theta + \cos(2\omega_c t + \theta)] + m_2(t)[-\sin\theta + \sin(2\omega_c t + \theta)] \xrightarrow{\text{LPF}} m_1(t)\cos\theta - m_2(t)\sin\theta$$

Advantages:

- Doubles the use of the band: two signals in the bandwidth of one DSB signal.
- For digital data, many bits per symbol (16-QAM carries 4), so high data rates in limited bandwidth.

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Drawbacks:

- Needs exact carrier synchronization. With a phase error θ, the upper output is $m_1(t)\cos\theta - m_2(t)\sin\theta$ (derived above), so m_2 leaks into m_1 (**crosstalk**), and the wanted signal weakens.
- More complex receiver, and it needs a linear power amplifier, since the amplitude varies.
- Denser QAM needs a higher SNR.

3. Notes and Remarks

- At $\theta = 90°$ the crosstalk is complete: the upper output is $-m_2(t)$, so the two outputs have swapped.

6.(c) Explain the principle of frequency hopping spread spectrum technique. Marks: 6

1. Theory

1. Why spread a signal? (Core Idea)

Normally we keep a signal's bandwidth as small as possible. **Spread spectrum** does the opposite: it spreads the signal across a much wider band than it needs, on purpose. The aim is protection. A jammer or an eavesdropper cannot cover or follow the whole band easily, and interference in one small part of the band has only a small effect.

2. The idea of hopping

Imagine two people talking on a walkie-talkie who keep changing channel every moment, following a secret list that they both have. Anyone listening on one channel hears only a fragment. A noisy channel spoils only a small part of the talk. This is **frequency hopping spread spectrum (FHSS)**.

2. Answer

In FHSS the carrier frequency hops from one value to another among M frequencies, following a pseudorandom pattern known to the transmitter and the receiver. The signal stays narrowband at each instant, but occupies a wide band over time.

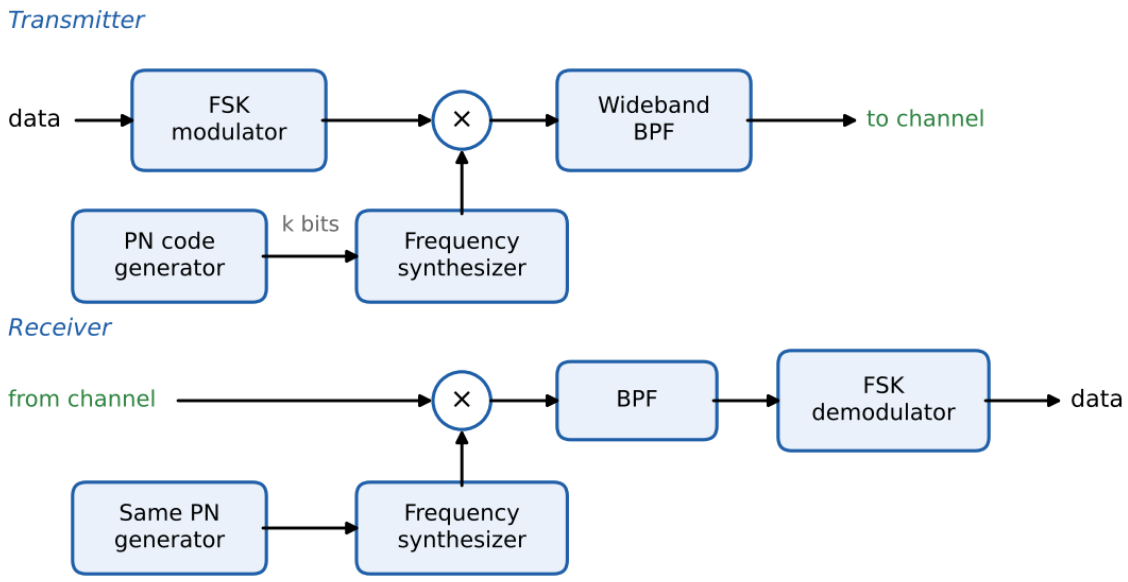


Figure 30: FHSS transmitter and receiver

Reading the figure. Top: the data is FSK modulated and mixed with a hopping tone, and the wideband filter passes the result. The tone comes from a frequency synthesizer set by the PN code generator. Bottom: an identical generator, in step with the transmitter, drives the same hops, so the mixer undoes them, and the filter and demodulator recover the data.

Transmitter:

1. The data modulates a fixed carrier (usually FSK), giving a narrowband signal.
2. A **pseudorandom noise (PN) generator** produces a k-bit pattern in each hopping period.
3. The **frequency synthesizer** turns the pattern into one of $M = 2^k$ frequencies.
4. The **mixer** shifts the narrowband signal to that frequency.

Receiver: the same PN generator, in synchronism, drives the same frequency sequence. Mixing brings the signal back to the fixed frequency (**dehopping**), and normal demodulation follows.

Example

with $k = 3$ there are 8 frequencies. The pattern 101, 111, 001, 000, 010, 110, 011, 100 selects the carrier for each hop, then repeats.

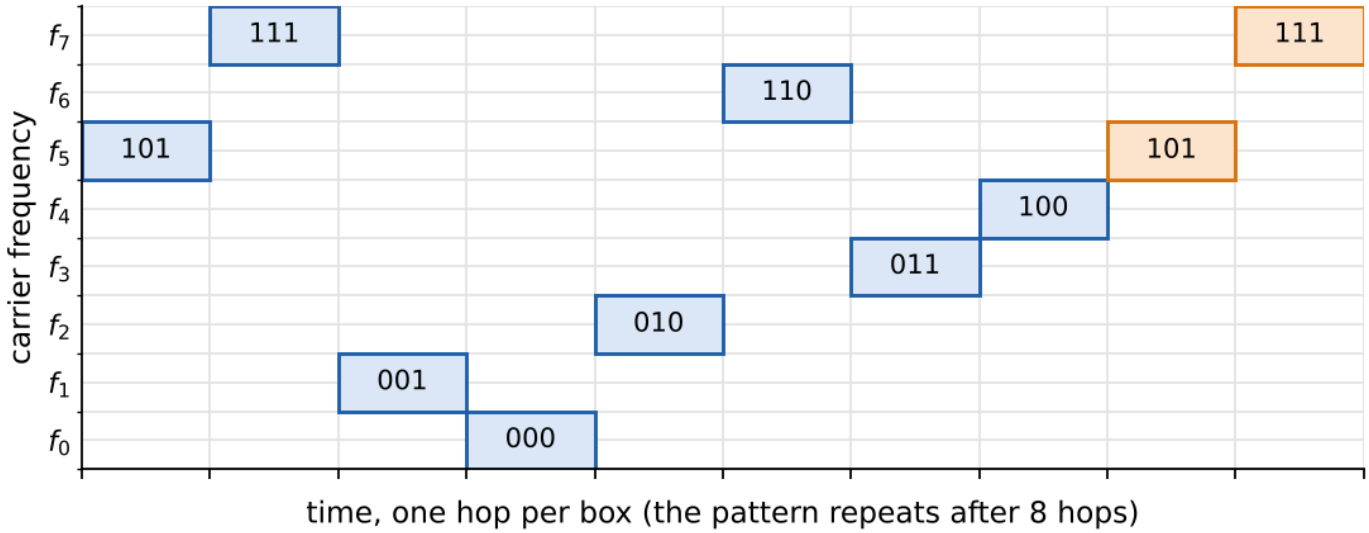


Figure 31: Hopping pattern of eight frequencies

Reading the figure. Each box is one hop, and its height shows the carrier in use, with the 3-bit pattern that chose it. The first 8 boxes are one full cycle, and the last 2 boxes show that it starts again.
Bandwidth: if each hop uses bandwidth B, the spread bandwidth is about $B_{SS} \approx MB$.
Types: **slow hopping** sends several bits per hop, and **fast hopping** uses several hops per bit.
Advantages: resists jamming (a jammer on one frequency hits only $1/M$ of the hops), resists eavesdropping, and lets many users share the band with different patterns. **Drawback:** needs a wide band and exact synchronization of the hop pattern.

3. Notes and Remarks

- Bluetooth is a well-known use: it hops among 79 channels of 1 MHz.

6.(d) Tabulate the values of signal at each stage of DPSK encoder and receiver for message signal '1010011100' Marks: 4

1. Theory

1. The idea of DPSK (Core Idea)

Ordinary PSK puts each bit in the absolute phase of the carrier, so the receiver needs a perfect local carrier. **Differential PSK (DPSK)** puts each bit in the *change* of phase: data 1 flips the phase, and data 0 keeps it. The receiver only compares each bit interval with the one before it, so it needs no local carrier.

2. The stages

- **Differential encoder:** $q_k = I_k \oplus q_{k-1}$, the XOR with the previous encoded bit. Since $x \oplus 1 = \bar{x}$ and $x \oplus 0 = x$, a 1 flips and a 0 keeps.
- **Polar coder:** $q_k = 1 \to a_k = +1$ (phase 0); $q_k = 0 \to a_k = -1$ (phase π).
- **PSK modulator:** $\varphi(t) = a_k A_c \cos \omega_c t$.
- **Receiver:** multiply the signal by itself delayed by one bit, then low-pass filter. The output is proportional to $a_k a_{k-1}$: positive (same phase) means 0, and negative (phase changed) means 1.

2. Answer

Given: message $I_k = 1\ 0\ 1\ 0\ 0\ 1\ 1\ 1\ 0\ 0$ (10 bits).

Convention (Lathi's differential encoding, where a 1 flips the phase): $q_k = I_k \oplus q_{k-1}$, with a reference bit $q_0 = 0$ sent first (phase π).

k	0 (ref)	1	2	3	4	5	6	7	8	9	10
Data I_k	–	1	0	1	0	0	1	1	1	0	0
q_{k-1} (delayed)	–	0	1	1	0	0	0	1	0	1	1
Encoded $q_k = I_k \oplus q_{k-1}$	0	1	1	0	0	0	1	0	1	1	1
Polar a_k	-1	$+1$	$+1$	-1	-1	-1	$+1$	-1	$+1$	$+1$	$+1$
Phase θ_k	π	0	0	π	π	π	0	π	0	0	0
Rx: delayed a_{k-1}	–	-1	$+1$	$+1$	-1	-1	-1	$+1$	-1	$+1$	$+1$
Rx: LPF output sign $a_k a_{k-1}$	–	$-$	$+$	$-$	$+$	$+$	$-$	$-$	$-$	$+$	$+$
Phase change?	–	yes	no	yes	no	no	yes	yes	yes	no	no
Detected bit	–	1	0	1	0	0	1	1	1	0	0

Table 6: DPSK encoder and receiver for the message 1010011100

Check: the detected bits 1 0 1 0 0 1 1 1 0 0 equal the message. ✔

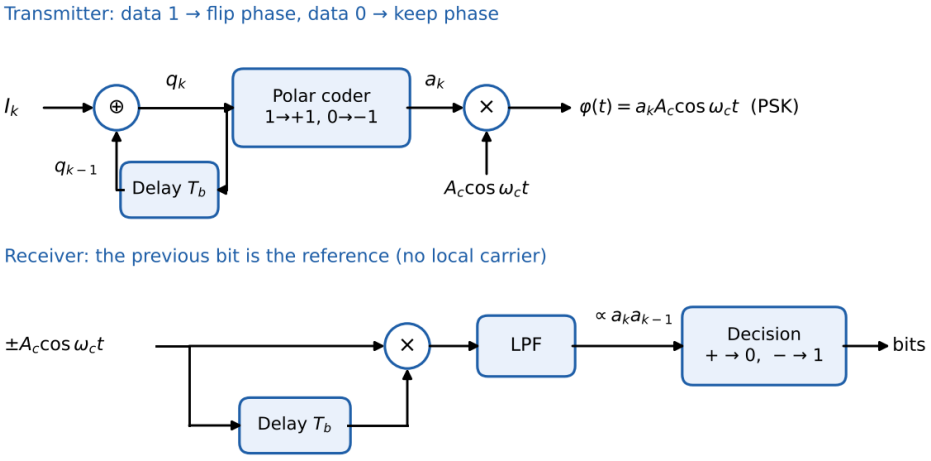


Figure 32: DPSK transmitter and receiver

Reading the figure. Top: the XOR of I_k and the delayed q_{k-1} gives q_k, then the polar coder and the multiplier give PSK. Bottom: the receiver multiplies the signal by its one-bit-delayed copy, and the LPF and the decision give the bits.

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3. Notes and Remarks

- Starting with $q_0 = 1$ inverts the q_k, a_k and phase rows, but the detected bits stay the same. State the convention used.
- A single wrong phase usually spoils *two* detected bits, because each interval is also the reference for the next.

4. Final Concise Answer

(a) Bits select $f_c \pm \Delta f$, by switching two oscillators or with a VCO. Spectrum: two ASK lobes, $B \approx 2\Delta f + 2R_b$. Advantages: constant envelope, simple detection. Disadvantages: wide bandwidth, less power-efficient than PSK.

(b) Multiply by $2\cos\omega_c t$ and $2\sin\omega_c t$, then LPF, to get m_1 and m_2. Doubles band use, but needs exact sync; a phase error gives crosstalk $m_1\cos\theta - m_2\sin\theta$.

(c) A PN code drives a synthesizer, so the carrier hops among $M = 2^k$ frequencies; the receiver hops in step. $B_{SS} \approx MB$. It resists jamming and eavesdropping.

(d) q_k = 1 1 0 0 0 1 0 1 1 1 (reference $q_0 = 0$); phases 0, 0, π, π, π, 0, π, 0, 0, 0; detected 1 0 1 0 0 1 1 1 0 0.

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7. (a) Find the entropy of single bit binary memoryless source. Also draw the curve of entropy vs probability for that source. Marks: 5

(b) Explain distortion as a transmission impairment in communication systems. What are the conditions required for distortionless transmission? Marks: 5

(c) We have a communication channel with a bandwidth of 5 MHz. A signal with an amplitude of 10V is transmitted, and the receiver detects a signal with an amplitude of 2V and noise with an amplitude of 5 mV.

(1) Calculate the attenuation.
(2) Determine the maximum channel capacity.
(3) To achieve a data rate of 100 Mbps, find the required signal level. Marks: 5

(d) Find the equivalent noise resistance for several amplifiers in cascade. Marks: 5

7.(a) Find the entropy of single bit binary memoryless source. Also draw the curve of entropy vs probability for that source. Marks: 5

1. Theory

1. What is a binary memoryless source? (Core Idea)

A **binary source** sends one of two symbols, 1 or 0, each time. If 1 comes out with probability p, then 0 comes out with probability $1-p$ (the two must add up to 1). The source is **memoryless** when each symbol is chosen independently of the ones before it, like tossing a (possibly unfair) coin.

2. Information in one symbol

A symbol of probability P carries $I = \log_2(1/P)$ bits: a sure symbol tells us nothing (0 bits), and a fair coin toss tells us exactly 1 bit. Rare symbols carry more information.

3. Entropy

The **entropy** is the average information per symbol, each symbol's information weighted by how often it occurs:

$$H = \sum_i P_i \log_2 \frac{1}{P_i} \quad \text{bits/symbol}$$

2. Answer

Step 1: The two probabilities

Let $P(1) = p$ and $P(0) = 1 - p$.

Step 2: Entropy

Put the two symbols into $H = \sum P_i \log_2(1/P_i)$:

$$\boxed{H(p) = -p \log_2 p - (1-p)\log_2(1-p) \quad \text{bits/symbol}}$$

Step 3: Special values

- $p = 0$ or $p = 1$: one symbol is certain, so $H = 0$ (using $0 \log_2 0 = 0$).
- $p = 0.5$: $H = -0.5 \log_2 0.5 - 0.5 \log_2 0.5 = 0.5 + 0.5 = 1$ bit, the largest value.
- $p = 0.1$: $H = -0.1 \log_2 0.1 - 0.9 \log_2 0.9 = 0.332 + 0.137 = 0.469$ bit; the same for $p = 0.9$.

Step 4: Where the maximum is

Differentiate with respect to p (using $\frac{d}{dp}[p \ln p] = \ln p + 1$):

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$$\frac{dH}{dp} = \frac{1}{\ln 2}\left[-\ln p - 1 + \ln(1-p) + 1\right] = \log_2 \frac{1-p}{p}$$

which is zero when $1 - p = p$, that is $p = 0.5$. So the maximum, 1 bit, is at $p = 0.5$.

Step 5: The curve

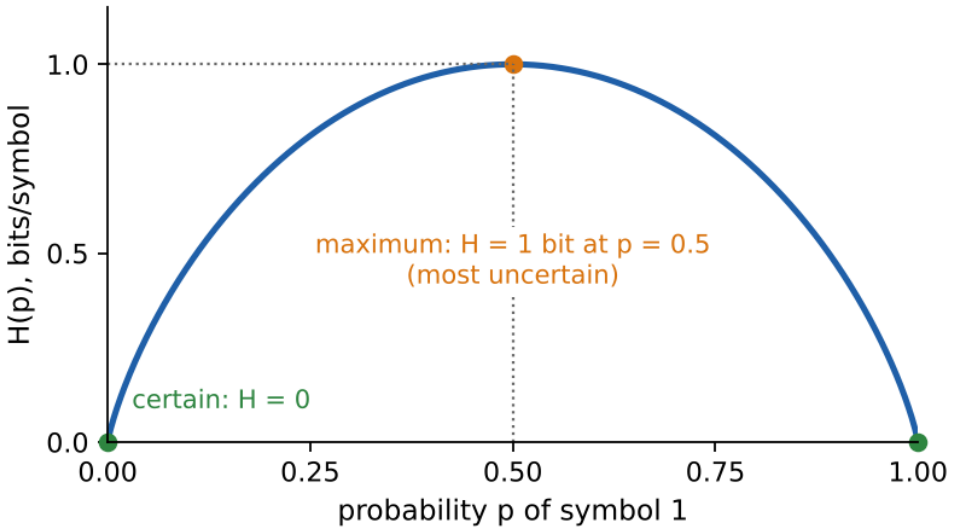


Figure 33: Entropy of a binary source against the probability p of symbol 1 (reused from 1(a))

Reading the figure. The solid blue curve starts at 0 for $p = 0$ (green dot), rises to its peak of 1 bit at $p = 0.5$ (orange dot, the most uncertain case), and falls back to 0 at $p = 1$. It is symmetric about $p = 0.5$, because swapping the names of 0 and 1 does not change the uncertainty.

Final Answer: $H(p) = -p\log_2 p - (1-p)\log_2(1-p)$; 0 at $p = 0, 1$; maximum 1 bit at $p = 0.5$; a symmetric arch.

3. Notes and Remarks

- The same question is solved as 1(a) of this paper.

7.(b) Explain distortion as a transmission impairment in communication systems. What are the conditions required for distortionless transmission? Marks: 5

1. Theory

1. What is distortion? (Core Idea)

A channel never delivers exactly what was sent. It weakens the signal (**attenuation**), adds unwanted random signals (**noise**), and changes its **shape** (**distortion**). Attenuation alone is harmless, because an amplifier can restore the size. A pure delay is harmless too: the signal arrives a little later but looks the same. Distortion is any change of shape beyond these two.

2. Why does a channel change the shape?

A signal is a sum of sinusoids of different frequencies (its Fourier components). The channel acts on each of them separately: it multiplies its amplitude by $|H(f)|$ and shifts its phase by $\theta_h(f)$. If different frequencies are treated differently, the components no longer add up to the original shape.

3. Two facts used below (Basic)

- **Fourier transform of an output:** a linear channel with transfer function $H(f)$ gives $Y(f) = X(f)H(f)$.
- **Time-shift property:** if $x(t) \leftrightarrow X(f)$, then $x(t - t_d) \leftrightarrow X(f)e^{-j2\pi f t_d}$. A delay multiplies the spectrum by a phase factor that grows linearly with frequency.

2. Answer

Distortion as a transmission impairment: distortion is the change in the shape of a signal as it passes through the channel, because the channel treats its frequency components differently. Its types are:

- **Amplitude distortion:** $|H(f)|$ is not constant, so some frequencies are amplified or attenuated more than others.
- **Phase (delay) distortion:** $\theta_h(f)$ is not a straight line through the origin, so components are delayed by different amounts and arrive out of step. (The ear barely notices it; video and data are very sensitive to it.)
- **Nonlinear distortion:** the channel is not linear (for example a saturating amplifier), which creates new frequencies (harmonics and intermodulation).

The conditions come from asking what a channel may do without changing the shape. Step 2 below is marked **Basic** (the time-shift property); a student who knows it can skip it.

Step 1: What "distortionless" means

The output may differ from the input only by a constant gain k and a constant delay t_d:

$$y(t) = k\,x(t - t_d)$$

Step 2 (Basic): Fourier transform of a delayed signal

From the definition, with $\tau = t - t_d$:

$$\int x(t-t_d)e^{-j2\pi ft}dt = \int x(\tau)e^{-j2\pi f(\tau+t_d)}d\tau = X(f)e^{-j2\pi ft_d}$$

Step 3: Transform the output

$$Y(f) = k\,X(f)\,e^{-j2\pi ft_d}$$

Step 4: The transfer function

Since $Y(f) = X(f)H(f)$:

$$\boxed{H(f) = \frac{Y(f)}{X(f)} = k\,e^{-j2\pi ft_d}}$$

Step 5: The two conditions

Write $H(f) = |H(f)|e^{j\theta_h(f)}$ and compare:

1. **Constant magnitude:** $|H(f)| = k$ for all frequencies in the signal band (no amplitude distortion).
2. **Linear phase:** $\theta_h(f) = -2\pi ft_d$, a straight line through the origin with slope $-2\pi t_d$; equivalently a constant time delay $t_g = -\frac{1}{2\pi}\frac{d\theta_h}{df} = t_d$ for every frequency (no phase distortion).

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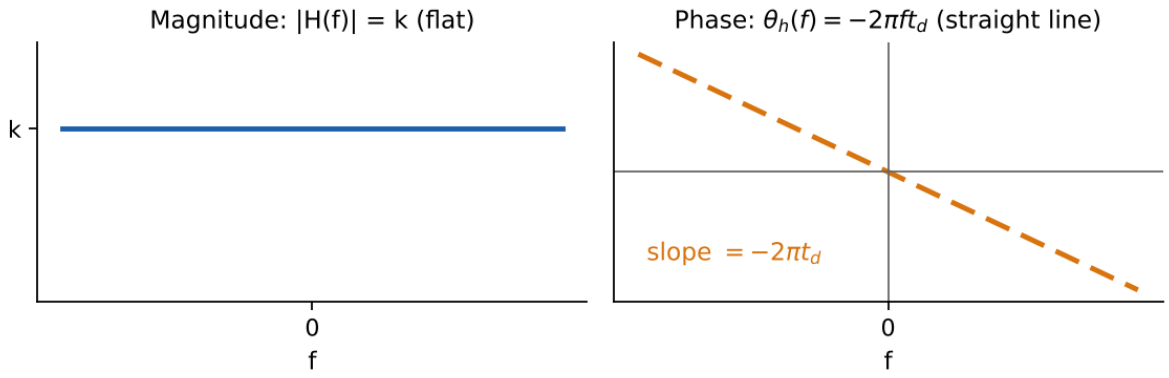


Figure 34: Distortionless channel: flat magnitude and linear phase (reused from the 2021-22 paper)

Reading the figure. On the left the solid blue magnitude response is flat at the height k for all f. On the right the dashed orange phase response is a straight line through the origin with slope $-2\pi t_d$.
Why linear phase means equal delay: a component $\cos 2\pi f t$ shifted in phase by $-2\pi f t_d$ becomes $\cos 2\pi f(t - t_d)$: it is delayed by t_d whatever its frequency. So all components move together and keep their alignment.

Steps to remember

$y = kx(t - t_d) \rightarrow$ time-shift property $\rightarrow Y = kXe^{-j2\pi f t_d} \rightarrow H = ke^{-j2\pi f t_d} \rightarrow |H| = k$ and $\theta_h = -2\pi f t_d$.

3. Notes and Remarks

- The conditions need to hold only over the band the signal occupies, not at all frequencies.
- Also solved as 1(d) of the 2021-22 paper (link), as extra reading.

7.(c) We have a communication channel with a bandwidth of 5 MHz. A signal with an amplitude of 10V is transmitted, and the receiver detects a signal with an amplitude of 2V and noise with an amplitude of 5 mV. (1) Calculate the attenuation. (2) Determine the maximum channel capacity. (3) To achieve a data rate of 100 Mbps, find the required signal level. Marks: 5

1. Theory

1. Attenuation (Core Idea)

Attenuation is the loss of signal strength along a channel. It is measured in decibels as a power ratio. Since power is proportional to the square of the amplitude (in the same resistance),

$$\text{attenuation} = 10 \log_{10} \frac{P_t}{P_r} = 20 \log_{10} \frac{V_t}{V_r} \text{ dB}$$

2. Signal-to-noise ratio

The **SNR** compares the signal power with the noise power at the receiver. With amplitudes in the same resistance, $\text{SNR} = (V_s/V_n)^2$.

3. Shannon capacity

The largest error-free bit rate of a channel of bandwidth B is

$$C = B \log_2(1 + \text{SNR})$$

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2. Answer

Given: $B = 5$ MHz, $V_t = 10$ V, $V_r = 2$ V, $V_n = 5$ mV $= 0.005$ V.

Step 1: Attenuation

$$\text{attenuation} = 20\log_{10}\frac{10}{2} = 20\log_{10} 5 = 20 \times 0.69897 = 13.98 \text{ dB}$$

$$\boxed{13.98 \text{ dB}}$$

Step 2: SNR at the receiver

$$\text{SNR} = \left(\frac{2}{0.005}\right)^2 = 400^2 = 160\,000 \ (52.04 \text{ dB})$$

Step 3: Maximum channel capacity

$$C = 5 \times 10^6 \log_2(1 + 160\,000) = 5 \times 10^6 \times 17.2877 = 86.44 \times 10^6 \text{ bits/s}$$

$$\boxed{C \approx 86.44 \text{ Mbps}}$$

Step 4: SNR needed for 100 Mbps

$$\log_2(1 + \text{SNR}) = \frac{100 \times 10^6}{5 \times 10^6} = 20 \quad \Rightarrow \quad \text{SNR} = 2^{20} - 1 = 1\,048\,575 \ (60.21 \text{ dB})$$

Step 5: Required signal level

With the same noise amplitude, $V_s = V_n\sqrt{\text{SNR}}$:

$$V_s = 0.005 \times \sqrt{1\,048\,575} = 0.005 \times 1024.0 = 5.12 \text{ V at the receiver}$$

With the same 13.98 dB attenuation, the transmitter must send $5.12 \times 5 = 25.6$ V.

$$\boxed{V_{received} = 5.12 \text{ V}, \quad V_{transmitted} = 25.6 \text{ V}}$$

Final Answer: attenuation 13.98 dB; capacity 86.44 Mbps; for 100 Mbps the received signal must be 5.12 V (25.6 V sent).

3. Notes and Remarks

- Forouzan writes the loss as a negative gain, -13.98 dB.
- The same numbers appear in 1(c) of the 2021-22 paper (link).

7.(d) Find the equivalent noise resistance for several amplifiers in cascade. Marks: 5

1. Theory

1. Thermal noise (Core Idea)

Every resistor makes a small random voltage because its electrons move with heat. Over a bandwidth B its mean-square value is $\overline{v_n^2} = 4kTBR$, proportional to the resistance R (k = Boltzmann's constant, T = temperature in kelvin).

2. Equivalent noise resistance

All the noise an amplifier stage adds can be pictured as coming from one imaginary resistor at its input, in front of a noiseless amplifier. The value of that resistor is the stage's **equivalent noise resistance**. Then noise is described by resistances, which are easy to combine.

3. Noise powers add

A voltage v through a gain A becomes Av, so its mean square becomes $A^2\overline{v^2}$. Independent noise sources add in **power** (mean square), not in voltage.

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2. Answer

Step 1 below is marked **Basic** (the resistor noise formula); a student who knows it can skip it.

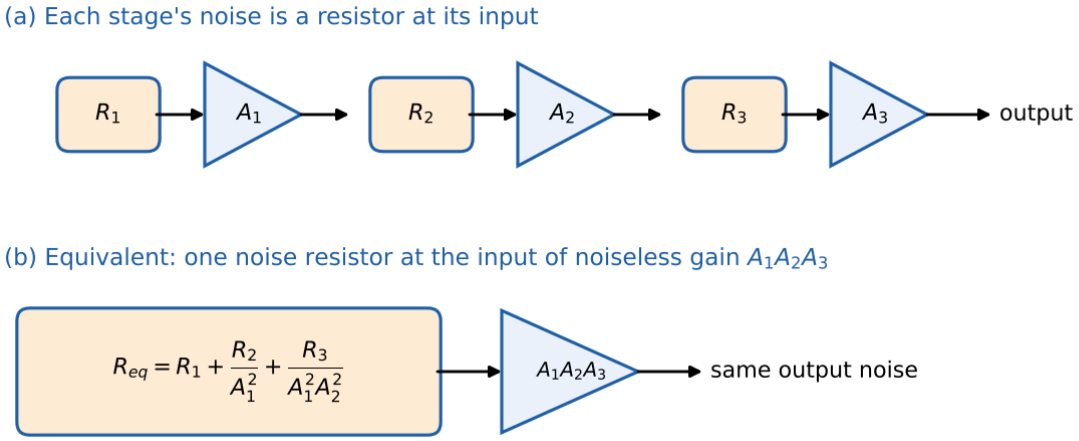


Figure 35: Amplifiers in cascade (a) and the single equivalent noise resistor at the input (b) (reused from the 2019-20 paper)

Reading the figure. In (a) each stage is a noiseless triangle of gain A_1, A_2, A_3 with an orange noise resistor R_1, R_2, R_3 in front. In (b) one orange resistor R_{eq} in front of a noiseless gain $A_1A_2A_3$ gives the same output noise.

Step 1 (Basic): Noise of a resistor

$\overline{v^2} = 4kTBR$.

Step 2: Model the stages

Stage i has voltage gain A_i and, at its input, a noise resistance R_i that stands for all the noise generated there (for stage 1: the source resistance plus its own noise resistance; for stage 2: the output resistance of stage 1 in parallel with the input resistance of stage 2, plus its own noise resistance).

Step 3: Output noise

Each noise voltage is amplified by all the stages after it, and the independent powers add:

$$\overline{v_{no}^2} = 4kTB\left[R_1(A_1A_2A_3)^2 + R_2(A_2A_3)^2 + R_3A_3^2\right]$$

Step 4: Refer it to the input

Set this equal to the noise of one resistor R_{eq} amplified by the whole chain, $4kTB\,R_{eq}(A_1A_2A_3)^2$, cancel $4kTB$ and divide by $(A_1A_2A_3)^2$:

$$\boxed{R_{eq} = R_1 + \frac{R_2}{A_1^2} + \frac{R_3}{A_1^2A_2^2} + \cdots}$$

Interpretation: a later stage's noise is divided by the square of all the gain before it, so the **first stage** sets the noise of the chain: it should be a low-noise, high-gain stage.

Example (illustrative values)

$R_1 = 2200\ \Omega$, $R_2 = 30.25\ \text{k}\Omega$, $R_3 = 1\ \text{M}\Omega$, $A_1 = 10$, $A_2 = 25$: $R_{eq} = 2200 + 302.5 + 16 = 2518.5\ \Omega$.

Steps to remember

$4kTBR \to$ each stage = noiseless gain with R_i in front $\to$ output powers add, each times the squared gain after it $\to$ divide by the total squared gain $\to R_{eq} = R_1 + R_2/A_1^2 + \ldots$

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3. Notes and Remarks

- Follows Kennedy and Davis, *Electronic Communication Systems* (noise chapter). Also solved as 1(d) of the 2019-20 paper.

4. Final Concise Answer

(a) $H(p) = -p\log_2 p - (1-p)\log_2(1-p)$; zero at $p = 0, 1$, maximum 1 bit at $p = 0.5$.

(b) Distortion = change of shape (amplitude, phase, nonlinear). Distortionless: $H(f) = ke^{-j2\pi f t_d}$, i.e. constant $|H|$ and linear phase over the signal band.

(c) 13.98 dB; 86.44 Mbps; 5.12 V received (25.6 V sent) for 100 Mbps.

(d) $R_{eq} = R_1 + R_2/A_1^2 + R_3/(A_1^2 A_2^2) + \ldots$

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8. (a) Amplitude spectrum of a message signal shown in figure-1. Draw the spectrum of its DSB-WC and SSB-SC modulated signal. Marks: 5

(b) Explain how phase and frequency error affect the coherent detection of AM. Marks: 5

(c) The antenna current of an AM transmitter is 8 Amps, when only the carrier is sent, but it increases to 8.93A, when the carrier is modulated by a single sine wave. Find percentage modulation. Determine the antenna current when the percentage modulation changes to 0.8 Marks: 5

(d) Explain principle of preemphasis and deemphasis Marks: 5

(e) Consider an FM signal obtained from a modulating signal frequency of 2000 Hz and maximum Amplitude of 5 volts. The frequency sensitivity of modulator is 2 kHz/V. Find the bandwidth of the FM using Carson's rule. Marks: 5

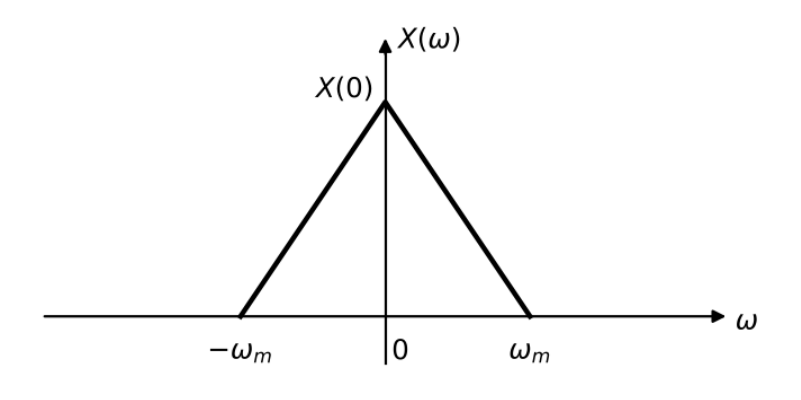


Figure 36: Figure-1 of the paper (redrawn): the amplitude spectrum X(ω) of the message

8.(a) Amplitude spectrum of a message signal shown in figure-1. Draw the spectrum of its DSB-WC and SSB-SC modulated signal. Marks: 5

1. Theory

1. Modulation shifts the spectrum (Core Idea)

Multiplying a message by $\cos\omega_c t$ moves its spectrum to the carrier frequency. With $\cos\omega_c t = \frac{1}{2}(e^{j\omega_c t} + e^{-j\omega_c t})$ and the frequency-shift property ($x(t)e^{j\omega_0 t} \leftrightarrow X(\omega - \omega_0)$):

$$x(t)\cos\omega_c t \;\leftrightarrow\; \tfrac{1}{2}[X(\omega-\omega_c) + X(\omega+\omega_c)]$$

So two copies of $X(\omega)$ appear, centred at $+\omega_c$ and $-\omega_c$, each *half* as tall.

2. A carrier gives impulses

A pure carrier $A\cos\omega_c t$ has the transform $\pi A[\delta(\omega-\omega_c) + \delta(\omega+\omega_c)]$: two impulses (spikes) at $\pm\omega_c$.

3. The three kinds of signal

- **DSB-WC** (double sideband with carrier, ordinary AM): $[A + x(t)]\cos\omega_c t$: both sidebands and the carrier.
- **DSB-SC:** $x(t)\cos\omega_c t$: both sidebands, no carrier.
- **SSB-SC:** only one sideband (the upper or the lower), no carrier.

The part of a shifted copy above ω_c (in $|\omega|$) is the **upper sideband (USB)**, the part below is the **lower sideband (LSB)**.

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2. Answer

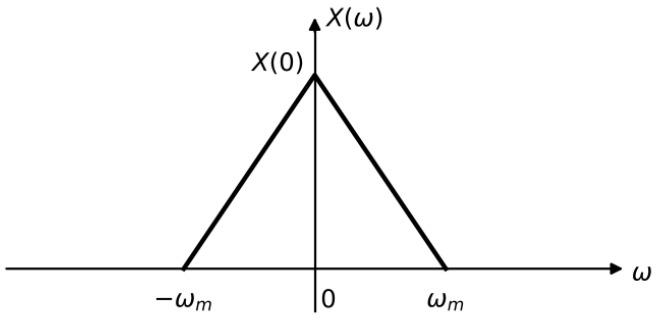


Figure 37: Repeated from the question: Figure-1 of the paper (redrawn)

The message spectrum is a triangle of height $X(0)$ between $-\omega_m$ and ω_m.

Step 1: DSB-WC (AM)

$$\varphi_{AM}(t) = [A + x(t)]\cos\omega_c t \;\leftrightarrow\; \pi A[\delta(\omega-\omega_c)+\delta(\omega+\omega_c)] + \tfrac{1}{2}[X(\omega-\omega_c)+X(\omega+\omega_c)]$$

- Carrier impulses of strength πA at $\pm\omega_c$.
- A triangle of height $X(0)/2$ centred at ω_c, from $\omega_c-\omega_m$ to $\omega_c+\omega_m$ (LSB below ω_c, USB above), and its mirror at $-\omega_c$.
- Bandwidth $2\omega_m$.

Step 2: SSB-SC (upper sideband)

Keep only the USB of each copy and drop the carrier:

- From ω_c to $\omega_c+\omega_m$: a half-triangle falling from $X(0)/2$ at ω_c to 0 at $\omega_c+\omega_m$.
- Its mirror from $-\omega_c-\omega_m$ to $-\omega_c$.
- Bandwidth ω_m, no impulses.

(For the LSB version, keep the halves from $\omega_c-\omega_m$ to ω_c and from $-\omega_c$ to $-\omega_c+\omega_m$ instead.)

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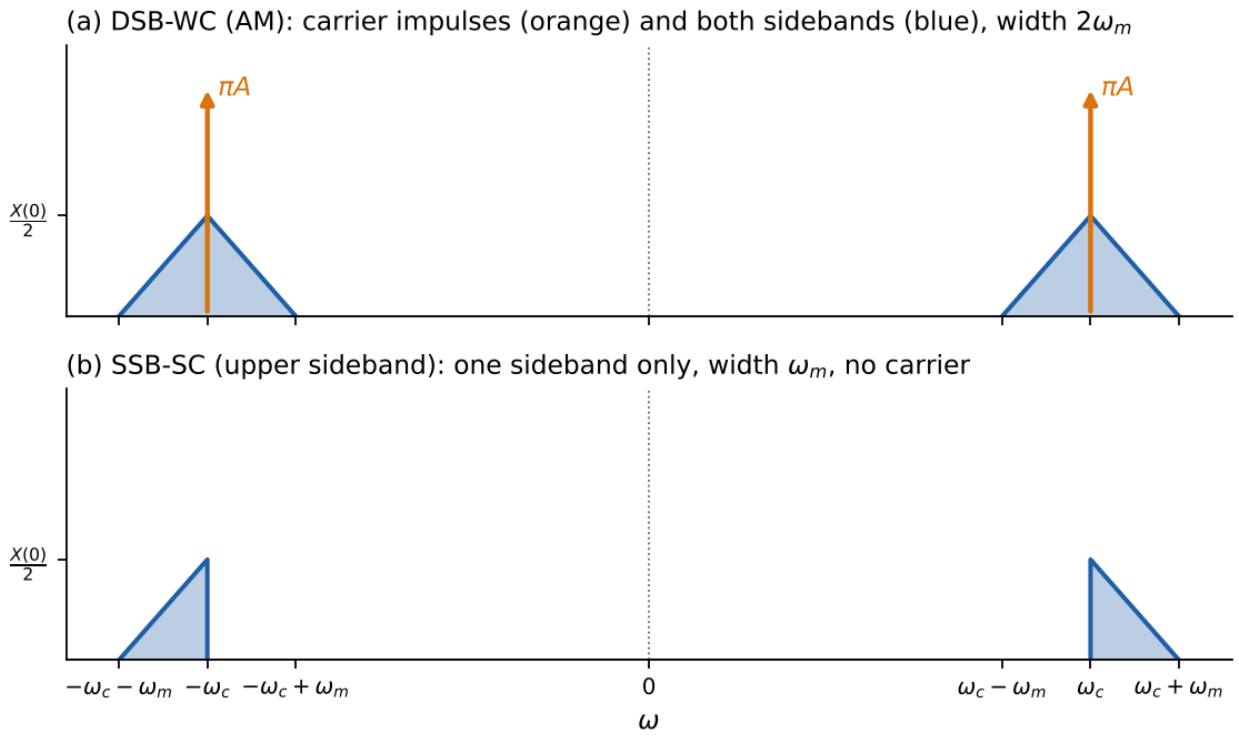


Figure 38: (a) DSB-WC spectrum: carrier impulses and both sidebands; (b) SSB-SC (USB): one sideband only

Reading the figure. In (a) the orange arrows at $\pm\omega_c$ are the carrier impulses of strength πA, and the shaded blue triangles of height $X(0)/2$ span $\omega_c \pm \omega_m$ and $-\omega_c \pm \omega_m$. In (b) only the outer halves remain: from ω_c up to $\omega_c + \omega_m$ and from $-\omega_c - \omega_m$ up to $-\omega_c$, with no carrier.
Final Answer: DSB-WC = impulses πA at $\pm\omega_c$ plus triangles of height $X(0)/2$ over $\pm\omega_c \pm \omega_m$ (bandwidth $2\omega_m$); SSB-SC (USB) = only the halves from ω_c to $\omega_c + \omega_m$ (and the mirror), no carrier (bandwidth ω_m).

3. Notes and Remarks

- The paper calls the amplitude spectrum $X(\omega)$; Lathi calls the message spectrum $M(\omega)$.

8.(b) Explain how phase and frequency error affect the coherent detection of AM. Marks: 5

1. Theory

1. Coherent detection (Core Idea)

A coherent (synchronous) detector multiplies the received signal by a local carrier and low-pass filters the product. It works for DSB-SC, SSB and also ordinary AM. It needs a local carrier with the **same frequency and phase** as the transmitter's carrier. A real local oscillator is never perfect: it is $\cos[(\omega_c + \Delta\omega)t + \delta]$, with a small frequency error $\Delta\omega$ and phase error δ.

2. Answer

Step 2 below is marked **Basic** (the product identity); a student who knows it can skip it.

Step 1: Signals

Received AM: $\varphi(t) = [A + m(t)]\cos\omega_c t$ (with $A = 0$ it is DSB-SC). Local carrier: $c(t) = 2\cos[(\omega_c + \Delta\omega)t + \delta]$ (the factor 2 only scales the output).

Step 2 (Basic): Product identity

$2\cos A\cos B = \cos(A - B) + \cos(A + B)$.

Step 3: Multiply

$$e(t) = [A + m(t)]\,\{\cos(\Delta\omega t + \delta) + \cos[(2\omega_c + \Delta\omega)t + \delta]\}$$

Step 4: Low-pass filter

The term around $2\omega_c$ is removed:

$$\boxed{e_o(t) = [A + m(t)]\cos(\Delta\omega t + \delta)}$$

The dc part $A\cos(\ldots)$ is removed by a blocking capacitor, leaving $m(t)\cos(\Delta\omega t + \delta)$.

Step 5: Phase error only ($\Delta\omega = 0$)

$e_o = m(t)\cos\delta$: the output is attenuated by $\cos\delta$ but not distorted if δ is constant. At $\delta = 90°$ it vanishes (**quadrature null effect**). A randomly varying δ makes the output fade.

Step 6: Frequency error only ($\delta = 0$)

$e_o = m(t)\cos\Delta\omega t$: the message is multiplied by a slowly varying cosine, so its loudness rises and falls at Δf (**beating**, a serious distortion even for Δf of a few hertz).

Final Summary: coherent detection gives $m(t)\cos(\Delta\omega t + \delta)$: a phase error attenuates the output (null at 90°), a frequency error makes it beat. For ordinary AM this is avoided entirely by using an envelope detector, which needs no local carrier; this is one reason AM keeps its carrier.

Steps to remember

Received × local carrier with $\Delta\omega$, δ → product identity → LPF → $m\cos(\Delta\omega t + \delta)$ → phase: $\cos\delta$ attenuation; frequency: beating.

3. Notes and Remarks

- The same effect for DSB-SC is derived in 2(b) of the 2021-22 paper (link), as extra reading.

8.(c) The antenna current of an AM transmitter is 8 Amps, when only the carrier is sent, but it increases to 8.93A, when the carrier is modulated by a single sine wave. Find percentage modulation. Determine the antenna current when the percentage modulation changes to 0.8 Marks: 5

1. Theory

1. Power in AM (Core Idea)

A tone-modulated AM signal is $A_c[1 + \mu\cos\omega_m t]\cos\omega_c t$. Using $\cos A\cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$, it is a carrier of amplitude A_c plus two side frequencies of amplitude $\mu A_c/2$. Adding the powers of the three sinusoids (each amplitude^2/2):

$$P_t = \frac{A_c^2}{2} + 2\cdot\frac{(\mu A_c/2)^2}{2} = P_c\left(1 + \frac{\mu^2}{2}\right)$$

2. From power to current

The antenna is a fixed resistance R, and $P = I^2R$, so

$$\frac{I_t^2}{I_c^2} = 1 + \frac{\mu^2}{2} \quad\Rightarrow\quad I_t = I_c\sqrt{1 + \frac{\mu^2}{2}}$$

2. Answer

Given: $I_c = 8$ A, $I_t = 8.93$ A; new modulation 0.8.

Step 1: Percentage modulation

$$\mu^2 = 2\left[\left(\frac{8.93}{8}\right)^2 - 1\right] = 2(1.24601 - 1) = 0.49203, \qquad \mu = 0.7015$$

$$\boxed{\mu \approx 70.15\%}$$

Step 2: Current at $\mu = 0.8$

$$I_t' = 8\sqrt{1 + \frac{0.8^2}{2}} = 8\sqrt{1.32} = 8 \times 1.14891 = 9.19 \text{ A}$$

$$\boxed{I_t' \approx 9.19 \text{ A}}$$

Final Answer: about 70 % modulation; 9.19 A at 80 %.

3. Notes and Remarks

- "Percentage modulation changes to 0.8" is read as $\mu = 0.8$ (80 %).

8.(d) Explain principle of preemphasis and deemphasis Marks: 5

1. Theory

1. FM noise grows with frequency (Core Idea)

At the output of an FM demodulator the noise is not flat: its power spectral density grows as f^2 (the demodulator differentiates the phase, and differentiation multiplies each noise component by its frequency). So the **high audio frequencies** are much noisier than the low ones. Meanwhile, speech and music have most of their power at low frequencies, and the high-frequency components are weak. The weak high frequencies therefore have a poor SNR.

2. Answer

Principle: boost the high-frequency part of the message *before* modulation (pre-emphasis), and cut it back by the same amount *after* demodulation (de-emphasis). The message is restored to its original shape, but the noise, which enters between the two filters, is cut by the de-emphasis filter where it is strongest.

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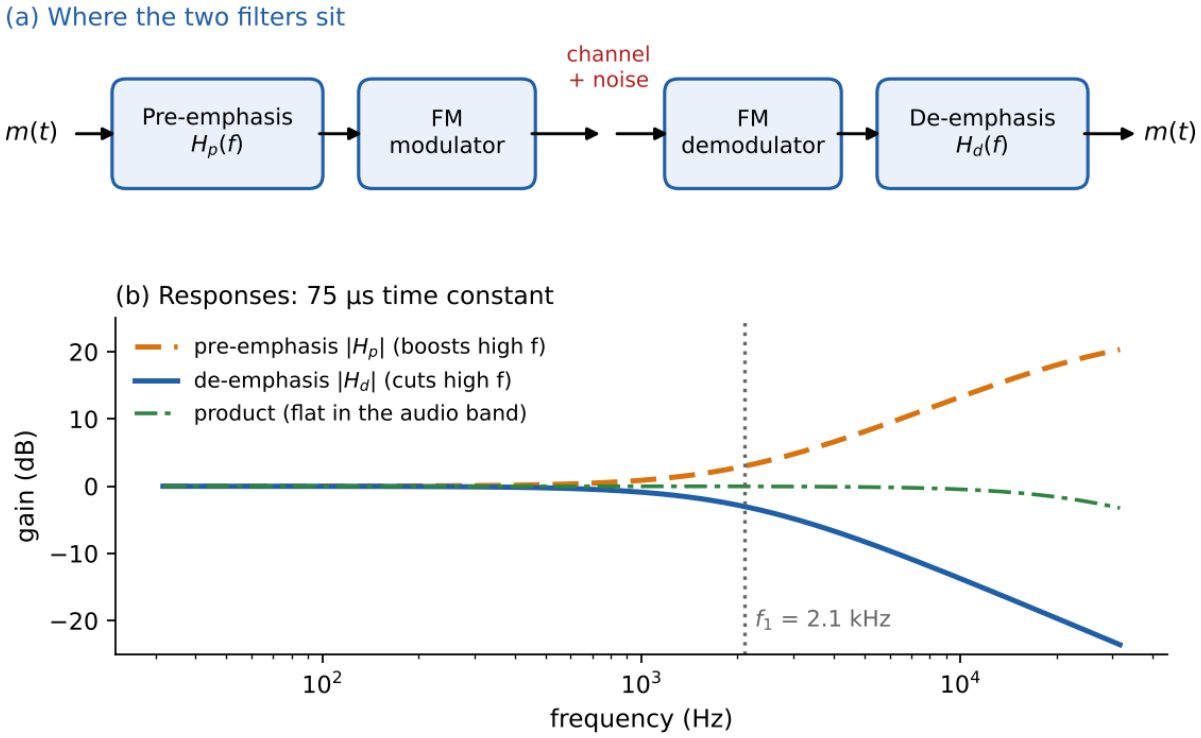


Figure 39: Pre-emphasis and de-emphasis: (a) where the filters sit; (b) their responses for a 75 μs time constant

Reading the figure. In (a) the message passes the pre-emphasis filter, the FM modulator, the noisy channel, the demodulator and the de-emphasis filter. In (b) the dashed orange pre-emphasis response rises above about 2.1 kHz, the solid blue de-emphasis response falls by the same amount, and their product (dash-dot green) stays flat over the audio band.

- **Pre-emphasis filter** (at the transmitter): a high-pass type network, $H_p(f) = 1 + j\dfrac{f}{f_1}$ (up to a second corner far above the audio band), which raises the components above f_1.
- **De-emphasis filter** (at the receiver): a simple RC low-pass, $H_d(f) = \dfrac{1}{1 + jf/f_1}$, the inverse of H_p over the audio band, so $H_p H_d = 1$ and the message is undistorted.
- **Time constant:** $f_1 = \dfrac{1}{2\pi RC}$; with the standard $RC = 75$ μs (FM broadcasting in the Americas), $f_1 = 2.12$ kHz.
- **Gain:** the de-emphasis filter reduces the high-frequency noise strongly, improving the output SNR by about 13 dB (more for a 15 kHz audio band), without extra transmitter power.
- Pre-emphasis raises the high-frequency deviation a little, so the peak deviation must still be kept within the allowed limit.

Final Summary: pre-emphasis boosts the weak high audio frequencies before FM; de-emphasis restores them after demodulation and, with them, strongly reduces the noise, which in FM is largest at high frequencies.

3. Notes and Remarks

- Lathi, Sec. 5.7 (pre-emphasis and de-emphasis in broadcasting); Dolby noise reduction uses the same idea for tape.

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8.(e) Consider an FM signal obtained from a modulating signal frequency of 2000 Hz and maximum Amplitude of 5 volts. The frequency sensitivity of modulator is 2 kHz/V. Find the bandwidth of the FM using Carson's rule. Marks: 5

1. Theory

1. Deviation from sensitivity (Core Idea)

The frequency sensitivity k_f (Hz/V) says how far the carrier frequency moves for each volt of message. The peak deviation is $\Delta f = k_f \times$ (peak message voltage).

2. Carson's rule

An FM signal has infinitely many sidebands, but almost all its power lies within

$$B_T \approx 2(\Delta f + f_m) = 2f_m(\beta + 1), \qquad \beta = \Delta f / f_m$$

2. Answer

Given: $f_m = 2000$ Hz, $A_m = 5$ V, $k_f = 2$ kHz/V.

Step 1: Deviation

$$\Delta f = k_f A_m = 2000 \times 5 = 10\,000 \text{ Hz} = 10 \text{ kHz}$$

Step 2: Modulation index

$$\beta = \frac{10\,000}{2000} = 5$$

Step 3: Bandwidth

$$B_T = 2(\Delta f + f_m) = 2(10 + 2) \text{ kHz} = 24 \text{ kHz}$$

$$\boxed{B_T = 24 \text{ kHz}}$$

Final Answer: $\Delta f = 10$ kHz, $\beta = 5$, Carson bandwidth 24 kHz.

3. Notes and Remarks

- With $\beta = 5$ this is wideband FM, so Carson's rule is a fair estimate.

4. Final Concise Answer

(a) DSB-WC: carrier impulses πA at $\pm\omega_c$ plus triangles of height $X(0)/2$ over $\pm\omega_c \pm \omega_m$; SSB-SC (USB): only the halves from ω_c to $\omega_c + \omega_m$ (and the mirror), no carrier.
(b) Output $m(t)\cos(\Delta\omega t + \delta)$: phase error attenuates by $\cos\delta$ (null at 90°), frequency error causes beating.
(c) $\mu \approx 70.15\%$; 9.19 A at $\mu = 0.8$.
(d) Boost high audio frequencies before FM, cut them after demodulation ($H_p H_d = 1$, 75 μs); the FM noise, which rises as f^2, is cut, improving SNR by about 13 dB.
(e) $\Delta f = 10$ kHz, $\beta = 5$, $B_T = 24$ kHz.

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FORMULA SHEET

EEE329

Basic Communication Engineering

Formula Sheet

Session 2020–21 paper

Every formula behind the 3rd Year 2nd Semester Final Examination, 2023 and Term Tests 1–2

TOPICS

1 Information and channel capacity — 12 formulas

2 Noise — 4 formulas

3 Amplitude modulation — 12 formulas

4 Angle modulation — 8 formulas

5 Sampling, PCM and delta modulation — 9 formulas

6 Multiplexing — 3 formulas

7 Receivers and digital modulation — 10 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

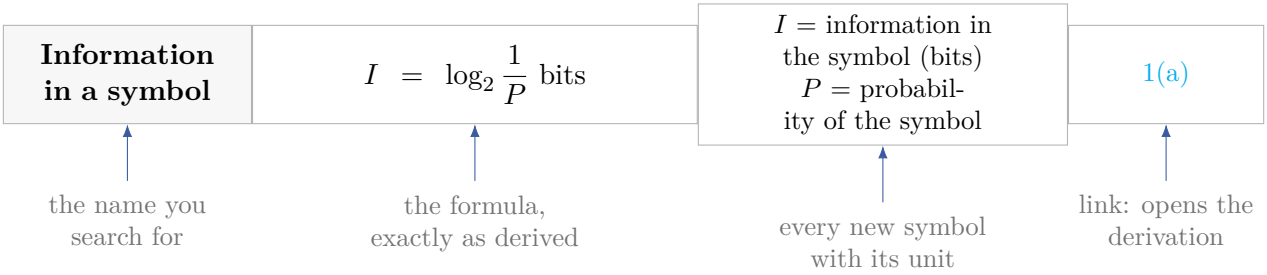
Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

Before you use the sheet

How to read a row



In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

Contents

1 Information and channel capacity 4

2 Noise 5

3 Amplitude modulation 5

4 Angle modulation 6

5 Sampling, PCM and delta modulation 7

6 Multiplexing 8

7 Receivers and digital modulation 8

8 Key formulas 10

9 Symbol index 11

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Conventions (Lathi and Ding, 4th ed.): $\omega = 2\pi f$; $m(t)$ is the message, m_p its peak value and B its bandwidth (Hz); μ is the AM modulation index and β the FM one; $\operatorname{sinc} x = \sin x/x$. Signal amplitudes are peak values.

Columns: Title · Equation · Notation · Ref. (reference).

1. Information and channel capacity:

Title	Equation	Notation	Ref.
Information in a symbol	$I = \log_2 \frac{1}{P}$ bits	I = information in the symbol (bits) P = probability of the symbol	1(a)
★ **Entropy**	$H = \sum_i P_i \log_2 \frac{1}{P_i}$ bits/symbol	H = entropy, the average information per symbol (bits/symbol) P_i = probability of symbol i	1(a)
★ **Binary entropy**	$H(p) = -p \log_2 p - (1-p)\log_2(1-p)$	p = probability of one of the two symbols $H_{max} = 1$ bit at $p = 0.5$	1(a)
Equally likely levels	$H = \log_2 L$	L = number of levels	3(d)
★ **Information rate**	$R = f_s H$ bit/s	R = information rate (bit/s) f_s = symbols per second	3(d)
★ **Shannon capacity**	$C = B \log_2 \left(1 + \frac{S}{N}\right)$	C = channel capacity (bit/s) B = bandwidth (Hz) S = signal power (W) N = noise power (W) S/N = plain power ratio; need $R \le C$	3(d)
Decibels to ratio	$\frac{S}{N} = 10^{\mathrm{dB}/10}$	20 dB = 100; 30 dB = 1000	5(b)
Minimum SNR	$\mathrm{SNR}_{min} = 2^{R/B} - 1$	at the limit $R = C$	5(b)
Bandwidth for a rate	$B = \frac{R}{\log_2(1 + S/N)}$	at the limit $R = C$	3(d)
Attenuation	$20 \log_{10} \frac{V_t}{V_r}$ dB	V_t, V_r = sent and received amplitudes (V)	7(c)
SNR from amplitudes	$\mathrm{SNR} = (V_s/V_n)^2$	V_s = signal amplitude (V) V_n = noise amplitude (V) same resistance	7(c)
Signal for a rate	$V_s = V_n \sqrt{2^{R/B} - 1}$	for a wanted rate R: 100 Mbps in 5 MHz needs 5.12 V	7(c)

Table 1: Information and channel capacity

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2. Noise:

Title	Equation	Notation	Ref.
★ **Thermal noise power**	$P_n = kTB$	P_n = thermal noise power (W) k = Boltzmann's constant, 1.38×10^{-23} J/K T = temperature (K) B = bandwidth (Hz)	1(c)
Thermal noise voltage	$V_n = \sqrt{4kTBR}$	V_n = r.m.s. noise voltage (V) R = resistance (Ω)	1(c)
★ **Cascade noise resistance**	$R_{eq} = R_1 + \frac{R_2}{A_1^2} + \frac{R_3}{A_1^2 A_2^2} + \cdots$	R_{eq} = equivalent noise resistance at the first input (Ω) R_1, R_2, R_3 = noise resistances at the inputs of stages 1, 2, 3 (Ω) A_1, A_2 = voltage gains of stages 1, 2	7(d)
★ **Distortionless channel**	$H(f) = k\, e^{-j2\pi f t_d}$	$H(f)$ = channel transfer function f = frequency (Hz) k = constant gain (here, not Boltzmann's constant) j = imaginary unit, $\sqrt{-1}$ t_d = time delay (s) $\lvert H\rvert = k$, $\theta_h = -2\pi f t_d$ over the band	7(b)

Table 2: Noise

3. Amplitude modulation:

Title	Equation	Notation	Ref.
AM wave	$s(t) = [E_c + E_m \sin \omega_m t] \sin \omega_c t$	$s(t)$ = AM wave (V) t = time (s) E_c, E_m = peak carrier and message amplitudes (V) ω_m = message angular frequency (rad/s) ω_c = carrier angular frequency (rad/s)	1(d)
★ **Modulation index**	$\mu = \frac{E_m}{E_c}$	μ = AM modulation index no distortion in an envelope detector if $\mu \leq 1$	1(d)
Side frequencies	$f_c \pm f_m$, amplitude $\frac{\mu E_c}{2}$	f_c = carrier frequency (Hz) f_m = message frequency (Hz) each side	1(d)
Carrier power	$P_c = \frac{E_c^2}{2R}$	P_c = carrier power (W) R = load (Ω); peak values	1(e)
★ **Total power**	$P_t = P_c \left(1 + \frac{\mu^2}{2}\right)$	P_t = total power (W) carrier plus two sidebands	1(e)
DSB-SC spectrum	$S(f) = \frac{A_c}{2}[M(f - f_c) + M(f + f_c)]$	$S(f)$ = DSB-SC spectrum A_c = carrier amplitude (V) $M(f)$ = message spectrum	5(d)
Coherent detector product	$V(f) = \frac{A_c}{2} M(f) + \frac{A_c}{4}[M(f - 2f_c) + M(f + 2f_c)]$	$V(f)$ = spectrum of the product local carrier $\cos 2\pi f_c t$	5(d)

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Title	Equation	Notation	Ref.
★ **Lowest carrier**	$f_c \geq W$	W = message bandwidth (Hz)	5(d)
★ **VSB receiver filter**	$H_o(f) = \dfrac{1}{H_i(f+f_c) + H_i(f-f_c)}$	$H_o(f)$ = receiver filter $H_i(f)$ = transmit filter B = message bandwidth (Hz); holds for $\lvert f\rvert \leq B$	4(b)
AM spectrum	$\pi A[\delta(\omega-\omega_c)+\delta(\omega+\omega_c)] + \frac{1}{2}[X(\omega-\omega_c)+X(\omega+\omega_c)]$	A = carrier amplitude (V) $\delta(\cdot)$ = impulse ω = angular frequency (rad/s) $X(\omega)$ = message spectrum SSB-SC keeps one half of each copy	8(a)
Coherent detector with errors	$e_o = [A + m(t)]\cos(\Delta\omega t + \delta)$	e_o = detector output (V) $m(t)$ = message signal (V) $\Delta\omega$ = LO frequency error (rad/s) δ = LO phase error (rad) (here, not an impulse)	8(b)
★ **AM antenna current**	$I_t = I_c\sqrt{1+\mu^2/2}$	I_t = antenna current with modulation (A) I_c = antenna current of the carrier alone (A) $\mu = 0.70$ for $8 \rightarrow 8.93$ A	8(c)

Table 3: Amplitude modulation

4. Angle modulation:

Title	Equation	Notation	Ref.
Instantaneous frequency	$\omega_i = \dfrac{d\theta}{dt}$	ω_i = instantaneous angular frequency (rad/s) $\theta(t)$ = carrier angle (rad) t = time (s)	2(c)
PM	$\varphi_{PM} = A\cos[\omega_c t + k_p m(t)]$	φ_{PM} = PM signal (V) A = carrier amplitude (V) ω_c = carrier angular frequency (rad/s) k_p = phase sensitivity (rad/V) $m(t)$ = message signal (V)	2(c)
FM	$\varphi_{FM} = A\cos\left[\omega_c t + k_f \int m\, d\alpha\right]$	φ_{FM} = FM signal (V) k_f = frequency sensitivity α = integration variable (s)	2(c)
★ **Frequency deviation**	$\Delta f = k_f V_m$	Δf = peak frequency deviation (Hz) V_m = message peak amplitude (V) k_f in Hz/V; independent of f_m	2(d)
★ **FM modulation index**	$\beta = \dfrac{\Delta f}{f_m}$	β = FM modulation index f_m = message (tone) frequency (Hz) can exceed 1; no unit	2(d)
★ **Carson's rule**	$B_T \approx 2(\Delta f + f_m) = 2(\beta+1)f_m$	B_T = transmission bandwidth (Hz) approximate	4(d)

Title	Equation	Notation	Ref.
Deviation from sensitivity	$\Delta f = k_f A_m$	A_m = message peak amplitude (V) 2 kHz/V × 5 V = 10 kHz	8(e)
De-emphasis	$H_d(f) = \dfrac{1}{1 + jf/f_1},\ f_1 = \dfrac{1}{2\pi RC}$	$H_d(f)$ = de-emphasis filter f = frequency (Hz) j = imaginary unit, $\sqrt{-1}$ f_1 = corner frequency (Hz) R, C = resistance (Ω) and capacitance (F) of the filter 75 μs: $f_1 = 2.12$ kHz; $H_p H_d = 1$	8(d)

Table 4: Angle modulation

5. Sampling, PCM and delta modulation:

Title	Equation	Notation	Ref.
★ **Nyquist rate**	$f_s \geq 2B$	f_s = sampling rate (samples/s) B = highest message frequency (Hz)	4(c)
Sampled spectrum	$\bar{G}(f) = \dfrac{1}{T_s}\sum_n G(f - nf_s)$	$G(f)$ = spectrum of the message $\bar{G}(f)$ = spectrum of the sampled signal f = frequency (Hz) T_s = sampling interval, $1/f_s$ (s) n = summation index	4(c)
★ **Interpolation formula**	$g(t) = \sum_k g(kT_s)\,\mathrm{sinc}(2\pi Bt - k\pi)$	$g(t)$ = message signal t = time (s) k = sample number $\mathrm{sinc}\,x = \sin x/x$; $T_s = 1/2B$	4(c)
Bits per sample	$n = \log_2 L$	n = bits per sample (here, not an index) L = number of levels, 2^n	3(c)
★ **PCM bit rate**	$R_b = nf_s$	R_b = PCM bit rate (bit/s)	3(c)
Quantization step	$\Delta = \dfrac{2m_p}{L}$	Δ = quantization step (V) m_p = peak signal (V)	5(c)
Largest error	$e_{max} = \dfrac{\Delta}{2} = \dfrac{m_p}{L}$	e_{max} = largest quantization error (V) half a step	5(c)
★ **DM largest slope**	σf_s	σ = DM step size (V)	3(a)
No slope overload	$\left\lvert\dfrac{dm}{dt}\right\rvert_{max} \leq \sigma f_s$	$m(t)$ = message signal (V) for a tone, $A\omega_m \leq \sigma f_s$	3(a)

Table 5: Sampling, PCM and delta modulation

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6. Multiplexing:

Title	Equation	Notation	Ref.
★ **FDM link bandwidth**	$B_{link} = NB_{ch} + (N-1)B_g$	B_{link} = link bandwidth (Hz) N = number of channels B_{ch} = bandwidth of one channel (Hz) B_g = guard band (Hz)	2(b)
★ **TDM bit rate**	$R_b = Nf_sn$	R_b = TDM bit rate (bit/s) f_s = sampling rate per channel (samples/s) n = bits per sample	5(c)
Minimum bandwidth	$B_T = \frac{R_b}{2}$	B_T = minimum transmission bandwidth (Hz) binary pulses	5(c)

Table 6: Multiplexing

7. Receivers and digital modulation:

Title	Equation	Notation	Ref.
★ **Local oscillator**	$f_{LO} = f_s + f_{IF}$	f_{LO} = local-oscillator frequency (Hz) f_s = signal (station) frequency (Hz) f_{IF} = intermediate frequency, 455 kHz for AM	4(a)
★ **Image frequency**	$f_{si} = f_s + 2f_{IF}$	f_{si} = image frequency (Hz) must be removed before the mixer	4(a)
FSK frequencies	$f_{1,0} = f_c \pm \Delta f$	$f_{1,0}$ = FSK frequencies: f_1 for bit 1, f_0 for bit 0 (Hz) f_c = carrier frequency (Hz) Δf = frequency deviation (Hz)	6(a)
★ **FSK bandwidth**	$B_{FSK} \approx 2\Delta f + 2R_b$	B_{FSK} = FSK bandwidth (Hz) R_b = bit rate (bit/s)	6(a)
FSK bandwidth (Forouzan)	$B = (1+d)S + 2\Delta f$	B = FSK bandwidth (Hz) S = baud rate (baud) d = modulation factor, $0 \leq d \leq 1$	6(a)
★ **QAM signal**	$\varphi_{QAM} = m_1 \cos \omega_c t + m_2 \sin \omega_c t$	φ_{QAM} = QAM signal (V) m_1, m_2 = the two messages (V) ω_c = carrier angular frequency (rad/s) t = time (s)	6(b)
QAM with phase error	$m_1 \cos\theta - m_2 \sin\theta$	θ = phase error of the local carrier (rad) upper output; crosstalk	6(b)
FHSS channels	$M = 2^k$, $B_{SS} \approx MB$	M = number of hop channels k = PN pattern bits B_{SS} = spread bandwidth (Hz) B = bandwidth of one channel (Hz) (here, not the FSK bandwidth)	6(c)

Title	Equation	Notation	Ref.
★ **DPSK encoder**	$q_k = I_k \oplus q_{k-1}$	q_k = encoded bit k q_{k-1} = previous encoded bit I_k = data bit k	6(d)
DPSK receiver	$y_k \propto a_k a_{k-1}$	y_k = receiver output for bit k a_k = sign (± 1) of pulse k a_{k-1} = sign of the previous pulse negative means 1	6(d)

Table 7: Receivers and digital modulation

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8. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Entropy	$H = \sum_i P_i \log_2 \dfrac{1}{P_i}$ bits/symbol
Binary entropy	$H(p) = -p \log_2 p - (1-p)\log_2(1-p)$
Information rate	$R = f_s H$ bit/s
Shannon capacity	$C = B \log_2 \left(1 + \dfrac{S}{N}\right)$
Thermal noise power	$P_n = kTB$
Cascade noise resistance	$R_{eq} = R_1 + \dfrac{R_2}{A_1^2} + \dfrac{R_3}{A_1^2 A_2^2} + \cdots$
Distortionless channel	$H(f) = k\, e^{-j2\pi f t_d}$
Modulation index	$\mu = \dfrac{E_m}{E_c}$
Total power	$P_t = P_c \left(1 + \dfrac{\mu^2}{2}\right)$
Lowest carrier	$f_c \geq W$
VSB receiver filter	$H_o(f) = \dfrac{1}{H_i(f+f_c) + H_i(f-f_c)}$
AM antenna current	$I_t = I_c \sqrt{1 + \mu^2/2}$
Frequency deviation	$\Delta f = k_f V_m$
FM modulation index	$\beta = \dfrac{\Delta f}{f_m}$
Carson's rule	$B_T \approx 2(\Delta f + f_m) = 2(\beta + 1) f_m$
Nyquist rate	$f_s \geq 2B$
Interpolation formula	$g(t) = \sum_k g(kT_s)\,\mathrm{sinc}(2\pi B t - k\pi)$
PCM bit rate	$R_b = n f_s$
DM largest slope	σf_s
FDM link bandwidth	$B_{link} = N B_{ch} + (N-1) B_g$
TDM bit rate	$R_b = N f_s n$
Local oscillator	$f_{LO} = f_s + f_{IF}$
Image frequency	$f_{si} = f_s + 2 f_{IF}$
FSK bandwidth	$B_{FSK} \approx 2\Delta f + 2R_b$
QAM signal	$\varphi_{QAM} = m_1 \cos \omega_c t + m_2 \sin \omega_c t$
DPSK encoder	$q_k = I_k \oplus q_{k-1}$

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9. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
I	information in the symbol (bits)	Information in a symbol
P	probability of the symbol	Information in a symbol
H	entropy, the average information per symbol (bits/symbol)	Entropy
P_i	probability of symbol i	Entropy
p	probability of one of the two symbols	Binary entropy
L	number of levels	Equally likely levels
R	information rate (bit/s)	Information rate
f_s	symbols per second	Information rate
C	channel capacity (bit/s)	Shannon capacity
B	bandwidth (Hz)	Shannon capacity
S	signal power (W)	Shannon capacity
N	noise power (W)	Shannon capacity
S/N	plain power ratio	Shannon capacity
20 dB	100	Decibels to ratio
30 dB	1000	Decibels to ratio
V_t, V_r	sent and received amplitudes (V)	Attenuation
V_s	signal amplitude (V)	SNR from amplitudes
V_n	noise amplitude (V)	SNR from amplitudes
P_n	thermal noise power (W)	Thermal noise power
k	Boltzmann's constant, 1.38×10^{-23} J/K	Thermal noise power
T	temperature (K)	Thermal noise power
V_n	r.m.s. noise voltage (V)	Thermal noise voltage
R	resistance (Ω)	Thermal noise voltage
R_{eq}	equivalent noise resistance at the first input (Ω)	Cascade noise resistance
R_1, R_2, R_3	noise resistances at the inputs of stages 1, 2, 3 (Ω)	Cascade noise resistance
A_1, A_2	voltage gains of stages 1, 2	Cascade noise resistance
$H(f)$	channel transfer function	Distortionless channel
f	frequency (Hz)	Distortionless channel
k	constant gain (here, not Boltzmann's constant)	Distortionless channel
j	imaginary unit, $\sqrt{-1}$	Distortionless channel
t_d	time delay (s)	Distortionless channel
$s(t)$	AM wave (V)	AM wave
t	time (s)	AM wave
E_c, E_m	peak carrier and message amplitudes (V)	AM wave
ω_m	message angular frequency (rad/s)	AM wave
ω_c	carrier angular frequency (rad/s)	AM wave

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Symbol	Meaning	First used in
μ	AM modulation index	Modulation index
f_c	carrier frequency (Hz)	Side frequencies
f_m	message frequency (Hz)	Side frequencies
P_c	carrier power (W)	Carrier power
R	load (Ω)	Carrier power
P_t	total power (W)	Total power
$S(f)$	DSB-SC spectrum	DSB-SC spectrum
A_c	carrier amplitude (V)	DSB-SC spectrum
$M(f)$	message spectrum	DSB-SC spectrum
$V(f)$	spectrum of the product	Coherent detector product
W	message bandwidth (Hz)	Lowest carrier
$H_o(f)$	receiver filter	VSB receiver filter
$H_i(f)$	transmit filter	VSB receiver filter
A	carrier amplitude (V)	AM spectrum
$\delta(\cdot)$	impulse	AM spectrum
ω	angular frequency (rad/s)	AM spectrum
$X(\omega)$	message spectrum	AM spectrum
e_o	detector output (V)	Coherent detector with errors
$m(t)$	message signal (V)	Coherent detector with errors
$\Delta\omega$	LO frequency error (rad/s)	Coherent detector with errors
δ	LO phase error (rad) (here, not an impulse)	Coherent detector with errors
I_t	antenna current with modulation (A)	AM antenna current
I_c	antenna current of the carrier alone (A)	AM antenna current
ω_i	instantaneous angular frequency (rad/s)	Instantaneous frequency
$\theta(t)$	carrier angle (rad)	Instantaneous frequency
φ_{PM}	PM signal (V)	PM
k_p	phase sensitivity (rad/V)	PM
φ_{FM}	FM signal (V)	FM
k_f	frequency sensitivity	FM
α	integration variable (s)	FM
Δf	peak frequency deviation (Hz)	Frequency deviation
V_m	message peak amplitude (V)	Frequency deviation
β	FM modulation index	FM modulation index
B_T	transmission bandwidth (Hz)	Carson's rule
A_m	message peak amplitude (V)	Deviation from sensitivity
2 kHz/V $\times$ 5 V	10 kHz	Deviation from sensitivity

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Symbol	Meaning	First used in
$H_d(f)$	de-emphasis filter	De-emphasis
f_1	corner frequency (Hz)	De-emphasis
R, C	resistance (Ω) and capacitance (F) of the filter	De-emphasis
f_s	sampling rate (samples/s)	Nyquist rate
B	highest message frequency (Hz)	Nyquist rate
$G(f)$	spectrum of the message	Sampled spectrum
$\bar{G}(f)$	spectrum of the sampled signal	Sampled spectrum
T_s	sampling interval, $1/f_s$ (s)	Sampled spectrum
n	summation index	Sampled spectrum
$g(t)$	message signal	Interpolation formula
k	sample number	Interpolation formula
n	bits per sample (here, not an index)	Bits per sample
R_b	PCM bit rate (bit/s)	PCM bit rate
Δ	quantization step (V)	Quantization step
m_p	peak signal (V)	Quantization step
e_{max}	largest quantization error (V)	Largest error
σ	DM step size (V)	DM largest slope
B_{link}	link bandwidth (Hz)	FDM link bandwidth
N	number of channels	FDM link bandwidth
B_{ch}	bandwidth of one channel (Hz)	FDM link bandwidth
B_g	guard band (Hz)	FDM link bandwidth
R_b	TDM bit rate (bit/s)	TDM bit rate
f_{LO}	local-oscillator frequency (Hz)	Local oscillator
f_s	signal (station) frequency (Hz)	Local oscillator
f_{IF}	intermediate frequency, 455 kHz for AM	Local oscillator
f_{si}	image frequency (Hz)	Image frequency
$f_{1,0}$	FSK frequencies: f_1 for bit 1, f_0 for bit 0 (Hz)	FSK frequencies
B_{FSK}	FSK bandwidth (Hz)	FSK bandwidth
S	baud rate (baud)	FSK bandwidth (Forouzan)
d	modulation factor, $0 \leq d \leq 1$	FSK bandwidth (Forouzan)
φ_{QAM}	QAM signal (V)	QAM signal
m_1, m_2	the two messages (V)	QAM signal
θ	phase error of the local carrier (rad)	QAM with phase error
M	number of hop channels	FHSS channels
k	PN pattern bits	FHSS channels
B_{SS}	spread bandwidth (Hz)	FHSS channels
q_k	encoded bit k	DPSK encoder

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Symbol	Meaning	First used in
q_{k-1}	previous encoded bit	DPSK encoder
I_k	data bit k	DPSK encoder
y_k	receiver output for bit k	DPSK receiver
a_k	sign (± 1) of pulse k	DPSK receiver
a_{k-1}	sign of the previous pulse	DPSK receiver

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